

21.30M TUG BOAT

FINAL INTACT STABILITY BOOKLET



APPROVED

DATE: 13 Jun 2017

**INDIAN REGISTER OF SHIPPING
MUMBAI**

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1 Introduction:

This Booklet consists of final intact stability analysis of the vessel "21.30M Tug Boat" with a bollard pull of 10.00 Tons complying the necessary stability requirements.

2 Main Particulars of the vessel:

2.1 General particulars

Name of the Vessel	:	KB-4
Client	:	KB SHIPPING
Type	:	Tug Boat
Flag	:	Indian
Class	:	Indian Register of Shipping
Date of keel laying	:	01-09-2016
Official Number	:	4394
Port of registry	:	Mumbai
Call Sign	:	AWSO
IMO Number	:	9823376
Yard Number	:	002
Engine	:	2 x 400 BHP
Complement	:	12 Persons
Class Notation	:	[SUL] TUG, [SUL] IV, INDIAN RIVER SEA VESSEL TYPE 4

2.2 Principal particulars

Length O.A.	:	21.30 M
Length B.P	:	19.76 m
Breadth (MLD)	:	06.50 M.
Depth (MLD)	:	02.80 m
Design Draft	:	1.968 m
Scantling Draft	:	2.00 m
Full Load Displacement	:	176.671 T

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Deadweight : 68.935 T

Design speed : 10 knots

Bollard Pull : 10 T

2.3 Tonnage

Gross Tonnage : 107

Net Tonnage : 32

3 Lightship weight particulars

Lightship weight : 107.736 T

Lightship LCG : 9.276 m forward of AP (Fr.2)

Lightship TCG : 0.018 m off centerline towards starboard

Lightship VCG : 2.548 m above baseline

Note:

1. The lightship weight and Centre of gravity used in this booklet is taken from the SBM-16-1011-706-0 Inclining Experiment Report with IRS ref. letter-E-49483-61968.
2. The above Lightship weight is without including solid ballast.

4 Solid Ballast details

The solid ballast of 845 kg is installed in Engine room between FR.17 & FR.18 for initial list correction. The detail information and the drawing of solid ballast are given in the Appendix.

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5 Notes on the Use of Stability Booklet

5.1 Conversion Tables

The Government of India strongly recommends use of S.I (Systems International Units). The following table shows conversion factors for converting British Units to Metric Units and vice versa.

Multiply by	To Convert from	To Obtain	
0.03937	Millimeters	Inches	25.4
0.3937	Centimeters	Inches	2.54
3.2808	Meters	Feet	0.3048
2.2046	Kilograms	Pound	0.45359
0.0009842	Kilograms	Tons (2240 lbs)	1016.047
0.9842	Metric Tonnes	Tones (2240 lbs)	1.016
2.4998	Metric Tonne per centimeter of immersion	Tones per inch of immersion	0.04
8.2014	Moment to change trim by one centimeter (tonne-metre)	Moment to change trim by one inch (Ton-foot)	0.122
0.0174532	Degree	Radians	57.2957795
	TO OBTAIN	TO CONVERT FROM	MULTIPLY BY

Relation between Weight and Volume:

10 ³ mm	=	1 cm ³
1 cm ³ of fresh water	=	1 Gram
1000 cm ³ of fresh water	=	1 Kilogram (1000 Grams)
1 m ³ of fresh water	=	1 Metric tonne (1000 Kilograms)
1 m ³ of salt water (Specific Gravity = 1.025 t/m ³)	=	1.025 Metric Tonnes
1 Metric Tonne of salt water	=	0.975 m ³
1 m ³	=	35.316 ft ³
1 ft ³	=	0.0283 m ³

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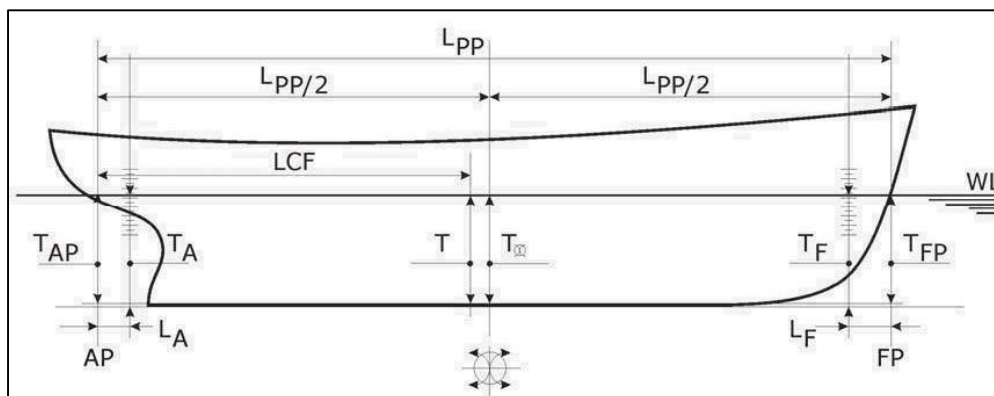
5.2 Symbols and Sign Convention

SYMBOL	DESCRIPTION	UNIT
Δ	Displacement	t
$\bar{v}$	Volume of Displacement	m ³
AP	Aft perpendicular	m
A _{WP}	Area of Water Plane	m ²
B	Breadth Moulded	m
D	Depth Moulded (Upto Main Deck at Side)	m
Dwt	Deadweight	tonnes
FP	Forward perpendicular	m
L _A	Distance from AP to aft draft mark (positive value if draft mark is positioned forward of AP)	m
LBP	Length between perpendiculars	m
LCB	Longitudinal Centre of Buoyancy (from A.P. i.e., Fr-2)	m
LCF	Longitudinal Centre of Flotation (from A.P. i.e., Fr-2)	m
LCG	Longitudinal centre of gravity (from A.P. i.e., Fr-2)	m
L _F	Distance from FP to forward draft mark (positive value if draft mark is positioned forward of FP)	m
TCG	Transverse centre of gravity from centre line	m
GG _o	Free surface correction	m
GM _T	Transverse metacentric height	m
GoM _T	Corrected transverse metacentric height	m
GM _L	Longitudinal metacentric height	m
GZ	Righting arm	m
GT	Gross Tonnage	-
GoZ	Corrected righting arm	m
GoZ _{Max}	Corrected maximum righting arm	m
KB	Vertical Centre of Buoyancy (from Base Line)	m
KG/VCG	Vertical centre of gravity from base line	m

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KGo	Corrected vertical centre of gravity from base line	m
KMT	Transverse Metacentric Height (from Base Line)	m
KN	Distance from base line to metacentric axis	m
MCT_{lcm}	Moment to change trim by one centimeter	tones x m/cm
NT	Net Tonnage	-
TPC	Tones per centimeter immersion	tonnes/cm
t	Total Trim ($T_{FP} - T_{AP}$)	m
T	Draft at LCF	m
T_{AP}	Draft Aft at AP	m
T_A	Draft aft draft marks from underside of keel	m
T_{FP}	Draft Fwd at FP	m
T_F	Draft forward draft marks from underside of keel	m
T_M	Draft Mean at Midship	m
W_{llool}	Wind heeling arm	m
θ	Angle of inclination in degrees	deg
ρ	Density of liquid	tonnes/m ³
θ_{Max}	Angle at which maximum G_0Z occurs	deg

5.3 Sign Convention:



LCB, LCG, LCF	Fwd of AP is positive Aft of AP is Negative
Trim by Aft	Negative
Trim by Forward	Positive

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5.4 General Instructions to Master against Capsizing

The fact that the vessel's stability has been calculated for several conditions of loading in which the vessel normally operates, and that such calculations have proven that the stability is good and in excess of the IMO requirements, and the fact that the calculations are checked for approval does not mean that the vessel is now immune against capsizing. A stamped and approved copy of this booklet must be on board at all times. The following measures should be considered as preliminary guidance on matters influencing safety and stability.

1. Compliance with the stability criteria does not ensure immunity against capsizing, regardless of the circumstances, or absolve the master from his responsibilities. Masters should therefore exercise prudence and good seamanship having regard to the season of the year weather forecasts and the navigational zone and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Before a voyage commences, care should be taken to ensure that the cargo and sizeable pieces of equipment have been properly stowed or lashed so as to minimize the possibility of both longitudinal and shifting, while at sea, under the effect of acceleration caused by rolling and pitching.
3. The stability criteria recommended as per IS Code 2008 and set minimum values, but no maximum values are recommended. It is advisable to avoid excessive values of metacentric height, since these might lead to acceleration forces which could be prejudicial to the vessel, its complement, its equipment and to safe carriage of the cargo.
4. All doorways and other openings through which can enter into the hull or deckhouses, forecabin, etc., should be suitably closed in adverse weather conditions and accordingly all appliances for this purpose should be maintained on board and in good condition.
5. Weather tight and watertight hatches, doors, etc., should be kept closed during navigation, except when necessarily opened for the working of the vessel and should always be ready for immediate closure and be clearly marked to indicate that these fittings are to be kept closed except for access. All portable deadlights should be maintained in good condition and securely closed in bad weather.
6. Any closing devices provided for vent pipes to fuel tanks should be secured in bad weather.
7. Reliance on automatic steering may be dangerous as this prevents ready changes to course which may be needed in bad weather.
8. In all conditions of loading necessary care should be taken to maintain a seaworthy freeboard.
9. In severe weather, the speed of the vessel should be reduced if excessive rolling, propeller emergence, shipping of water on deck or heavy slamming occurs. Six heavy slamming or 25 propeller emergences during 100 pitching motions should be considered dangerous.

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10. Special attention should be paid when a vessel is sailing in following or quartering seas because dangerous phenomena such as parametric resonance, broaching to reduction of stability on the wave crest, and excessive rolling may occur singularly in sequence or simultaneously in a multiple combination, creating a threat of capsizing. Particularly dangerous is the situation when the wave length is of the order of 1.0-1.5 vessel's length. A vessel's speed and / or course should be altered appropriately to avoid the above mentioned phenomena.
11. Water trapping in deck wells should be avoided. If freeing ports are not sufficient for draining of the well, either speed of the vessel should be reduced, or course should be changed, or both. Freeing ports provided with closing appliances should always be capable of functioning and are not to be locked.
12. Masters should be aware that steep or breaking waves may occur in certain areas, or in certain wind and current combinations.

5.5 Notes on Stability

Stability

The stability of a vessel in general is its ability to maintain an upright position or to re-establish this after disturbing by an external force or moment. For an undamaged vessel at small angle of heel up to 5° , it is sufficient to evaluate the stability in transverse direction. This depends on position of two points relative to each other, centre of gravity (G) and transverse metacenter (M), see figure. 2.4.1. The metacentric height GM, the distance between the points G and M, means the stability at small angles and is given by the following equation:

$$GM_T = KM_T - KG.$$

The centre of gravity of the vessel (KG) depends on distribution of cargo and liquids in tanks in the vessel. Vessel KG for a particular loading condition is obtained by taking vertical moments (sum of weight of an item x centre of gravity of that item) and by division of total moments with the total weights.

The position of the transverse metacentre varies with the draft. For a certain draft corresponding to a certain displacement, the transverse metacentric position (M) above the keel point K is fixed and is denoted as KM_T . The location of the metacentre has neither to do with the nature nor the distribution of cargo & other weights onboard. The transverse metacentre above keel, only dependent on the lines of the vessel, shall be obtained from hydrostatics curves or tables.

Cross curves of stability

Position of transverse metacentre (M) is not fixed as the angle of heel exceeds beyond 5° and GM is no longer a stability criterion for measuring the stability of the vessel. Therefore righting lever or arm (GZ) shall be used for this purpose.

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These curves show the variation of the righting arms (GZ) with displacement and are drawn for various angles of inclination, each curve being drawn for a particular angle of inclination. When these curves are drawn for a fixed value of the vertical center of gravity position, they are called “GZ Curves”. When the vertical center of gravity is assumed to be zero, these curves are called “KN Curves” and the same are used to derive the statical stability curve.

When the angle of inclination exceeds small angles, 4° or 5° the intersection of the vertical through the center of buoyancy in the inclined condition with the center line of vessel, point “M” can no longer be regarded as a fixed point relative to the vessel. The quantum “GM” no longer a suitable criterion for measuring the stability of the vessel and it is usual to use the value of the righting arm “GZ” for this purpose. The Cross Curves of stability provide a means of presenting the righting arm “GZ” for any probable value of displacement and for several angles of heel. The cross curves of stability depends on the geometry of the vessel and is independent of the loading of the vessel.

Curve of Statical Stability:

A curve of righting arm GZ to a base of angle of inclination for a fixed displacement is known as a curve of statical stability. This curve is provided for every intended loading condition in the booklet. To illustrate the righting levers at various angles of heel in a condition, the effective righting lever GZ is derived from:

$$GZ = KN - KG \times \sin \theta$$

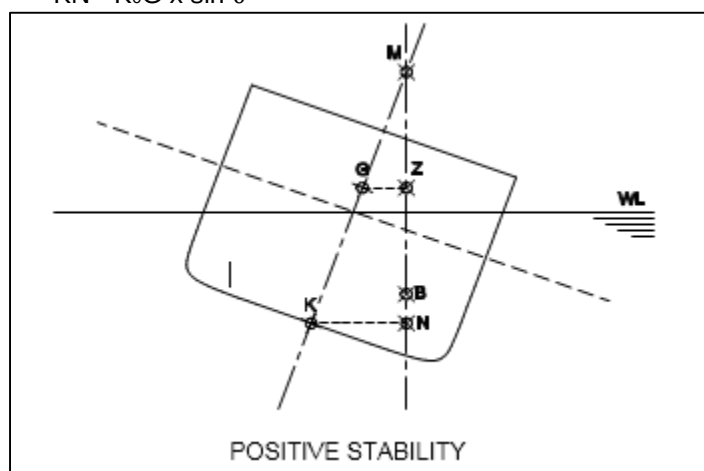
KN = Horizontal distance from keel to the vertical line passing through centre of Buoyancy.

KG = Centre of gravity from base line.

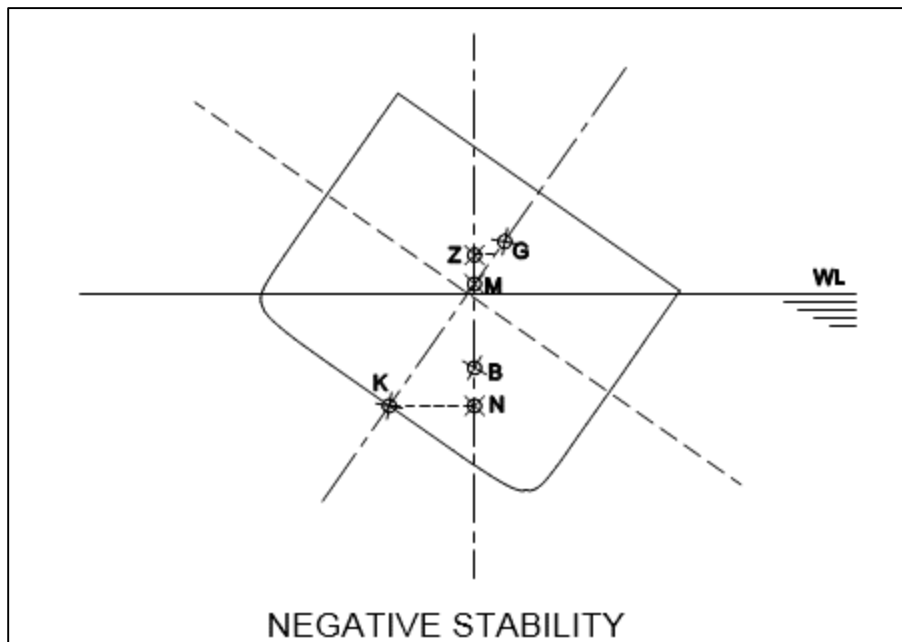
θ = Angle of inclination (heel).

Righting arm after correction of free surface effects is given by G_0Z .

$$G_0Z = KN - K_0G \times \sin \theta$$



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Use of Tables:

The tables for hydrostatics, cross curves and maximum permissible KG should be entered with the ship's moulded draught amidships, T_M , measured from the upper side of the keel plate.

$$T_M = \frac{T_{AP} + T_{FP}}{2}$$

Draft and Trim Calculation:

For calculation of trim and draft based on a loading condition calculation, the following data are required for ascertaining trim and draft.

- Longitudinal centre of gravity of the ship calculated as follows.

$$LCG = \frac{\text{SUMMATION OF LONGITUDINAL WEIGHT MOMENTS}}{\text{DISPLACEMENT}}$$

- Longitudinal centre of buoyancy, LCB
- Displacement, Δ
- Moment to change trim, MCT
- Longitudinal centre of Floatation, LCF.

LCB, LCF and MCT are taken from Hydrostatic tables (trim = 0 metres) corresponding to the actual displacement.

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Trim is calculated according to the following formula:

$$\text{Trim (t)} = \frac{LBP \times (LCG - LCB)}{GM_L}$$

The Moulded Drafts at AP (T_{AP}) and FP (T_{FP}) are calculated to following formulas:

$$\text{Draft at LCF (m)} T_{LCF} = T_m - \left(\frac{\frac{LBP}{2} - LCF}{LBP} \right) \times TRIM$$

$$T_{AP} = T_{LCF} - \frac{LCF \times TRIM}{LBP}$$

$$T_{FP} = T_{AP} + TRIM$$

Where T is the draft taken from Hydrostatic Tables ($t = 0$ metres) corresponding to the actual displacement.

The draft at draft marks is as calculated as follows:

$$T_A = T_{AP} + \frac{L_A \times TRIM}{LBP} + \text{keel plate thickness}$$

$$T_F = T_{FP} + \frac{L_F \times TRIM}{LBP} + \text{keel plate thickness}$$

If the righting lever curve (GZ Curve) has to be established, the KM_T and KN values to be used for this purpose are calculated for the calculated trim value. This is done by interpolation for the KMT and KN values, respectively, between the two tabulated trim values nearest to calculated trim.

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6 Stability Criteria

6.1 IMO Intact Stability Criteria

6.1.1 General Criteria

The vessel is required to comply with the general intact stability criteria laid down in MSC.267 (85) (IS Code 2008) Part A, Chapter 2.

Recommended General Criteria:

2.2.1 The area under the righting lever curve (GZ curve) should not be less than 0.05 meter Radians up to an angle of 30° and not be less than 0.09 meter radians up to $\theta = 40^\circ$ or the angle of flooding θ_f . If this angle is less than 40° . Additionally, the area under the righting lever curve (GZ curve) between the angles of heel of 30° and 40° or between 30° and θ_f , if this angle is less than 40° , should not be less than 0.030 meter-radians.

2.2.2 The right lever GZ should be at least 0.20 m at an angle of heel equal to or greater than 30° .

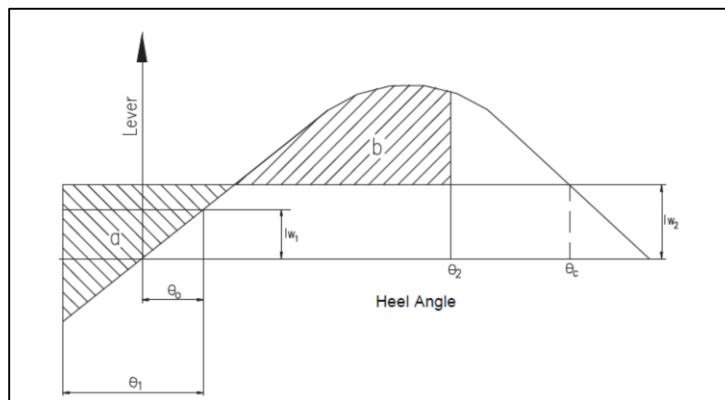
2.2.3 The maximum righting lever shall occur at an angle of heel not less than 25° . If this is not practicable, alternative criteria, based on an equivalent level of safety, may be applied subject to the approval of the Administration.

2.2.4 The initial metacentric height GM_0 should not be less than 0.15 m.

6.1.2 Severe Wind and Rolling Criteria (Weather Criteria)

The following criteria is extracted from IS Code 2008, Part A Chapter 2.3

The ability of a ship to withstand the combined effects of beam wind & rolling should be demonstrated for each standard condition of loading, with reference to the figure as following:



Area- B should be greater than Area- A

Maximum allowable static heel angle should be less than 16° or an angle corresponding to 80% of the deck immersion angle whichever is less.

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Where,

Area-B is the area under the righting curve up to an angle of 50° or second intercept of lw_2 or Down Flooding angle whichever is less.

Area- A is the area under the curve up to the angle of roll back measured backward from equilibrium angle due to steady wind as shown in figure.

Intermediate value in tables 1-4 should be obtained by liner interpolation

$$\text{Rolling period } T = \frac{2XCXB}{\sqrt{GM}} \text{ seconds}$$

$$\text{Where, } C = 0.373 + 0.023X \left(\frac{B}{d} \right) - 0.043X \left(\frac{L}{100} \right)$$

The symbols in the above tables and formula for the rolling period are defined as follows:

L	=	waterline length of the ship (m)
B	=	moulded breadth of the ship (m)
D	=	moulded draft of the ship (m)
C_b	=	block coefficient
A_k	=	total overall area of bilge keels, or area of the lateral projection of the bar keel, or sum of these areas. (m ²)
GM	=	metacentric height corrected for free surface effect (m)

The wind heeling levers lw_1 and lw_2 referred to are constant values at all angles of inclination and are calculated as follows:

$$lw_1 = \frac{P \times A \times Z}{1000 \times g \times Disp} \text{ and}$$

$$lw_2 = 1.5 \times lw_1$$

Where,

P	=	Pressure, 504 N/m ²
A	=	Projected lateral area of the portion of the ship and deck cargo if any above the waterline (m ²)
Z	=	vertical distance from the center of A to the center of the underwater lateral area or approximately to a point at one half the draft (m).
Disp	=	Displacement (t)

$$\text{Roll back angle} = 109 \times X_1 \times X_2 \times \sqrt{r \times s} \text{ (degrees)}$$

Where:

X1	=	factor as shown in table 1
X2	=	factor as shown in table 2
K	=	factor as follows

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K = 1.0 for round-bilged ship having no bilge or bar keels;
 K = 0.7 for ship having sharp bilge
 K = as shown in table 3 for a ship having bilge keels, a bar keel or both
 r = $0.73 \pm 0.6 \text{ OG/d}$

With:

OG = distance between the center of gravity and the waterline (-ve if it is below).
 D = mean moulded draft of the ship (m)
 S = factor as shown in table 4.

TABLE-1	
B/d	X1
≤ 2.4	1
2.5	0.98
2.6	0.96
2.7	0.95
2.8	0.93
2.9	0.91
3	0.9
3.1	0.88
3.2	0.86
3.3	0.84
3.4	0.82
≥ 3.5	0.8

TABLE-2	
Cb	X2
≤ 0.45	0.75
0.5	0.82
0.55	0.89
0.6	0.95
0.65	0.97
≥ 0.70	1
3.5	0.72
≥ 4.0	0.7

TABLE-3	
(Akx100)/(LxB)	k
0	1
1	0.98
1.5	0.95
2	0.88
2.5	0.79
3	0.74
18	0.038
≥ 20	0.035

TABLE-4	
T	s
≤ 6	0.1
7	0.098
8	0.093
12	0.065
14	0.053
16	0.044

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6.2 Towing criteria:

The vessel is required to comply with the general intact stability criteria laid down in *Rules and Regulations for the Construction and Classification of Steel Ships – 2016, Part 5 section 4.1*.

The stability of tugs is to be assessed considering the effect of transverse heeling force caused when the tow rope is not in line with the tug's longitudinal centerline.

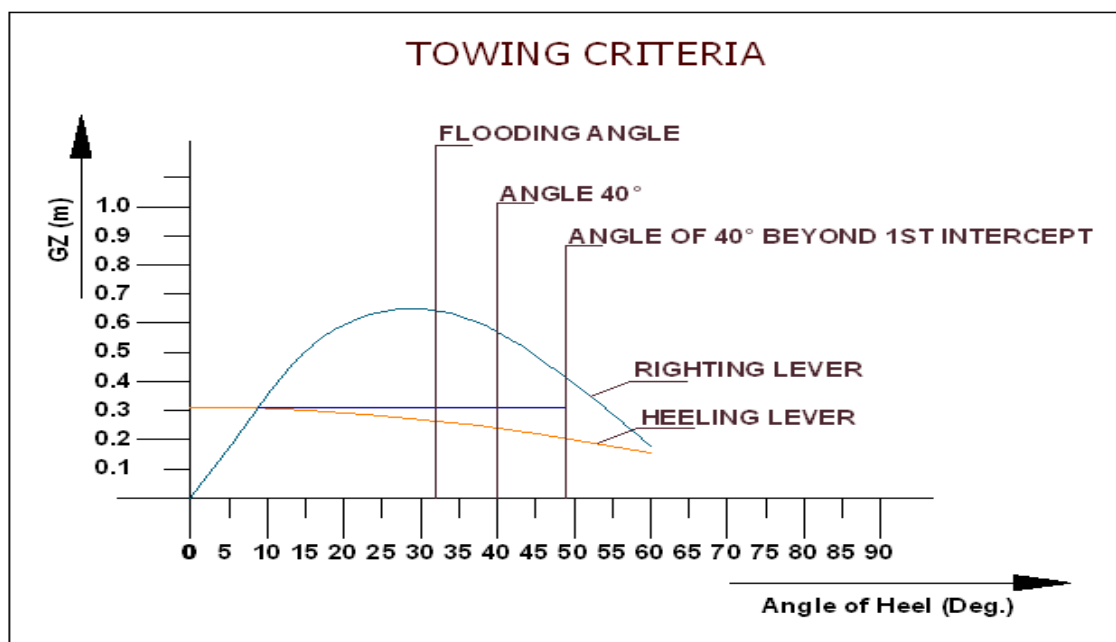
Compliance with the following criteria is recommended, as a minimum:

- (i) The residual area between a righting lever curve and a heeling lever curve developed from 70% of the maximum bollard pull force acting in 90° to the ship length direction is not to be less than 0.09 [m-rad]; The residual areas under the curves are to be calculated from the 1st intercept up to an angle of 40° beyond the first intercept or up to the down-flooding angle, whichever is less.

OR

- (ii) Alternatively, the area under a righting lever curve should not be less than 1.4 times the area under a heeling lever curve developed from 70% of the maximum bollard pull force acting in 90° to ship-length direction.

In this case the areas under the curves are to be calculated from upright condition up to an angle of 40° or the down-flooding angle, whichever is less.



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The heeling lever curve is to be derived by using the following formula:

$$b_h = 0.7 TH \cos\theta / 9.81\Delta$$

Where,

b_h = heeling arm [m];

T = maximum bollard pull [kN];

H = vertical distance [m] between the towing hook and the VCB of the vessel at loading draft;

Δ = loading condition displacement [t];

θ = angle of heel.

7 Closures & Down flooding Points

7.1 Down flooding Points:

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive flooding can occur as the vessel heels.
2. If this opening has a weather tight closing device which cannot be kept closed and secured at all times while the vessel is underway or has no weather tight closing device, progressive flooding will occur when the sill of the opening immerses. This angle has to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve terminates at this angle i.e. the actual angle of flooding becomes the range of the GZ curve.
3. If the lower sill of the opening through which Down flooding can occur immerses at 40° or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea.

Description	Down flooding points			
	X coordinate (m)	Y coordinate (m)	Z coordinate (m)	Symmetric
Engine Room Funnel	7.052	2.100	7.700	Yes

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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8 Hydrostatics:

Trim = 1.00 m by aft

Trim: -1.000 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t*m/cm)	TpCm (t/cm)
0.500	25.952	26.601	7.875	0.358	8.645	72.50	7.292	30.38	0.404	0.698
0.550	29.440	30.176	7.973	0.384	8.757	76.79	6.830	31.02	0.468	0.734
0.600	33.097	33.924	8.065	0.412	8.849	80.95	6.422	31.51	0.534	0.767
0.650	36.912	37.835	8.150	0.439	8.913	84.96	6.064	31.82	0.601	0.799
0.700	40.876	41.898	8.225	0.466	8.946	88.83	5.749	31.90	0.667	0.828
0.750	44.979	46.103	8.292	0.494	8.950	92.65	5.474	31.94	0.734	0.856
0.800	49.220	50.450	8.348	0.523	8.927	96.55	5.233	32.09	0.806	0.885
0.850	53.604	54.944	8.393	0.551	8.867	100.68	5.025	32.53	0.889	0.915
0.900	58.151	59.605	8.425	0.581	8.705	105.76	4.849	34.15	1.013	0.955
0.950	62.948	64.521	8.433	0.612	8.425	112.06	4.716	37.14	1.193	1.006
1.000	67.941	69.640	8.427	0.645	8.284	116.86	4.614	37.90	1.313	1.041
1.050	73.066	74.893	8.415	0.677	8.271	120.15	4.474	36.94	1.374	1.060
1.100	78.264	80.220	8.406	0.709	8.282	123.06	4.334	35.68	1.420	1.074
1.150	83.530	85.618	8.399	0.740	8.291	125.93	4.210	34.50	1.463	1.087
1.200	88.854	91.076	8.393	0.771	8.298	128.75	4.099	33.39	1.504	1.099
1.250	94.237	96.593	8.388	0.802	8.304	131.52	3.999	32.33	1.541	1.111
1.300	99.674	102.165	8.383	0.832	8.313	134.22	3.905	31.29	1.575	1.121
1.350	105.155	107.784	8.380	0.862	8.337	136.79	3.809	30.20	1.600	1.128
1.400	110.673	113.439	8.378	0.891	8.358	139.34	3.724	29.19	1.625	1.136
1.450	116.221	119.127	8.378	0.920	8.378	141.89	3.648	28.26	1.648	1.143
1.500	121.807	124.852	8.378	0.949	8.396	144.43	3.581	27.41	1.672	1.150
1.550	127.429	130.615	8.379	0.978	8.412	146.97	3.521	26.61	1.694	1.157
1.600	133.083	136.410	8.381	1.006	8.427	149.50	3.467	25.87	1.717	1.164
1.650	138.768	142.237	8.383	1.035	8.441	152.04	3.419	25.18	1.738	1.170
1.700	144.485	148.097	8.386	1.063	8.454	154.57	3.377	24.54	1.760	1.177
1.750	150.237	153.993	8.389	1.091	8.466	157.10	3.339	23.94	1.780	1.183
1.800	156.020	159.921	8.392	1.119	8.477	159.63	3.306	23.37	1.801	1.190
1.850	161.829	165.875	8.395	1.147	8.487	162.17	3.277	22.84	1.821	1.196
1.900	167.669	171.861	8.398	1.175	8.496	164.71	3.251	22.35	1.841	1.202
1.950	173.539	177.877	8.402	1.202	8.506	167.25	3.229	21.88	1.861	1.208
2.000	179.444	183.930	8.405	1.230	8.514	169.79	3.210	21.44	1.881	1.214

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m).

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Trim = 0.75 m by aft

Trim: -0.750 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t*m/cm)	TpCm (t/cm)
0.500	25.059	25.685	8.247	0.333	9.058	73.34	7.536	33.78	0.435	0.710
0.550	28.605	29.321	8.354	0.360	9.165	77.45	7.008	33.86	0.497	0.744
0.600	32.306	33.114	8.452	0.388	9.235	81.26	6.550	33.50	0.555	0.774
0.650	36.145	37.048	8.537	0.416	9.279	84.87	6.153	32.96	0.610	0.801
0.700	40.111	41.114	8.612	0.443	9.305	88.38	5.807	32.43	0.666	0.826
0.750	44.198	45.303	8.677	0.471	9.314	91.84	5.506	31.94	0.721	0.851
0.800	48.402	49.612	8.732	0.500	9.307	95.26	5.244	31.51	0.779	0.874
0.850	52.720	54.038	8.778	0.528	9.282	98.71	5.015	31.18	0.838	0.897
0.900	57.153	58.582	8.815	0.557	9.238	102.24	4.816	31.00	0.902	0.921
0.950	61.704	63.247	8.844	0.586	9.168	105.95	4.643	31.04	0.975	0.946
1.000	66.386	68.045	8.863	0.615	9.053	110.07	4.495	31.54	1.065	0.976
1.050	71.241	73.022	8.868	0.646	8.779	115.83	4.375	33.94	1.230	1.021
1.100	76.310	78.218	8.855	0.677	8.586	120.79	4.287	35.06	1.361	1.057
1.150	81.528	83.566	8.834	0.709	8.483	124.73	4.203	35.01	1.450	1.082
1.200	86.833	89.004	8.812	0.741	8.474	127.58	4.090	33.87	1.492	1.094
1.250	92.196	94.501	8.793	0.772	8.469	130.37	3.988	32.76	1.530	1.106
1.300	97.615	100.055	8.775	0.803	8.465	133.12	3.897	31.74	1.567	1.117
1.350	103.087	105.664	8.758	0.834	8.462	135.85	3.815	30.78	1.601	1.127
1.400	108.606	111.321	8.743	0.864	8.462	138.54	3.740	29.87	1.634	1.137
1.450	114.166	117.020	8.730	0.894	8.471	141.14	3.665	28.94	1.661	1.145
1.500	119.768	122.762	8.718	0.924	8.487	143.66	3.593	28.01	1.683	1.152
1.550	125.396	128.531	8.708	0.953	8.501	146.18	3.530	27.15	1.704	1.158
1.600	131.060	134.337	8.699	0.982	8.514	148.71	3.473	26.36	1.725	1.165
1.650	136.752	140.171	8.692	1.011	8.526	151.23	3.423	25.62	1.746	1.171
1.700	142.477	146.039	8.685	1.040	8.537	153.75	3.378	24.94	1.766	1.178
1.750	148.234	151.940	8.680	1.068	8.548	156.27	3.339	24.30	1.786	1.184
1.800	154.018	157.869	8.675	1.097	8.557	158.80	3.304	23.70	1.806	1.190
1.850	159.833	163.829	8.671	1.125	8.567	161.33	3.274	23.15	1.826	1.196
1.900	165.678	169.819	8.667	1.153	8.576	163.86	3.247	22.63	1.846	1.202
1.950	171.553	175.842	8.664	1.181	8.585	166.40	3.224	22.14	1.865	1.208
2.000	177.462	181.898	8.662	1.209	8.594	168.95	3.204	21.69	1.885	1.214

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m).

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Trim = 0.50 m by aft

Trim: -0.500 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t*m/cm)	TpCm (t/cm)
0.500	24.512	25.125	8.674	0.314	9.455	74.35	7.723	37.10	0.468	0.724
0.550	28.113	28.816	8.779	0.343	9.520	78.06	7.140	35.99	0.520	0.753
0.600	31.850	32.647	8.869	0.371	9.568	81.58	6.641	34.89	0.570	0.779
0.650	35.711	36.604	8.946	0.399	9.599	84.95	6.212	33.85	0.620	0.804
0.700	39.687	40.679	9.012	0.428	9.616	88.22	5.842	32.88	0.668	0.826
0.750	43.770	44.864	9.069	0.456	9.620	91.41	5.523	32.01	0.716	0.848
0.800	47.954	49.153	9.117	0.484	9.612	94.57	5.246	31.23	0.765	0.868
0.850	52.238	53.544	9.157	0.513	9.594	97.71	5.005	30.56	0.814	0.888
0.900	56.620	58.035	9.189	0.542	9.564	100.86	4.795	30.00	0.865	0.908
0.950	61.096	62.623	9.215	0.570	9.521	104.06	4.611	29.55	0.918	0.928
1.000	65.670	67.312	9.235	0.599	9.463	107.33	4.451	29.22	0.975	0.948
1.050	70.346	72.105	9.247	0.628	9.385	110.75	4.312	29.07	1.038	0.969
1.100	75.130	77.009	9.253	0.658	9.280	114.45	4.191	29.22	1.113	0.993
1.150	80.044	82.045	9.250	0.687	9.115	118.79	4.090	30.00	1.217	1.023
1.200	85.138	87.266	9.234	0.718	8.850	124.25	4.012	31.88	1.376	1.065
1.250	90.404	92.664	9.206	0.749	8.694	128.60	3.951	32.41	1.485	1.093
1.300	95.783	98.177	9.175	0.780	8.630	131.93	3.880	31.96	1.549	1.111
1.350	101.224	103.755	9.146	0.811	8.619	134.68	3.798	30.99	1.585	1.122
1.400	106.720	109.388	9.118	0.842	8.608	137.40	3.724	30.09	1.619	1.132
1.450	112.266	115.072	9.093	0.872	8.599	140.10	3.657	29.24	1.652	1.142
1.500	117.855	120.801	9.069	0.903	8.592	142.77	3.597	28.44	1.683	1.151
1.550	123.486	126.573	9.047	0.932	8.589	145.41	3.540	27.65	1.712	1.159
1.600	129.152	132.381	9.028	0.962	8.597	147.94	3.483	26.83	1.733	1.166
1.650	134.856	138.228	9.009	0.991	8.608	150.45	3.430	26.05	1.753	1.172
1.700	140.587	144.102	8.993	1.021	8.618	152.96	3.384	25.33	1.772	1.178
1.750	146.346	150.005	8.979	1.050	8.627	155.48	3.343	24.65	1.792	1.184
1.800	152.132	155.935	8.966	1.078	8.637	158.00	3.307	24.03	1.811	1.190
1.850	157.952	161.901	8.954	1.107	8.646	160.53	3.275	23.45	1.831	1.196
1.900	163.805	167.900	8.943	1.136	8.655	163.07	3.248	22.92	1.851	1.203
1.950	169.682	173.924	8.933	1.164	8.665	165.62	3.224	22.42	1.871	1.209
2.000	175.591	179.981	8.924	1.192	8.675	168.18	3.203	21.96	1.892	1.215

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m).

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Trim = 0.25 m by aft

Trim: -0.250 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t*m/cm)	TpCm (t/cm)
0.500	24.288	24.895	9.146	0.304	9.800	75.10	7.836	39.13	0.489	0.733
0.550	27.930	28.628	9.234	0.333	9.841	78.61	7.213	37.39	0.537	0.760
0.600	31.698	32.490	9.308	0.361	9.869	81.92	6.684	35.79	0.583	0.784
0.650	35.577	36.466	9.370	0.390	9.885	85.09	6.234	34.35	0.627	0.806
0.700	39.560	40.549	9.422	0.419	9.891	88.16	5.849	33.07	0.670	0.827
0.750	43.640	44.731	9.466	0.448	9.887	91.16	5.518	31.94	0.713	0.846
0.800	47.812	49.008	9.502	0.476	9.875	94.13	5.232	30.93	0.755	0.864
0.850	52.073	53.375	9.532	0.505	9.855	97.06	4.984	30.04	0.798	0.882
0.900	56.416	57.827	9.556	0.533	9.826	99.99	4.769	29.25	0.840	0.899
0.950	60.845	62.367	9.574	0.562	9.788	102.92	4.580	28.56	0.884	0.916
1.000	65.354	66.988	9.588	0.590	9.743	105.90	4.415	27.99	0.929	0.933
1.050	69.949	71.697	9.596	0.619	9.690	108.94	4.271	27.55	0.977	0.951
1.100	74.627	76.493	9.600	0.648	9.627	112.07	4.144	27.24	1.029	0.968
1.150	79.398	81.383	9.599	0.677	9.550	115.33	4.034	27.07	1.087	0.988
1.200	84.265	86.372	9.594	0.706	9.455	118.79	3.938	27.11	1.154	1.008
1.250	89.240	91.471	9.583	0.735	9.326	122.61	3.856	27.47	1.237	1.033
1.300	94.353	96.712	9.564	0.765	9.120	127.29	3.789	28.59	1.362	1.066
1.350	99.640	102.131	9.534	0.795	8.905	132.12	3.739	29.75	1.497	1.099
1.400	105.061	107.688	9.498	0.826	8.786	135.95	3.695	29.90	1.584	1.122
1.450	110.568	113.333	9.461	0.856	8.747	138.92	3.638	29.29	1.631	1.135
1.500	116.124	119.027	9.427	0.887	8.731	141.62	3.579	28.50	1.663	1.145
1.550	121.732	124.775	9.394	0.917	8.718	144.31	3.526	27.76	1.695	1.154
1.600	127.385	130.570	9.364	0.947	8.706	146.98	3.479	27.05	1.725	1.163
1.650	133.078	136.405	9.335	0.976	8.698	149.62	3.435	26.38	1.753	1.171
1.700	138.809	142.279	9.309	1.006	8.697	152.22	3.393	25.71	1.779	1.179
1.750	144.570	148.185	9.285	1.035	8.706	154.74	3.351	25.01	1.798	1.185
1.800	150.367	154.126	9.263	1.064	8.716	157.26	3.313	24.37	1.818	1.191
1.850	156.191	160.096	9.243	1.093	8.726	159.80	3.281	23.78	1.838	1.197
1.900	162.047	166.099	9.224	1.122	8.737	162.35	3.252	23.23	1.859	1.203
1.950	167.927	172.125	9.208	1.150	8.749	164.91	3.227	22.73	1.880	1.210
2.000	173.845	178.191	9.192	1.179	8.760	167.47	3.206	22.26	1.901	1.216

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m)

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Trim = 0.00 m

Trim: 0.000 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t*m/cm)	TpCm (t/cm)
0.500	24.361	24.970	9.647	0.301	10.117	75.73	7.867	40.21	0.504	0.740
0.550	28.034	28.735	9.710	0.330	10.134	79.05	7.221	38.01	0.548	0.765
0.600	31.821	32.616	9.761	0.360	10.142	82.19	6.677	36.10	0.590	0.787
0.650	35.712	36.605	9.802	0.388	10.143	85.21	6.216	34.43	0.631	0.808
0.700	39.699	40.691	9.836	0.417	10.137	88.14	5.825	32.95	0.670	0.827
0.750	43.773	44.868	9.864	0.446	10.124	91.00	5.489	31.64	0.708	0.844
0.800	47.932	49.131	9.886	0.474	10.104	93.80	5.201	30.46	0.745	0.861
0.850	52.170	53.474	9.902	0.503	10.078	96.59	4.951	29.41	0.782	0.876
0.900	56.481	57.893	9.915	0.531	10.048	99.37	4.735	28.51	0.820	0.892
0.950	60.871	62.393	9.923	0.560	10.014	102.17	4.545	27.74	0.858	0.907
1.000	65.334	66.967	9.928	0.588	9.976	104.99	4.379	27.08	0.898	0.923
1.050	69.872	71.619	9.930	0.617	9.932	107.83	4.233	26.52	0.939	0.938
1.100	74.485	76.347	9.928	0.645	9.882	110.72	4.104	26.05	0.982	0.953
1.150	79.174	81.153	9.924	0.673	9.826	113.67	3.991	25.68	1.027	0.969
1.200	83.941	86.040	9.917	0.702	9.761	116.70	3.892	25.41	1.076	0.986
1.250	88.789	91.008	9.906	0.730	9.686	119.83	3.805	25.26	1.130	1.003
1.300	93.729	96.072	9.892	0.759	9.597	123.12	3.730	25.24	1.190	1.022
1.350	98.761	101.230	9.875	0.788	9.486	126.66	3.665	25.42	1.262	1.043
1.400	103.907	106.504	9.852	0.817	9.337	130.64	3.611	25.95	1.355	1.068
1.450	109.192	111.922	9.822	0.846	9.110	135.48	3.569	27.19	1.492	1.102
1.500	114.636	117.501	9.784	0.876	8.956	139.62	3.537	27.71	1.596	1.128
1.550	120.184	123.189	9.743	0.906	8.868	143.07	3.505	27.63	1.666	1.146
1.600	125.800	128.945	9.704	0.936	8.843	145.85	3.462	27.03	1.703	1.156
1.650	131.457	134.743	9.667	0.966	8.828	148.54	3.422	26.40	1.734	1.166
1.700	137.171	140.600	9.631	0.995	8.816	151.23	3.385	25.80	1.765	1.175
1.750	142.923	146.496	9.598	1.025	8.808	153.90	3.352	25.24	1.795	1.183
1.800	148.718	152.436	9.567	1.054	8.805	156.55	3.322	24.69	1.824	1.192
1.850	154.545	158.408	9.539	1.083	8.810	159.14	3.291	24.14	1.848	1.199
1.900	160.407	164.417	9.512	1.112	8.822	161.70	3.261	23.58	1.870	1.205
1.950	166.303	170.461	9.488	1.141	8.835	164.27	3.235	23.07	1.891	1.212
2.000	172.232	176.538	9.466	1.169	8.847	166.84	3.213	22.59	1.914	1.218

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m)

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Trim = 0.25 m by forward:

Trim: 0.250 (m)										
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t*m/cm)	TpCm (t/cm)
0.500	24.713	25.331	10.158	0.307	10.406	76.20	7.815	40.33	0.513	0.745
0.550	28.404	29.114	10.191	0.336	10.404	79.36	7.164	37.91	0.554	0.768
0.600	32.200	33.005	10.215	0.365	10.394	82.38	6.618	35.84	0.593	0.789
0.650	36.094	36.996	10.234	0.394	10.379	85.28	6.159	34.04	0.630	0.808
0.700	40.076	41.078	10.247	0.423	10.358	88.09	5.770	32.44	0.666	0.825
0.750	44.139	45.242	10.256	0.451	10.334	90.84	5.439	31.03	0.700	0.841
0.800	48.279	49.486	10.262	0.479	10.308	93.56	5.154	29.81	0.734	0.857
0.850	52.494	53.806	10.264	0.507	10.281	96.27	4.907	28.74	0.769	0.871
0.900	56.779	58.198	10.265	0.536	10.250	98.96	4.692	27.81	0.803	0.886
0.950	61.136	62.664	10.262	0.564	10.217	101.66	4.504	27.00	0.838	0.900
1.000	65.561	67.200	10.258	0.592	10.180	104.36	4.339	26.29	0.874	0.914
1.050	70.054	71.806	10.252	0.620	10.140	107.08	4.194	25.67	0.910	0.928
1.100	74.615	76.481	10.244	0.648	10.096	109.83	4.066	25.13	0.948	0.942
1.150	79.244	81.225	10.234	0.676	10.048	112.61	3.953	24.68	0.987	0.956
1.200	83.943	86.042	10.222	0.704	9.996	115.43	3.854	24.29	1.027	0.970
1.250	88.713	90.931	10.208	0.732	9.938	118.30	3.766	23.99	1.070	0.985
1.300	93.556	95.895	10.192	0.760	9.873	121.25	3.689	23.76	1.116	1.000
1.350	98.473	100.935	10.175	0.788	9.800	124.30	3.622	23.62	1.166	1.016
1.400	103.471	106.058	10.155	0.816	9.716	127.47	3.564	23.59	1.222	1.034
1.450	108.558	111.272	10.132	0.845	9.618	130.83	3.514	23.69	1.287	1.052
1.500	113.745	116.588	10.106	0.873	9.496	134.48	3.472	24.01	1.365	1.074
1.550	119.042	122.018	10.075	0.902	9.334	138.60	3.439	24.66	1.467	1.101
1.600	124.491	127.603	10.038	0.931	9.136	143.18	3.415	25.64	1.596	1.131
1.650	130.066	133.317	9.997	0.961	9.017	146.99	3.395	25.94	1.686	1.154
1.700	135.733	139.126	9.954	0.990	8.959	150.17	3.372	25.76	1.744	1.169
1.750	141.455	144.991	9.914	1.019	8.944	152.91	3.342	25.26	1.779	1.178
1.800	147.225	150.906	9.876	1.048	8.934	155.62	3.315	24.77	1.812	1.188
1.850	153.045	156.871	9.840	1.077	8.927	158.33	3.290	24.30	1.844	1.197
1.900	158.907	162.880	9.806	1.106	8.924	161.01	3.268	23.85	1.875	1.206
1.950	164.806	168.926	9.774	1.135	8.926	163.68	3.247	23.41	1.904	1.214
2.000	170.741	175.009	9.745	1.164	8.937	166.28	3.225	22.94	1.929	1.221
2.050	176.715	181.133	9.718	1.193	8.951	168.88	3.205	22.49	1.952	1.228

Note: Density of water = 1.025 tonnes/m³

Draft, VCB, KMt, KMI is from moulded base line.

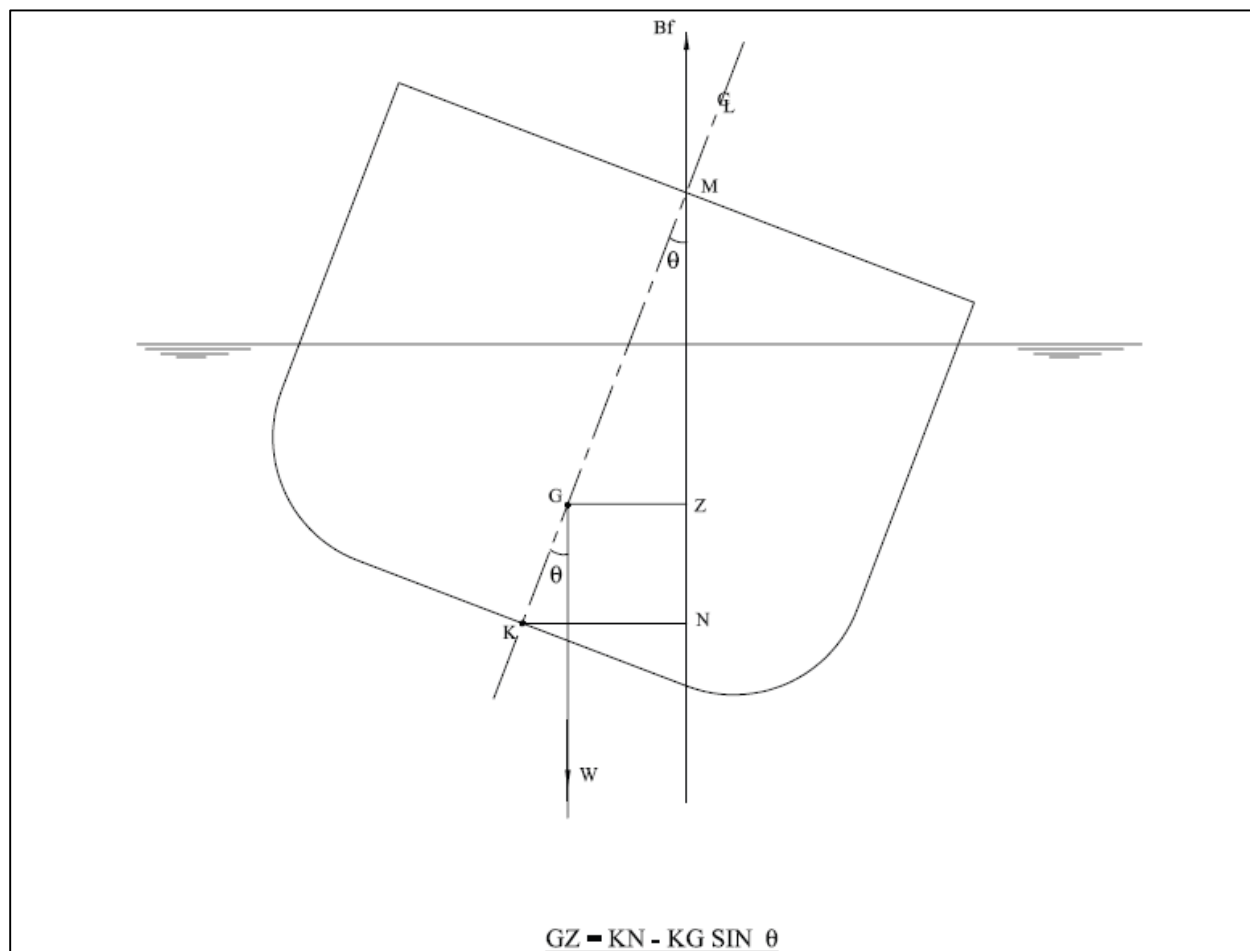
LCB and LCF values are given relative to A.P. (Fr. No-2)

Trim is taken according to LBP (=19.76m)

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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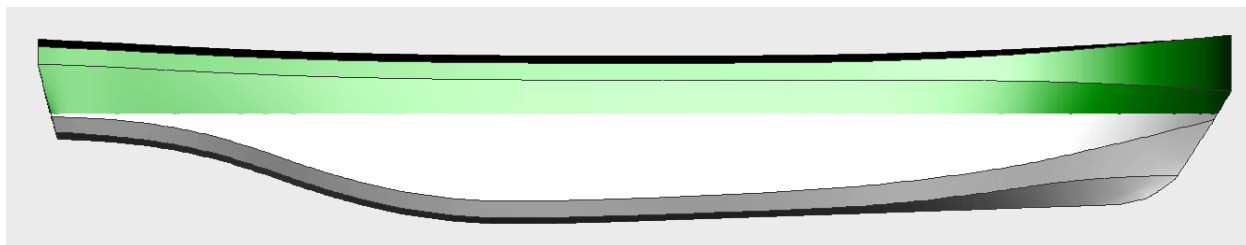
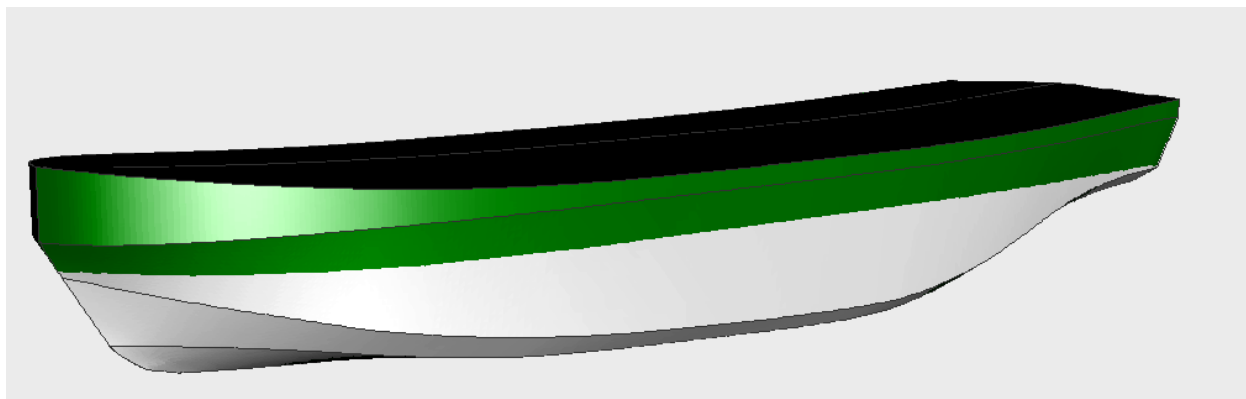
9 Cross curves of stability:

9.1 KN Concept diagram:



K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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Volume considered for calculation of KN is depicted below:



K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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9.2 KN Values:

KN Values for Initial trim: 1.00 m by aft:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.457	0.902	1.297	1.612	1.849	2.033	2.297	2.423	2.401	2.263
55.000	0.000	0.438	0.868	1.256	1.575	1.820	2.013	2.291	2.415	2.390	2.252
60.000	0.000	0.424	0.838	1.219	1.541	1.793	1.994	2.283	2.404	2.378	2.241
65.000	0.000	0.411	0.811	1.185	1.509	1.769	1.977	2.273	2.389	2.363	2.229
70.000	0.000	0.399	0.788	1.155	1.479	1.746	1.961	2.260	2.373	2.347	2.216
75.000	0.000	0.388	0.766	1.127	1.451	1.725	1.946	2.243	2.354	2.329	2.202
80.000	0.000	0.378	0.747	1.101	1.425	1.704	1.931	2.225	2.333	2.310	2.187
85.000	0.000	0.368	0.729	1.077	1.401	1.684	1.915	2.204	2.311	2.291	2.172
90.000	0.000	0.359	0.712	1.055	1.378	1.664	1.897	2.181	2.288	2.270	2.156
95.000	0.000	0.350	0.697	1.035	1.357	1.645	1.878	2.157	2.263	2.249	2.140
100.000	0.000	0.343	0.682	1.016	1.336	1.626	1.857	2.132	2.238	2.228	2.124
105.000	0.000	0.336	0.669	0.998	1.317	1.607	1.835	2.105	2.213	2.206	2.108
110.000	0.000	0.329	0.657	0.982	1.299	1.587	1.811	2.078	2.186	2.183	2.091
115.000	0.000	0.323	0.646	0.967	1.281	1.566	1.786	2.050	2.160	2.161	2.075
120.000	0.000	0.318	0.636	0.953	1.265	1.545	1.761	2.021	2.132	2.138	2.058
125.000	0.000	0.313	0.626	0.940	1.250	1.523	1.734	1.992	2.105	2.115	2.041
130.000	0.000	0.308	0.618	0.928	1.234	1.500	1.706	1.962	2.077	2.092	2.024
135.000	0.000	0.304	0.610	0.917	1.218	1.477	1.678	1.932	2.049	2.068	2.008
140.000	0.000	0.300	0.603	0.907	1.202	1.453	1.649	1.901	2.021	2.045	1.991
145.000	0.000	0.297	0.596	0.897	1.186	1.428	1.619	1.870	1.992	2.022	1.974
150.000	0.000	0.294	0.591	0.889	1.169	1.404	1.590	1.839	1.964	1.998	1.957
155.000	0.000	0.291	0.585	0.880	1.152	1.379	1.560	1.807	1.935	1.975	1.940
160.000	0.000	0.289	0.580	0.872	1.134	1.353	1.530	1.774	1.906	1.952	1.924
165.000	0.000	0.287	0.576	0.863	1.116	1.327	1.500	1.742	1.877	1.928	1.907
170.000	0.000	0.285	0.572	0.853	1.098	1.301	1.469	1.710	1.848	1.905	1.890
175.000	0.000	0.283	0.568	0.843	1.079	1.275	1.439	1.678	1.819	1.881	1.873
180.000	0.000	0.282	0.564	0.831	1.059	1.249	1.408	1.646	1.790	1.857	1.857

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KN Values for Initial trim: 0.75 m by aft:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.454	0.893	1.285	1.599	1.834	2.018	2.282	2.413	2.393	2.257
55.000	0.000	0.433	0.857	1.244	1.562	1.805	1.997	2.277	2.405	2.384	2.247
60.000	0.000	0.415	0.825	1.206	1.527	1.778	1.977	2.270	2.395	2.371	2.236
65.000	0.000	0.401	0.798	1.172	1.495	1.753	1.960	2.260	2.381	2.357	2.224
70.000	0.000	0.389	0.775	1.141	1.465	1.729	1.943	2.247	2.365	2.341	2.211
75.000	0.000	0.379	0.754	1.114	1.437	1.708	1.928	2.232	2.347	2.325	2.198
80.000	0.000	0.371	0.735	1.088	1.411	1.688	1.914	2.216	2.328	2.307	2.184
85.000	0.000	0.362	0.719	1.065	1.388	1.669	1.900	2.197	2.308	2.288	2.170
90.000	0.000	0.355	0.703	1.044	1.365	1.650	1.885	2.177	2.286	2.269	2.155
95.000	0.000	0.347	0.689	1.024	1.344	1.632	1.869	2.155	2.263	2.249	2.140
100.000	0.000	0.340	0.676	1.006	1.325	1.615	1.851	2.131	2.239	2.229	2.124
105.000	0.000	0.333	0.664	0.989	1.306	1.597	1.832	2.107	2.215	2.208	2.109
110.000	0.000	0.327	0.652	0.974	1.289	1.580	1.811	2.082	2.190	2.186	2.093
115.000	0.000	0.321	0.642	0.960	1.272	1.562	1.788	2.055	2.164	2.164	2.077
120.000	0.000	0.316	0.632	0.946	1.257	1.543	1.765	2.028	2.138	2.142	2.061
125.000	0.000	0.311	0.623	0.934	1.242	1.524	1.740	2.000	2.111	2.120	2.044
130.000	0.000	0.307	0.615	0.923	1.229	1.503	1.714	1.971	2.085	2.098	2.028
135.000	0.000	0.303	0.607	0.913	1.215	1.482	1.687	1.942	2.058	2.075	2.012
140.000	0.000	0.299	0.600	0.903	1.201	1.460	1.659	1.913	2.031	2.052	1.995
145.000	0.000	0.296	0.594	0.894	1.187	1.437	1.631	1.883	2.003	2.030	1.979
150.000	0.000	0.293	0.588	0.885	1.173	1.413	1.603	1.853	1.976	2.007	1.962
155.000	0.000	0.290	0.583	0.878	1.158	1.390	1.574	1.822	1.948	1.984	1.946
160.000	0.000	0.288	0.578	0.870	1.142	1.366	1.545	1.791	1.920	1.962	1.929
165.000	0.000	0.286	0.574	0.863	1.126	1.341	1.516	1.760	1.892	1.939	1.913
170.000	0.000	0.284	0.570	0.856	1.109	1.317	1.486	1.728	1.864	1.916	1.897
175.000	0.000	0.282	0.566	0.848	1.091	1.292	1.457	1.697	1.836	1.893	1.881
180.000	0.000	0.281	0.563	0.839	1.073	1.267	1.427	1.665	1.808	1.870	1.864

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

KN Values for Initial trim: 0.50 m by aft:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.452	0.888	1.275	1.584	1.817	2.001	2.267	2.400	2.383	2.249
55.000	0.000	0.429	0.849	1.232	1.547	1.788	1.980	2.263	2.393	2.374	2.239
60.000	0.000	0.411	0.816	1.193	1.513	1.762	1.961	2.255	2.383	2.362	2.229
65.000	0.000	0.395	0.787	1.159	1.481	1.737	1.943	2.245	2.369	2.348	2.217
70.000	0.000	0.382	0.762	1.128	1.451	1.714	1.927	2.232	2.354	2.333	2.205
75.000	0.000	0.370	0.741	1.100	1.423	1.693	1.912	2.218	2.336	2.316	2.191
80.000	0.000	0.361	0.722	1.074	1.397	1.672	1.897	2.201	2.317	2.299	2.178
85.000	0.000	0.354	0.706	1.051	1.373	1.654	1.884	2.184	2.298	2.281	2.164
90.000	0.000	0.347	0.692	1.031	1.351	1.635	1.871	2.165	2.277	2.262	2.150
95.000	0.000	0.341	0.679	1.012	1.331	1.618	1.856	2.145	2.256	2.243	2.135
100.000	0.000	0.335	0.667	0.995	1.312	1.602	1.841	2.124	2.233	2.224	2.120
105.000	0.000	0.330	0.656	0.979	1.294	1.585	1.823	2.102	2.210	2.204	2.105
110.000	0.000	0.324	0.646	0.964	1.278	1.569	1.804	2.078	2.186	2.183	2.090
115.000	0.000	0.319	0.636	0.951	1.262	1.553	1.784	2.053	2.162	2.162	2.075
120.000	0.000	0.314	0.627	0.938	1.247	1.536	1.762	2.028	2.137	2.141	2.059
125.000	0.000	0.310	0.619	0.927	1.234	1.518	1.739	2.001	2.112	2.120	2.043
130.000	0.000	0.306	0.611	0.917	1.221	1.500	1.715	1.974	2.087	2.098	2.028
135.000	0.000	0.302	0.604	0.907	1.209	1.481	1.689	1.947	2.061	2.077	2.012
140.000	0.000	0.298	0.597	0.898	1.197	1.461	1.663	1.919	2.035	2.055	1.996
145.000	0.000	0.295	0.591	0.889	1.185	1.440	1.637	1.890	2.009	2.033	1.980
150.000	0.000	0.292	0.586	0.882	1.172	1.418	1.609	1.861	1.983	2.012	1.964
155.000	0.000	0.289	0.581	0.874	1.159	1.396	1.582	1.831	1.956	1.990	1.948
160.000	0.000	0.287	0.576	0.867	1.145	1.373	1.554	1.801	1.929	1.968	1.933
165.000	0.000	0.285	0.572	0.861	1.131	1.350	1.526	1.771	1.902	1.946	1.917
170.000	0.000	0.283	0.568	0.855	1.115	1.326	1.498	1.740	1.874	1.924	1.901
175.000	0.000	0.281	0.565	0.849	1.099	1.303	1.469	1.710	1.847	1.902	1.885
180.000	0.000	0.280	0.562	0.842	1.082	1.278	1.440	1.679	1.820	1.879	1.870

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

KN Values for Initial trim: 0.25 m by aft:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.450	0.884	1.267	1.570	1.800	1.984	2.253	2.387	2.371	2.239
55.000	0.000	0.427	0.844	1.222	1.533	1.772	1.963	2.248	2.379	2.362	2.230
60.000	0.000	0.407	0.809	1.181	1.499	1.746	1.944	2.240	2.368	2.350	2.220
65.000	0.000	0.391	0.779	1.146	1.467	1.722	1.927	2.229	2.355	2.337	2.208
70.000	0.000	0.377	0.753	1.114	1.437	1.699	1.912	2.217	2.339	2.321	2.196
75.000	0.000	0.365	0.730	1.086	1.409	1.678	1.897	2.202	2.322	2.305	2.183
80.000	0.000	0.355	0.711	1.061	1.383	1.658	1.883	2.186	2.304	2.288	2.169
85.000	0.000	0.346	0.694	1.038	1.359	1.640	1.870	2.168	2.284	2.270	2.155
90.000	0.000	0.339	0.680	1.018	1.337	1.622	1.856	2.150	2.264	2.251	2.141
95.000	0.000	0.333	0.667	0.999	1.317	1.604	1.842	2.131	2.243	2.233	2.127
100.000	0.000	0.328	0.656	0.982	1.298	1.588	1.826	2.111	2.222	2.214	2.112
105.000	0.000	0.324	0.646	0.967	1.281	1.572	1.810	2.090	2.200	2.195	2.098
110.000	0.000	0.320	0.637	0.954	1.266	1.557	1.792	2.068	2.178	2.175	2.083
115.000	0.000	0.315	0.629	0.941	1.251	1.542	1.774	2.045	2.155	2.155	2.069
120.000	0.000	0.311	0.621	0.929	1.237	1.526	1.754	2.021	2.132	2.136	2.054
125.000	0.000	0.308	0.613	0.919	1.224	1.509	1.732	1.997	2.108	2.115	2.039
130.000	0.000	0.304	0.607	0.909	1.212	1.492	1.709	1.971	2.084	2.095	2.024
135.000	0.000	0.300	0.600	0.900	1.201	1.475	1.686	1.945	2.060	2.075	2.009
140.000	0.000	0.297	0.594	0.892	1.190	1.456	1.661	1.918	2.035	2.054	1.994
145.000	0.000	0.294	0.589	0.884	1.179	1.437	1.636	1.891	2.010	2.033	1.978
150.000	0.000	0.291	0.584	0.877	1.168	1.417	1.610	1.863	1.984	2.012	1.963
155.000	0.000	0.289	0.579	0.871	1.156	1.396	1.584	1.834	1.958	1.991	1.948
160.000	0.000	0.287	0.575	0.864	1.143	1.375	1.557	1.805	1.932	1.970	1.933
165.000	0.000	0.285	0.571	0.859	1.130	1.353	1.530	1.776	1.906	1.949	1.918
170.000	0.000	0.283	0.567	0.853	1.116	1.330	1.503	1.746	1.880	1.927	1.903
175.000	0.000	0.281	0.564	0.848	1.102	1.308	1.476	1.717	1.853	1.906	1.888
180.000	0.000	0.280	0.561	0.842	1.086	1.284	1.448	1.686	1.826	1.885	1.873

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KN Values for Initial trim: 0.00 m

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.448	0.879	1.259	1.557	1.785	1.967	2.239	2.374	2.358	2.228
55.000	0.000	0.424	0.839	1.213	1.520	1.756	1.947	2.233	2.365	2.348	2.220
60.000	0.000	0.405	0.803	1.172	1.485	1.730	1.929	2.224	2.353	2.336	2.209
65.000	0.000	0.388	0.772	1.135	1.453	1.707	1.912	2.213	2.339	2.323	2.198
70.000	0.000	0.374	0.746	1.103	1.423	1.685	1.897	2.200	2.323	2.308	2.186
75.000	0.000	0.361	0.723	1.075	1.395	1.664	1.883	2.185	2.305	2.291	2.173
80.000	0.000	0.351	0.702	1.049	1.369	1.645	1.869	2.169	2.287	2.274	2.159
85.000	0.000	0.342	0.685	1.026	1.346	1.626	1.856	2.152	2.268	2.257	2.145
90.000	0.000	0.334	0.670	1.006	1.324	1.609	1.842	2.134	2.249	2.238	2.131
95.000	0.000	0.327	0.657	0.987	1.304	1.592	1.828	2.115	2.228	2.220	2.117
100.000	0.000	0.322	0.646	0.971	1.286	1.575	1.812	2.095	2.207	2.201	2.102
105.000	0.000	0.317	0.636	0.956	1.269	1.559	1.796	2.075	2.186	2.182	2.088
110.000	0.000	0.313	0.627	0.942	1.254	1.544	1.778	2.054	2.165	2.164	2.074
115.000	0.000	0.310	0.620	0.931	1.240	1.529	1.759	2.032	2.143	2.145	2.060
120.000	0.000	0.307	0.613	0.920	1.227	1.513	1.740	2.009	2.121	2.126	2.046
125.000	0.000	0.304	0.607	0.910	1.215	1.498	1.720	1.986	2.099	2.107	2.032
130.000	0.000	0.301	0.601	0.901	1.203	1.482	1.698	1.962	2.076	2.088	2.018
135.000	0.000	0.298	0.595	0.893	1.192	1.465	1.676	1.937	2.053	2.068	2.003
140.000	0.000	0.296	0.590	0.886	1.182	1.447	1.653	1.912	2.029	2.049	1.989
145.000	0.000	0.293	0.586	0.879	1.172	1.429	1.629	1.885	2.005	2.029	1.975
150.000	0.000	0.291	0.581	0.872	1.161	1.410	1.604	1.859	1.980	2.009	1.960
155.000	0.000	0.288	0.577	0.866	1.150	1.390	1.579	1.831	1.956	1.989	1.946
160.000	0.000	0.286	0.573	0.861	1.138	1.370	1.554	1.803	1.931	1.968	1.931
165.000	0.000	0.284	0.570	0.856	1.126	1.350	1.529	1.775	1.905	1.948	1.917
170.000	0.000	0.283	0.566	0.851	1.113	1.328	1.502	1.746	1.880	1.927	1.902
175.000	0.000	0.281	0.564	0.846	1.099	1.307	1.476	1.717	1.854	1.906	1.887
180.000	0.000	0.280	0.561	0.840	1.085	1.285	1.449	1.688	1.828	1.885	1.873

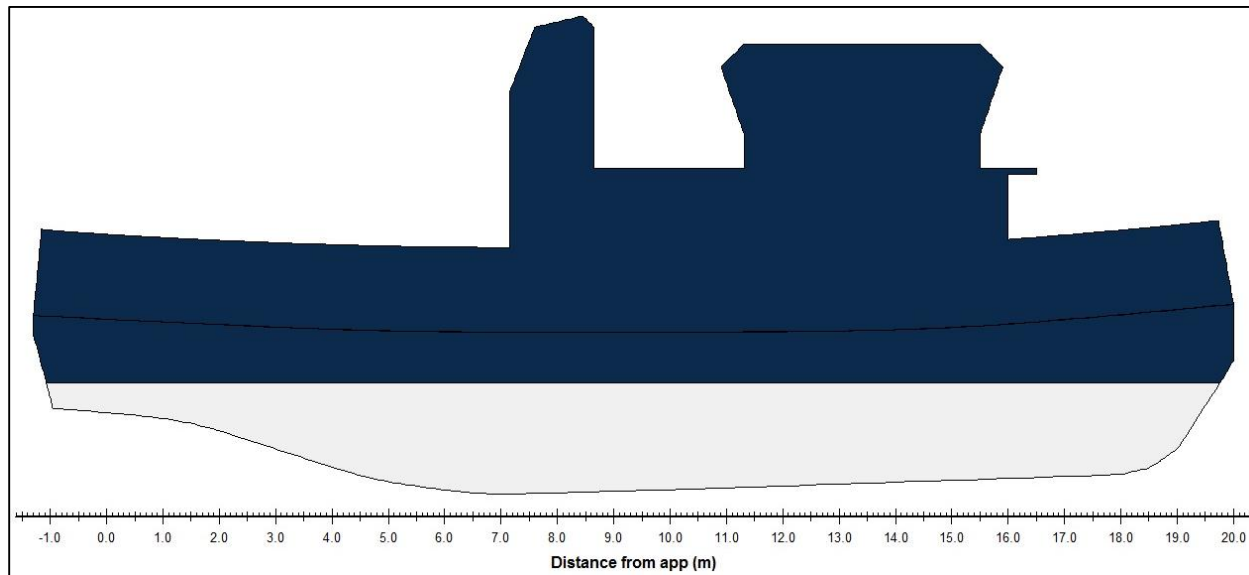
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KN Values for Initial trim: 0.25 m by forward:

Displ. Deg.	0.0° Deg.	5.0° Deg.	10.0° Deg.	15.0° Deg.	20.0° Deg.	25.0° Deg.	30.0° Deg.	40.0° Deg.	50.0° Deg.	60.0° Deg.	70.0° Deg.
50.000	0.000	0.445	0.873	1.251	1.545	1.770	1.952	2.226	2.360	2.345	2.217
55.000	0.000	0.422	0.833	1.204	1.508	1.742	1.932	2.218	2.350	2.334	2.208
60.000	0.000	0.402	0.798	1.163	1.473	1.716	1.914	2.209	2.337	2.322	2.197
65.000	0.000	0.385	0.767	1.126	1.441	1.693	1.898	2.197	2.322	2.307	2.186
70.000	0.000	0.371	0.740	1.094	1.411	1.671	1.883	2.184	2.305	2.292	2.174
75.000	0.000	0.358	0.716	1.065	1.383	1.651	1.869	2.169	2.288	2.276	2.161
80.000	0.000	0.348	0.696	1.040	1.357	1.632	1.856	2.153	2.270	2.259	2.148
85.000	0.000	0.338	0.678	1.016	1.333	1.614	1.843	2.135	2.251	2.241	2.134
90.000	0.000	0.330	0.662	0.996	1.312	1.596	1.829	2.117	2.232	2.224	2.120
95.000	0.000	0.323	0.649	0.977	1.293	1.580	1.814	2.098	2.212	2.206	2.106
100.000	0.000	0.317	0.637	0.960	1.275	1.563	1.798	2.078	2.192	2.188	2.091
105.000	0.000	0.312	0.627	0.946	1.259	1.547	1.781	2.058	2.171	2.169	2.077
110.000	0.000	0.308	0.619	0.933	1.244	1.532	1.763	2.037	2.150	2.151	2.064
115.000	0.000	0.304	0.611	0.921	1.230	1.516	1.744	2.015	2.129	2.132	2.050
120.000	0.000	0.301	0.605	0.911	1.217	1.500	1.724	1.993	2.107	2.114	2.036
125.000	0.000	0.299	0.599	0.901	1.205	1.485	1.704	1.971	2.085	2.095	2.022
130.000	0.000	0.297	0.594	0.893	1.194	1.468	1.683	1.947	2.063	2.077	2.009
135.000	0.000	0.295	0.590	0.886	1.183	1.452	1.661	1.924	2.041	2.058	1.995
140.000	0.000	0.293	0.586	0.879	1.173	1.435	1.639	1.899	2.018	2.039	1.981
145.000	0.000	0.291	0.582	0.873	1.163	1.417	1.616	1.874	1.995	2.020	1.968
150.000	0.000	0.289	0.578	0.867	1.152	1.399	1.593	1.848	1.971	2.001	1.954
155.000	0.000	0.288	0.575	0.862	1.141	1.380	1.569	1.822	1.947	1.982	1.940
160.000	0.000	0.286	0.572	0.857	1.130	1.361	1.545	1.795	1.923	1.962	1.926
165.000	0.000	0.284	0.569	0.852	1.118	1.341	1.521	1.768	1.899	1.942	1.912
170.000	0.000	0.283	0.566	0.848	1.106	1.320	1.496	1.740	1.874	1.922	1.898
175.000	0.000	0.281	0.563	0.843	1.093	1.299	1.470	1.712	1.849	1.902	1.883
180.000	0.000	0.280	0.561	0.838	1.079	1.278	1.444	1.683	1.823	1.881	1.869

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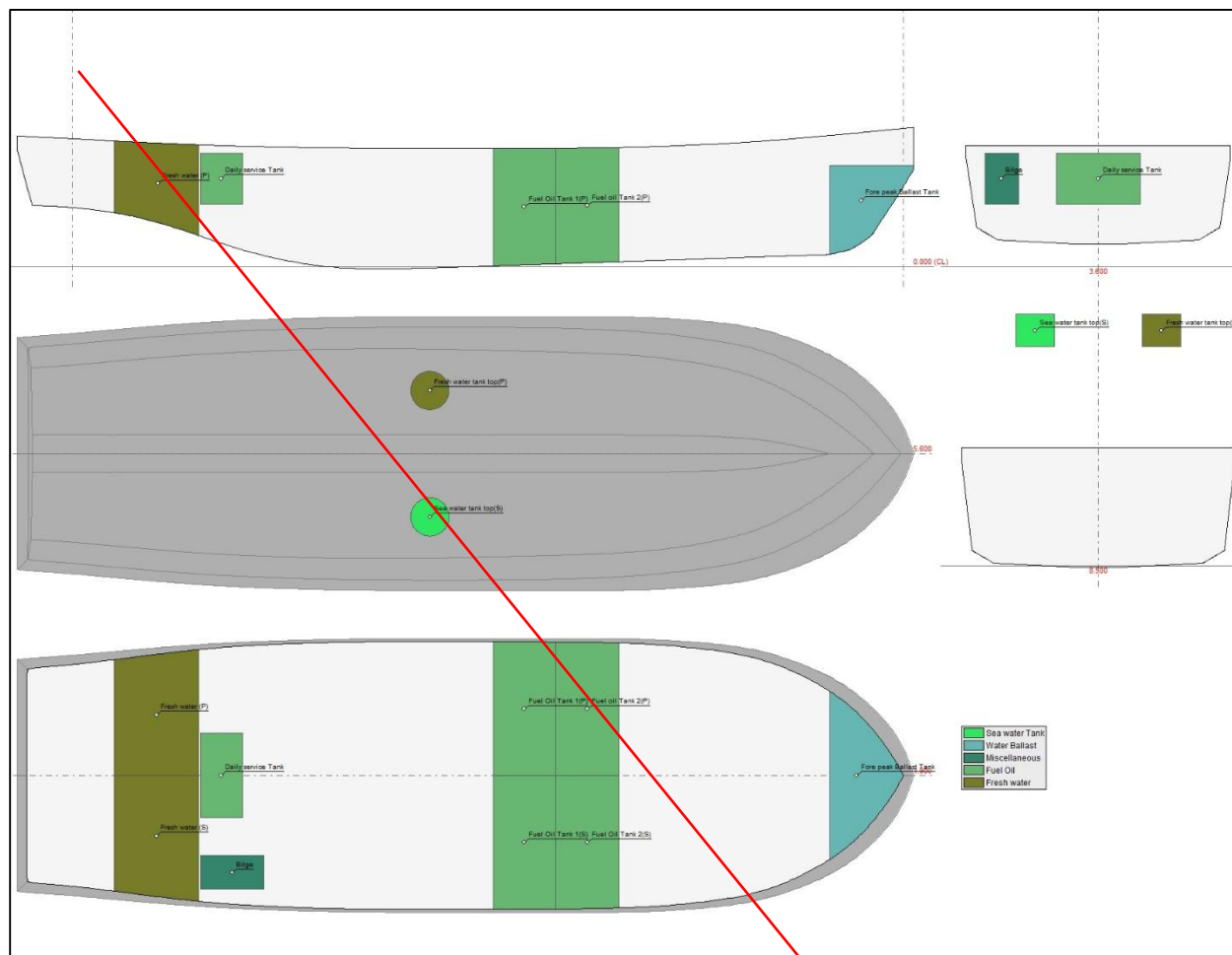
10 Wind Moments:



Wind pressure: 51.40 (kg/m ²) Bilge type: Sharp bilges								
Draft (m)	Lateral area (m ²)	LCE (m)	VCE (m)	Wind area (m ²)	LCE (m)	VCE (m)	Wind lever (m)	Wind moment (Tonnes*m)
1.400	20.907	10.612	0.783	89.025	10.084	3.797	3.014	13.791
1.450	21.910	10.558	0.812	87.603	10.120	3.835	3.023	13.611
1.500	22.933	10.500	0.842	87.000	10.101	3.852	3.010	13.458
1.550	23.957	10.447	0.871	85.975	10.111	3.879	3.008	13.293
1.600	24.984	10.400	0.900	84.948	10.121	3.907	3.007	13.130
1.650	26.013	10.356	0.929	83.919	10.131	3.935	3.006	12.968
1.700	27.045	10.316	0.957	82.888	10.142	3.963	3.006	12.807
1.750	28.078	10.279	0.985	81.854	10.152	3.991	3.006	12.647
1.800	29.112	10.245	1.014	80.817	10.163	4.020	3.006	12.489
1.850	30.153	10.214	1.042	79.780	10.174	4.048	3.007	12.330
1.900	31.193	10.185	1.069	78.740	10.185	4.077	3.008	12.173
1.950	32.235	10.157	1.097	77.697	10.196	4.106	3.009	12.017
2.000	33.278	10.132	1.125	76.654	10.207	4.135	3.011	11.862

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11 Tank Capacities



Sea water Tank							
Tank description	Relative density	Volume (m ³)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	Max FSM (Tonnes*m)
Sea water tank top(S)	1.025	0.500	0.513	5.585	8.500	-1.500 (SB)	0.034
Total		0.500	0.513	5.585	8.500	-1.500 (SB)	0.034

Water Ballast							
Tank description	Relative density	Volume (m ³)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	Max FSM (Tonnes*m)
Fore peak Ballast Tank	1.025	5.343	5.477	1.768	18.604	0.000 (CL)	6.636
Total		5.343	5.477	1.768	18.604	0.000 (CL)	6.636

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Miscellaneous							
Tank description	Relative density	Volume (m³)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	Max FSM (Tonnes*m)
Bilge	1.000	1.411	1.411	2.087	3.800	-2.300 (SB)	0.063
Total		1.411	1.411	2.087	3.800	-2.300 (SB)	0.063

Fuel Oil							
Tank description	Relative density	Volume (m³)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	Max FSM (Tonnes*m)
Fuel Oil Tank 1(P)	0.850	12.299	10.454	1.481	10.747	1.536 (PS)	3.573
Fuel Oil Tank 1(S)	0.850	12.299	10.454	1.481	10.747	-1.536 (SB)	3.573
Fuel oil Tank 2(P)	0.850	12.011	10.210	1.518	12.247	1.525 (PS)	3.569
Fuel Oil Tank 2(S)	0.850	12.012	10.210	1.518	12.247	-1.525 (SB)	3.569
Daily service Tank	0.850	2.352	1.999	2.087	3.550	0.000 (CL)	0.555
Total		50.973	43.327	1.527	11.122	0.000 (CL)	14.839

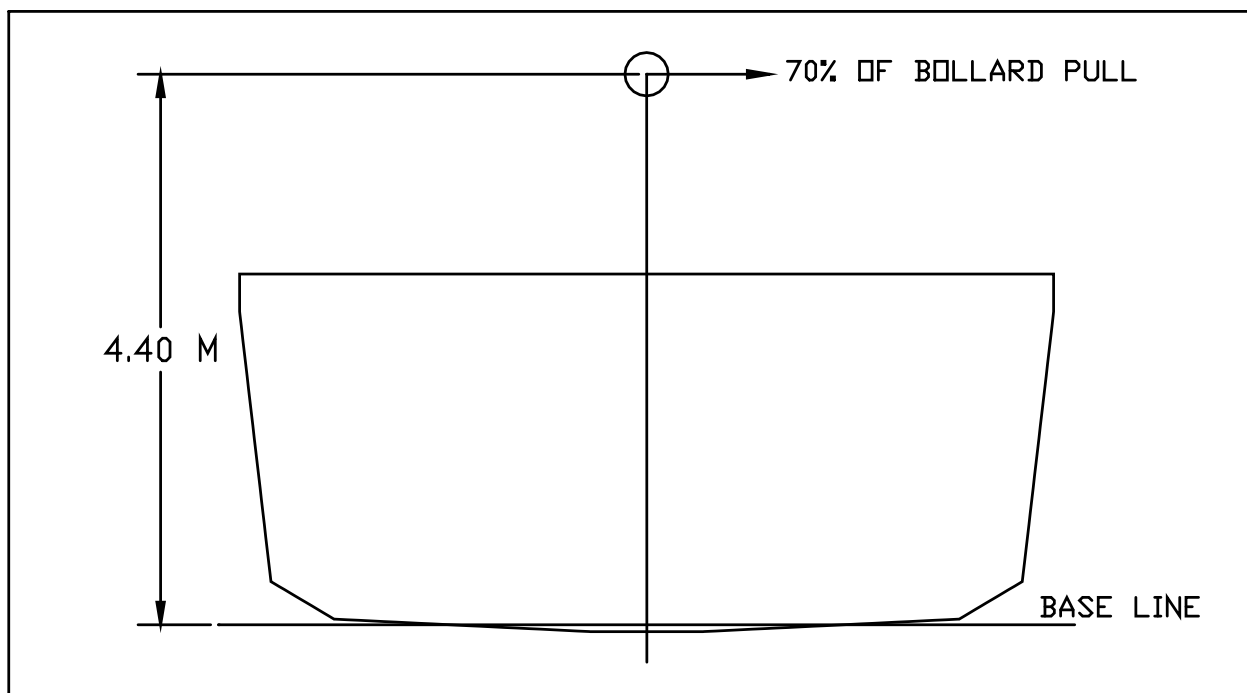
Fresh water							
Tank description	Relative density	Volume (m³)	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	Max FSM (Tonnes*m)
Fresh water (P)	1.000	10.340	10.340	2.028	2.051	1.421 (PS)	4.534
Fresh water (S)	1.000	10.340	10.340	2.028	2.051	-1.421 (SB)	4.534
Fresh water tank top(P)	1.000	0.500	0.500	5.585	8.500	1.500 (PS)	0.034
Total		21.181	21.181	2.112	2.204	0.035 (PS)	9.101

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12 Bollard Pull:

Maximum Bollard pull = 10.00 Ton

Towing hook height from baseline = 04.40 m



13 Intact stability:

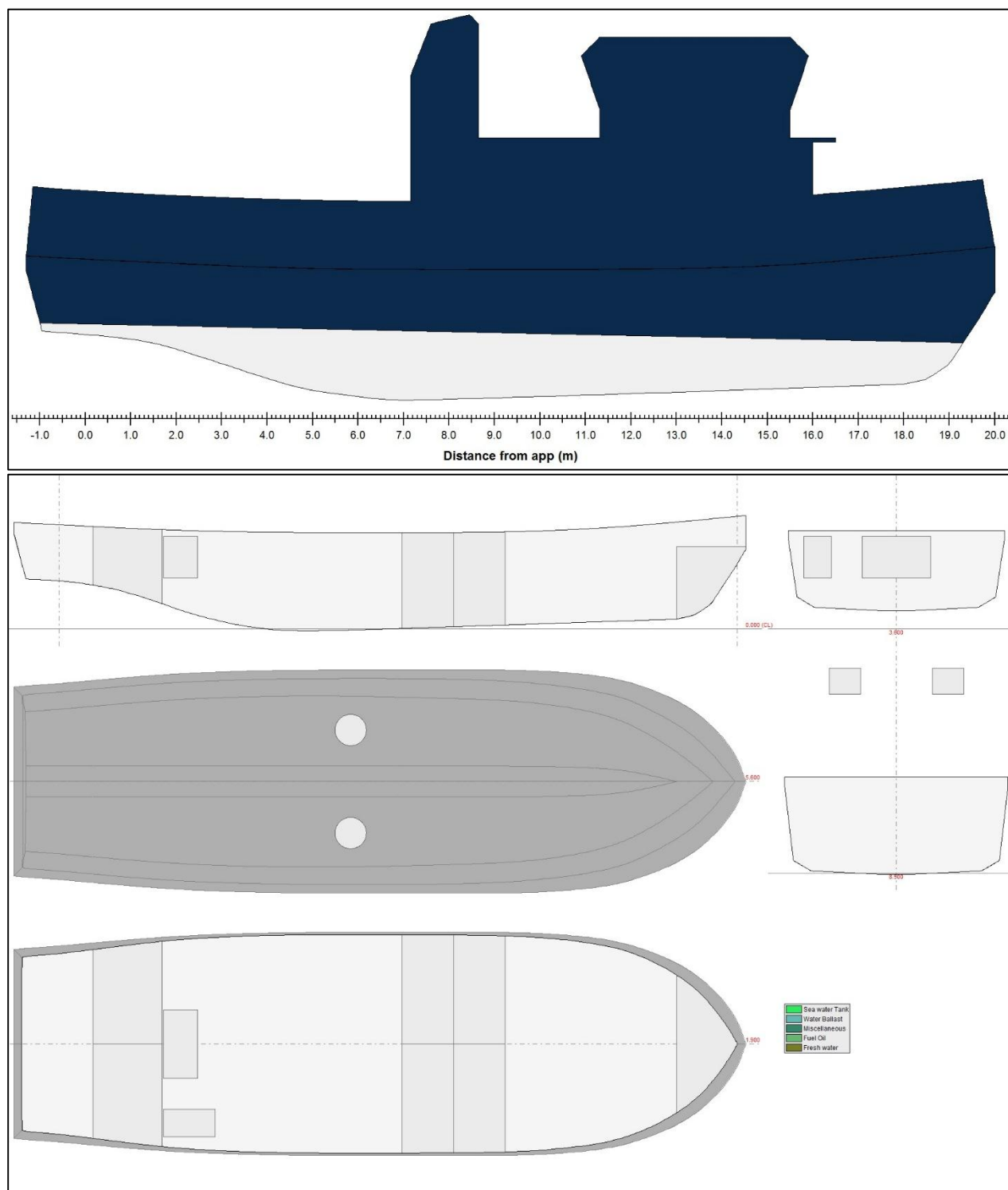
For the purpose of Intact Stability Calculations, the following Loading Conditions are considered.

1. Lightship condition
2. Fully Loaded Departure condition
3. Fully Loaded Arrival condition
4. Ballast Departure condition
5. Ballast Arrival condition

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13.1 Lightship condition

The below load case for the Intact Stability Analysis has been done by considering the Lightship weight of the vessel and solid ballast added for list correction.

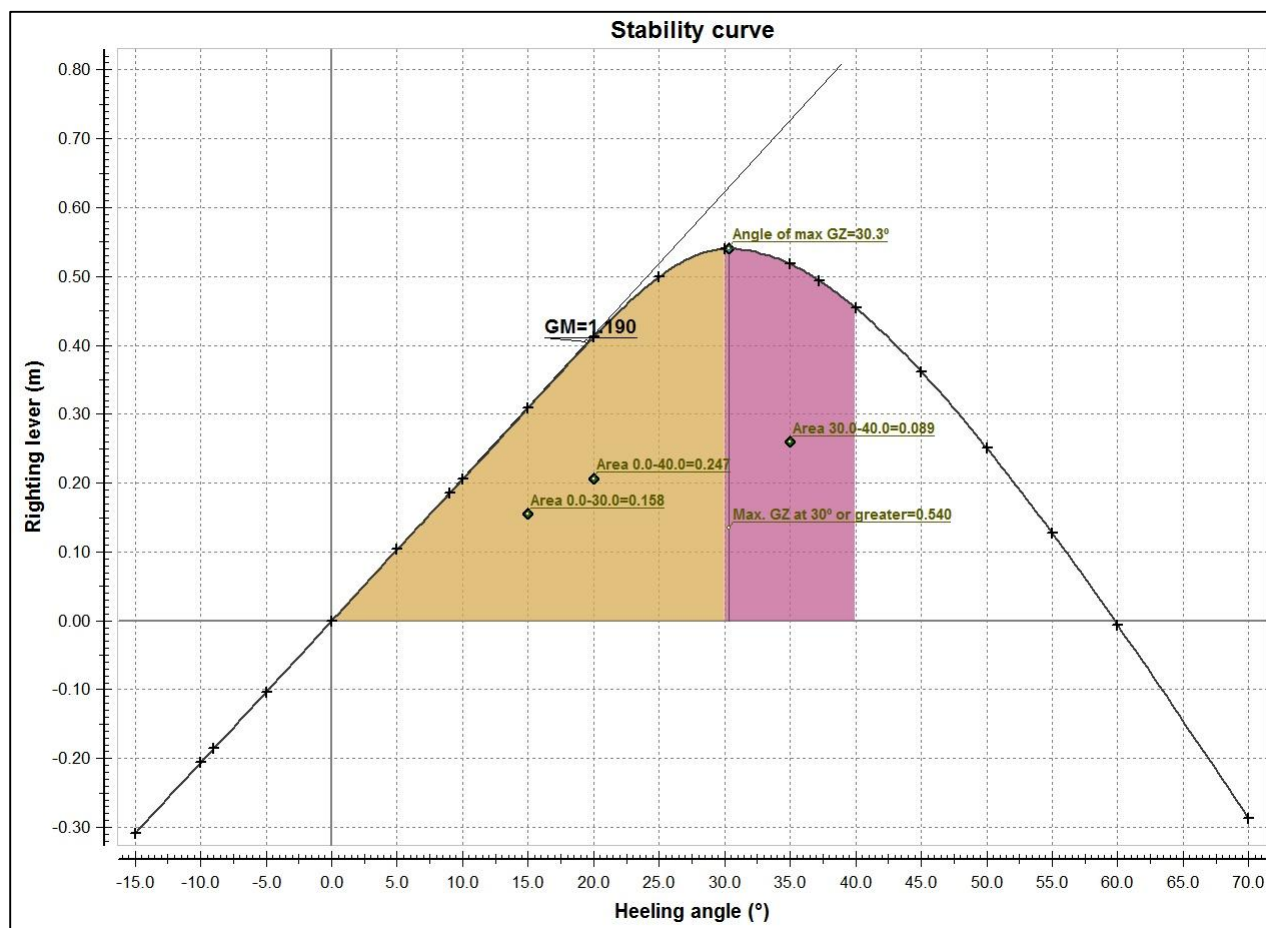


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Hydrostatic particulars			
List	0.01 (PS) (Degr.)	KM	3.721 (m)
Draft aft pp	1.611 (m)	GM solid	1.190 (m)
Mean draft	1.398 (m)	GG'	0.000 (m)
Draft forward pp	1.184 (m)	GM liquid	1.190 (m)
Trim	-0.427 (m)	MCT	1.611 (t*m/cm)
Volume	105.93 (m ³)	Relative water density	1.0250
LCF	8.654 (m)	Immersion rate	1.129 (t/cm)

GZ values		
Heeling angle (Degr.)		GZ (m)
0.00 (CL)		0.000
5.00 (PS)		0.104
9.00 (PS)		0.185
10.00 (PS)		0.206
15.00 (PS)		0.309
20.00 (PS)		0.412
25.00 (PS)		0.499
30.00 (PS)		0.540
35.00 (PS)		0.518
37.20 (PS)		0.494
40.00 (PS)		0.454
45.00 (PS)		0.362
50.00 (PS)		0.251
55.00 (PS)		0.127
60.00 (PS)		-0.006
70.00 (PS)		-0.286

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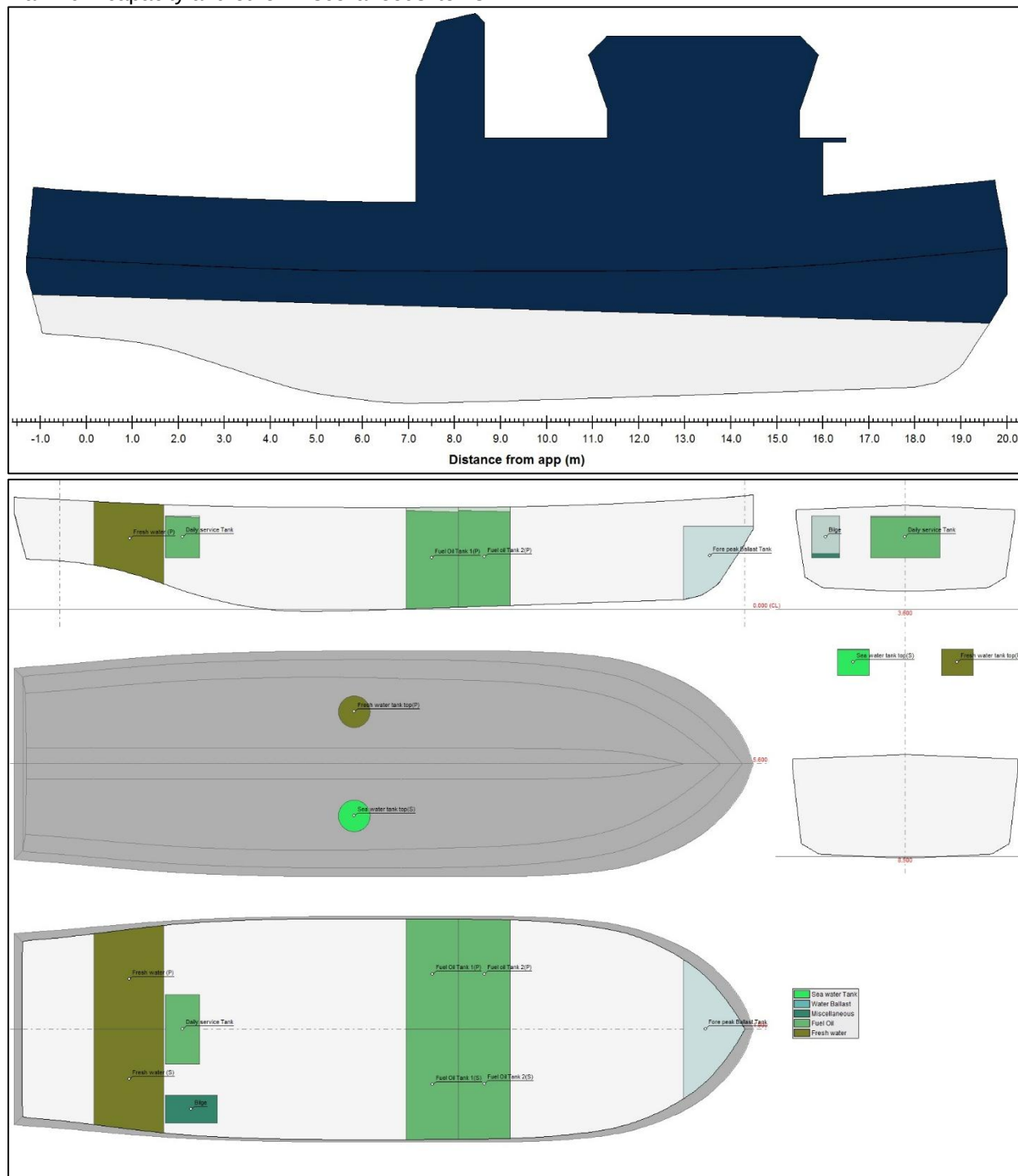
Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Miscellaneous							
Solid Ballast for list correction			0.845	0.326	8.250	2.313 (PS)	
Totals for Miscellaneous			0.845	0.326	8.250	2.313 (PS)	
Lightship			107.736	2.548	9.276	-0.018 (SB)	
Deadweight			0.845	0.326	8.250	2.313 (PS)	0.000
Displacement			108.581	2.531	9.268	0.000 (PS)	0.000

Evaluation of criteria				
Minimum design criteria applicable to Tugs				
Description	Attained value	Criterion	Required value	Complies
Area 0° - 30°	0.1577 (mrad)	>=	0.0550 (mrad)	YES
Area 0° - 40°	0.2469 (mrad)	>=	0.0900 (mrad)	YES
Area 30° - 40°	0.0892 (mrad)	>=	0.0300 (mrad)	YES
Max. GZ at 30° or greater	0.540 (m)	>=	0.200 (m)	YES
Angle of max GZ	30.3 (Degr.)	>=	25.0 (Degr.)	YES
Initial metacentric height	1.190 (m)	>	0.150 (m)	YES
The vessel complies with the stability criteria				

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13.2 Fully Loaded Departure condition

The below load case gives the Intact Stability Analysis of the vessel loaded with consumables at their maximum capacity and other miscellaneous items.

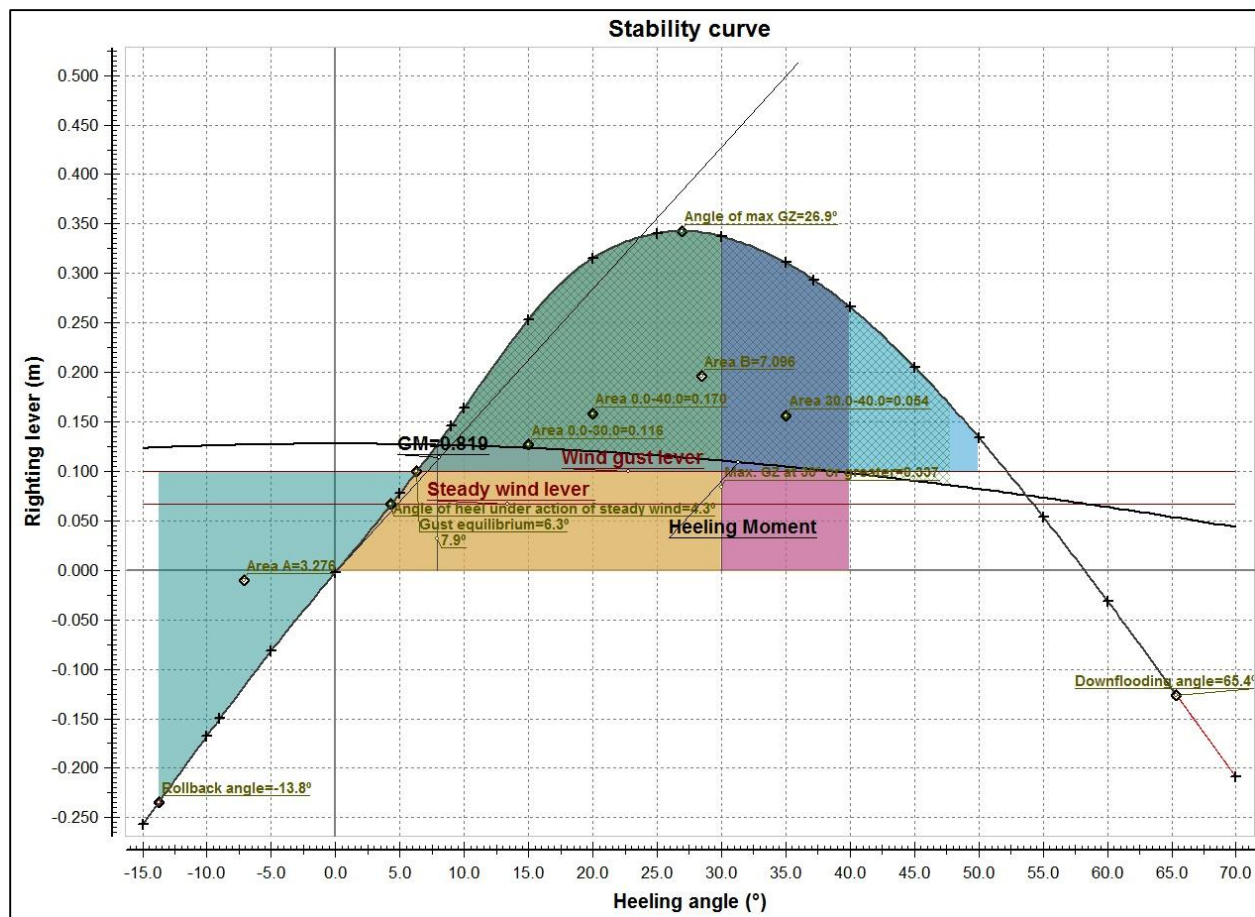


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Hydrostatic particulars			
List	-0.11 (SB) (Degr.)	KM	3.216 (m)
Draft aft pp	2.258 (m)	GM solid	0.955 (m)
Mean draft	1.968 (m)	GG'	0.136 (m)
Draft forward pp	1.677 (m)	GM liquid	0.819 (m)
Trim	-0.581 (m)	MCT	1.876 (t*m/cm)
Volume	172.33 (m ³)	Relative water density	1.0250
LCF	8.642 (m)	Immersion rate	1.211 (t/cm)

GZ values	
Heeling angle (Degr.)	GZ (m)
0.00 (CL)	-0.002
-5.00 (SB)	0.078
-9.00 (SB)	0.146
-10.00 (SB)	0.164
-15.00 (SB)	0.254
-20.00 (SB)	0.316
-25.00 (SB)	0.341
-30.00 (SB)	0.337
-35.00 (SB)	0.311
-37.20 (SB)	0.293
-40.00 (SB)	0.266
-45.00 (SB)	0.206
-50.00 (SB)	0.134
-55.00 (SB)	0.054
-60.00 (SB)	-0.031
-70.00 (SB)	-0.209

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Critical points			
Description	Type	Dist. to wl (m)	Submersion angle (Degr.)
Engine Room Funnel	Downflooding	5.643	-65.37

Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Sea water Tank							
Sea water tank top(S)	1.0250	98.0	0.503	5.577	8.500	-1.500 (SB)	0.034
Totals for Sea water Tank			0.503	5.577	8.500	-1.500 (SB)	0.034
Water Ballast							
Fore peak Ballast Tank	1.0250	0.0	0.000	0.000	0.000	0.000 (CL)	0.000
Totals for Water Ballast			0.000	0.000	0.000	0.000 (CL)	0.000
Miscellaneous							
Bilge	1.0000	10.0	0.141	1.547	3.800	-2.300 (SB)	0.063
Crew and Misc.			2.000	4.000	12.300	0.000 (CL)	
Solid Ballast for list correction			0.845	0.326	8.250	2.313 (PS)	
Totals for Miscellaneous			2.986	2.844	10.752	0.546 (PS)	0.063
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	98.0	10.504	1.488	10.747	1.536 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	98.0	10.504	1.488	10.747	-1.536 (SB)	3.573

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Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Fuel oil Tank 2(P)	0.8500	98.0	10.264	1.525	12.246	1.525 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	98.0	10.264	1.525	12.246	-1.525 (SB)	3.569
Daily service Tank	0.8500	98.0	1.959	2.075	3.550	0.000 (CL)	0.555
Totals for Fuel Oil			43.494	1.532	11.130	0.000 (CL)	14.839
Fresh water							
Fresh water (P)	1.0000	100.0	10.726	2.062	2.050	1.406 (PS)	4.534
Fresh water (S)	1.0000	100.0	10.726	2.062	2.050	-1.406 (SB)	4.534
Fresh water tank top(P)	1.0000	100.0	0.500	5.585	8.500	1.500 (PS)	0.034
Totals for Fresh water			21.952	2.142	2.197	0.034 (PS)	9.101
Lightship			107.736	2.548	9.276	-0.018 (SB)	
Deadweight			68.935	1.812	8.250	0.024 (PS)	24.037
Displacement			176.671	2.261	8.876	-0.002 (SB)	24.037

Evaluation of criteria				
Minimum design criteria applicable to Tugs				
Description	Attained value	Criterion	Required value	Complies
Area 0° - 30°	0.1158 (mrad)	>=	0.0550 (mrad)	YES
Area 0° - 40°	0.1696 (mrad)	>=	0.0900 (mrad)	YES
Area 30° - 40°	0.0537 (mrad)	>=	0.0300 (mrad)	YES
Max. GZ at 30° or greater	0.337 (m)	>=	0.200 (m)	YES
Angle of max GZ	26.9 (Degr.)	>=	25.0 (Degr.)	YES
Initial metacentric height	0.819 (m)	>	0.150 (m)	YES
Severe wind and rolling criterion				YES
Wind pressure	51.4 (kg/m ²)			
Wind area	76.97 (m ²)			
Steady wind lever	0.066 (m)			
Deck immersion angle	12.99 (Degr.)			
Wind gust lever	0.100 (m)			
Ratio of areaA/areaB	0.462	<=	1.000	YES
Maximum allowed static heeling angle	4.3 (Degr.)	<=	16.0 (Degr.)	YES
Max allowed ratio static angle/deck immersion angle	0.332	<=	0.800	YES
The vessel complies with the stability criteria				

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13.2.1 Evaluation of towing criteria:

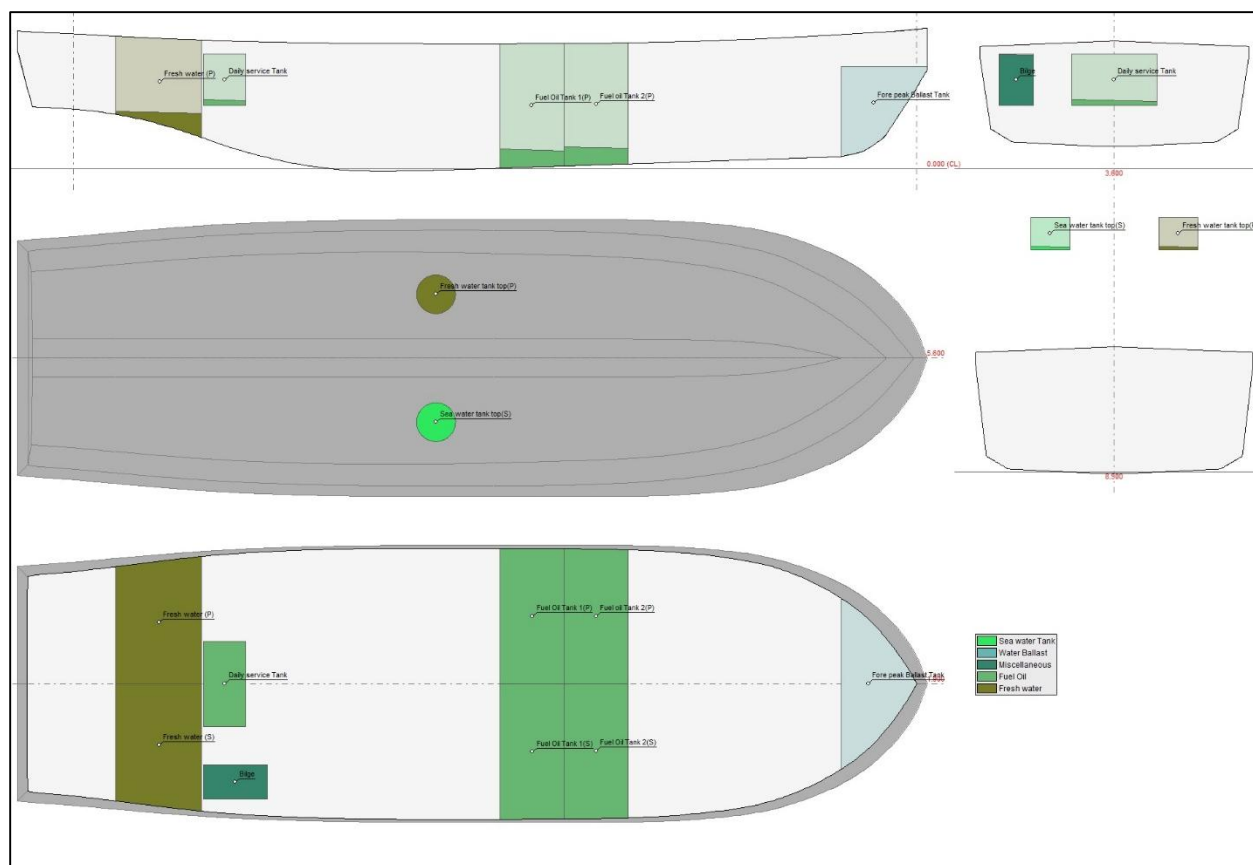
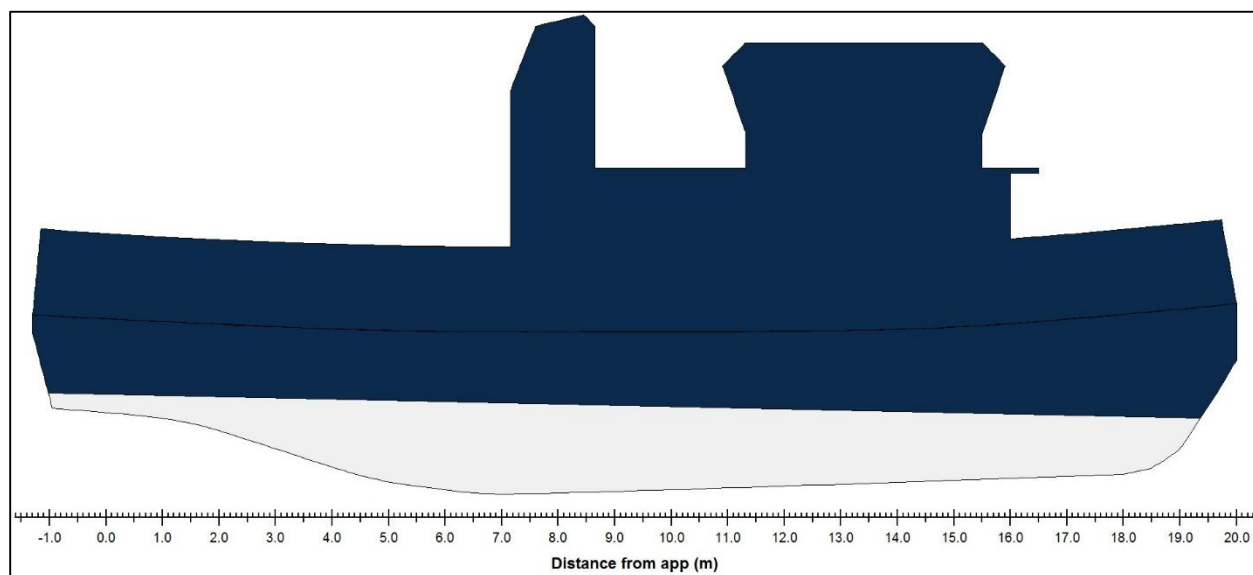
Maximum Bollard pull (T)	=	10.00	Ton
C.G. of Towline pull (C.G. _T)	=	04.40	m
V.C.B.	=	1.179	m
Lever (H = C.G. _T – V.C.B.)	=	3.221	m
Heeling Moment (0.7*T*H)	=	22.547	T-m
Displacement (Δ)	=	176.671	T
Heeling lever (0.7*T*H*cosθ/Δ)	=	0.128* Cosθ	m

Towing criteria				
Description	Attained value	Criterion	Required value	Complies
Area between heeling lever and righting lever calculated from the 1st intercept up to an angle 40° beyond the first intercept or up to the down flooding angle, whichever is less.	0.114	m-rad >=	0.09	m-rad YES
The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the down-flooding angle, whichever is less	2.067	- >=	1.40	YES
Note: The Towing Criteria for a Tug complies if one of the above two criteria is satisfied				
THE VESSEL COMPLIES WITH THE CRITERIA				

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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13.3 Fully Loaded Arrival condition

The below load case gives the Intact Stability Analysis of the vessel loaded with 10% consumables and other miscellaneous items.



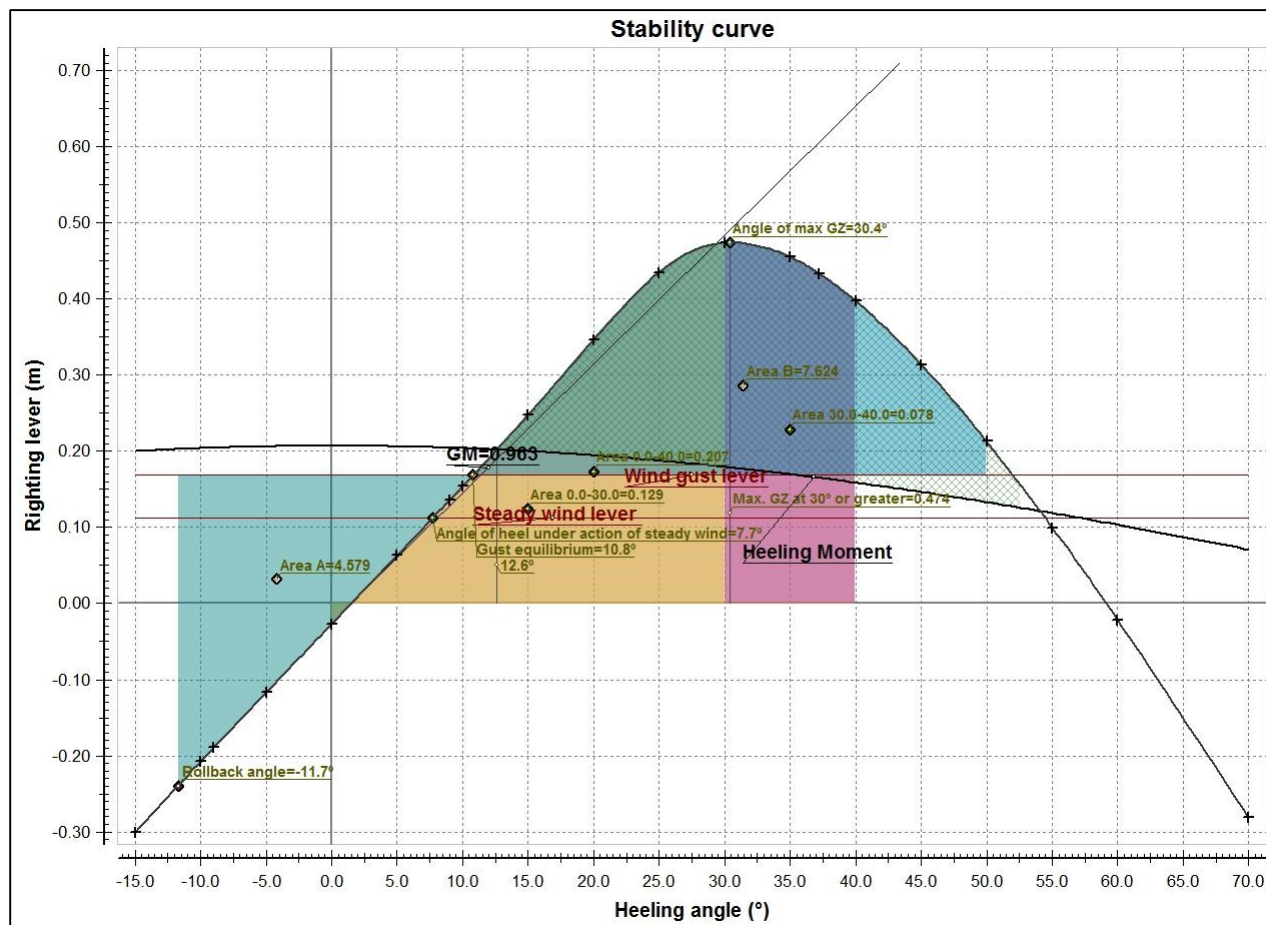
K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Hydrostatic particulars

List	-1.48 (SB) (Degr.)	KM	3.610 (m)
Draft aft pp	1.704 (m)	GM solid	1.166 (m)
Mean draft	1.485 (m)	GG'	0.203 (m)
Draft forward pp	1.266 (m)	GM liquid	0.963 (m)
Trim	-0.438 (m)	MCT	1.669 (t*m/cm)
Volume	115.75 (m ³)	Relative water density	1.0250
LCF	8.630 (m)	Immersion rate	1.147 (t/cm)

GZ values		
Heeling angle (Degr.)		GZ (m)
0.00 (CL)		-0.027
-5.00 (SB)		0.063
-9.00 (SB)		0.136
-10.00 (SB)		0.154
-15.00 (SB)		0.248
-20.00 (SB)		0.347
-25.00 (SB)		0.435
-30.00 (SB)		0.474
-35.00 (SB)		0.456
-37.20 (SB)		0.434
-40.00 (SB)		0.398
-45.00 (SB)		0.314
-50.00 (SB)		0.213
-55.00 (SB)		0.100
-60.00 (SB)		-0.022
-70.00 (SB)		-0.281

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Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Sea water Tank							
Sea water tank top(S)	1.0250	10.0	0.051	5.238	8.500	-1.500 (SB)	0.034
Totals for Sea water Tank			0.051	5.238	8.500	-1.500 (SB)	0.034
Water Ballast							
Fore peak Ballast Tank	1.0250	0.0	0.000	0.000	0.000	0.000 (CL)	0.000
Totals for Water Ballast			0.000	0.000	0.000	0.000 (CL)	0.000
Miscellaneous							
Bilge	1.0000	98.0	1.383	2.075	3.800	-2.300 (SB)	0.063
Crew and Misc.			2.000	4.000	12.300	0.000 (CL)	
Solid Ballast for list correction			0.845	0.326	8.250	2.313 (PS)	
Totals for Miscellaneous			4.228	2.636	8.710	-0.290 (SB)	0.063
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	10.0	1.072	0.245	10.726	1.255 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	10.0	1.072	0.245	10.726	-1.255 (SB)	3.573
Fuel oil Tank 2(P)	0.8500	10.0	1.047	0.298	12.222	1.221 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	10.0	1.047	0.298	12.222	-1.221 (SB)	3.569
Daily service Tank	0.8500	10.0	0.200	1.547	3.550	0.000 (CL)	0.555
Totals for Fuel Oil			4.438	0.329	11.109	0.000 (CL)	14.839
Fresh water							

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Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Fresh water (P)	1.0000	10.0	1.073	1.129	2.429	1.171 (PS)	4.534
Fresh water (S)	1.0000	10.0	1.073	1.129	2.429	-1.171 (SB)	4.534
Fresh water tank top(P)	1.0000	10.0	0.050	5.238	8.500	1.500 (PS)	0.034
Totals for Fresh water			2.195	1.222	2.567	0.034 (PS)	9.101
Lightship			107.736	2.548	9.276	-0.018 (SB)	
Deadweight			10.913	1.426	8.449	-0.113 (SB)	24.037
Displacement			118.649	2.445	9.200	-0.027 (SB)	24.037

Evaluation of criteria Minimum design criteria applicable to Tugs				
Description	Attained value	Criterion	Required value	Complies
Area 0° - 30°	0.1290 (mrad)	>=	0.0550 (mrad)	YES
Area 0° - 40°	0.2073 (mrad)	>=	0.0900 (mrad)	YES
Area 30° - 40°	0.0783 (mrad)	>=	0.0300 (mrad)	YES
Max. GZ at 30° or greater	0.474 (m)	>=	0.200 (m)	YES
Angle of max GZ	30.4 (Degr.)	>=	25.0 (Degr.)	YES
Initial metacentric height	0.963 (m)	>	0.150 (m)	YES
Severe wind and rolling criterion				YES
Wind pressure	51.4 (kg/m ²)			
Wind area	87.01 (m ²)			
Steady wind lever	0.112 (m)			
Deck immersion angle	22.72 (Degr.)			
Wind gust lever	0.168 (m)			
Ratio of area A/area B	0.601	<=	1.000	YES
Maximum allowed static heeling angle	7.7 (Degr.)	<=	16.0 (Degr.)	YES
Max allowed ratio static angle/deck immersion angle	0.339	<=	0.800	YES
The vessel complies with the stability criteria				

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13.3.1 Evaluation of towing criteria:

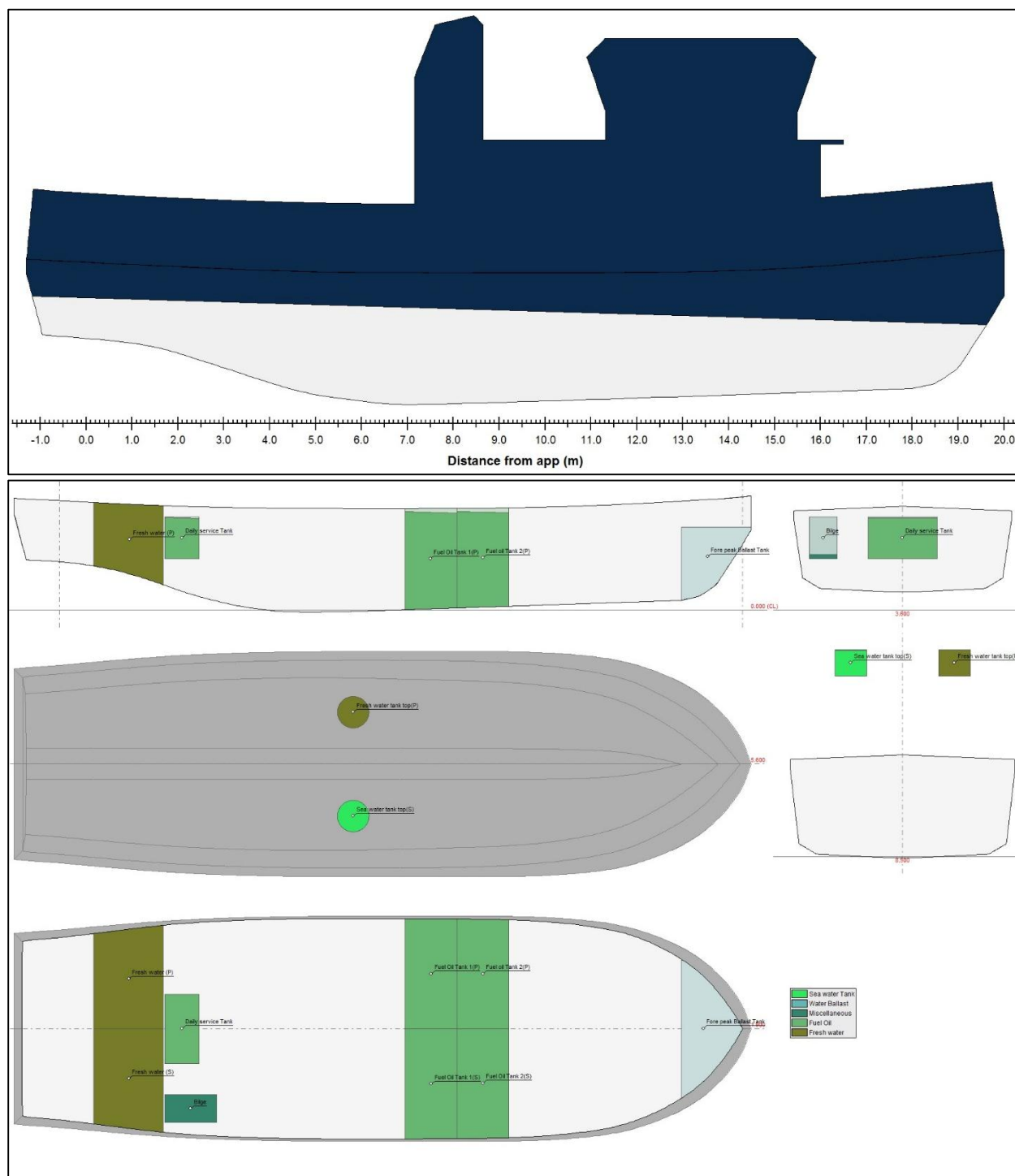
Maximum Bollard pull (T)	=	10.00	Ton
C.G. of Towline pull (C.G. _T)	=	04.40	m
V.C.B.	=	0.890	m
Lever (H = C.G. _T – V.C.B.)	=	3.510	m
Heeling Moment (0.7*T*H)	=	24.57	T-m
Displacement (Δ)	=	118.649	T
Heeling lever (0.7*T*H*cosθ/Δ)	=	0.207* Cosθ	m

Towing criteria				
Description	Attained value	Criterion	Required value	Complies
Area between heeling lever and righting lever calculated from the 1st intercept upto an angle 40° beyond the first intercept or upto the down flooding angle, whichever is less.	0.1315	m-rad >=	0.09	m-rad YES
The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the downflooding angle, whichever is less	1.555	- >=	1.4	YES
Note: The Towing Criteria for a Tug complies if one of the above two criteria is satisfied				
THE VESSEL COMPLIES WITH THE CRITERIA				

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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13.4 Ballast Departure condition

The below load case gives the Intact Stability Analysis of the vessel loaded with consumables at their maximum capacity and other miscellaneous items.

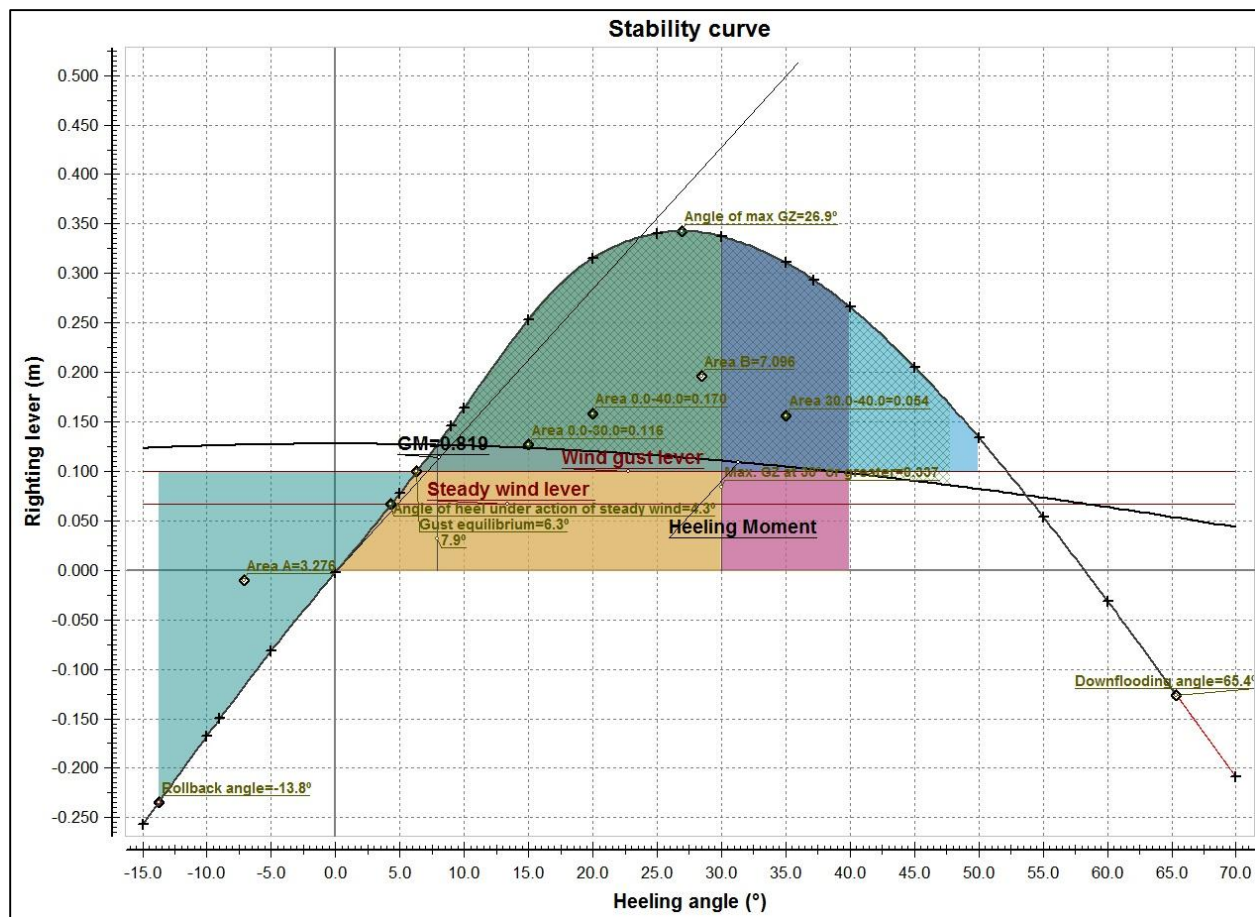


K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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Hydrostatic particulars			
List	-0.11 (SB) (Degr.)	KM	3.216 (m)
Draft aft pp	2.258 (m)	GM solid	0.955 (m)
Mean draft	1.968 (m)	GG'	0.136 (m)
Draft forward pp	1.677 (m)	GM liquid	0.819 (m)
Trim	-0.581 (m)	MCT	1.876 (t*m/cm)
Volume	172.33 (m ³)	Relative water density	1.0250
LCF	8.642 (m)	Immersion rate	1.211 (t/cm)

GZ values	
Heeling angle (Degr.)	GZ (m)
0.00 (CL)	-0.002
-5.00 (SB)	0.078
-9.00 (SB)	0.146
-10.00 (SB)	0.164
-15.00 (SB)	0.254
-20.00 (SB)	0.316
-25.00 (SB)	0.341
-30.00 (SB)	0.337
-35.00 (SB)	0.311
-37.20 (SB)	0.293
-40.00 (SB)	0.266
-45.00 (SB)	0.206
-50.00 (SB)	0.134
-55.00 (SB)	0.054
-60.00 (SB)	-0.031
-70.00 (SB)	-0.209

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Critical points			
Description	Type	Dist. to wl (m)	Submersion angle (Degr.)
Engine Room Funnel	Downflooding	5.643	-65.37

Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Sea water Tank							
Sea water tank top(S)	1.0250	98.0	0.503	5.577	8.500	-1.500 (SB)	0.034
Totals for Sea water Tank			0.503	5.577	8.500	-1.500 (SB)	0.034
Water Ballast							
Fore peak Ballast Tank	1.0250	0.0	0.000	0.000	0.000	0.000 (CL)	0.000
Totals for Water Ballast			0.000	0.000	0.000	0.000 (CL)	0.000
Miscellaneous							
Bilge	1.0000	10.0	0.141	1.547	3.800	-2.300 (SB)	0.063
Crew and Misc.			2.000	4.000	12.300	0.000 (CL)	
Solid Ballast for list correction			0.845	0.326	8.250	2.313 (PS)	
Totals for Miscellaneous			2.986	2.844	10.752	0.546 (PS)	0.063
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	98.0	10.504	1.488	10.747	1.536 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	98.0	10.504	1.488	10.747	-1.536 (SB)	3.573

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Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Fuel oil Tank 2(P)	0.8500	98.0	10.264	1.525	12.246	1.525 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	98.0	10.264	1.525	12.246	-1.525 (SB)	3.569
Daily service Tank	0.8500	98.0	1.959	2.075	3.550	0.000 (CL)	0.555
Totals for Fuel Oil			43.494	1.532	11.130	0.000 (CL)	14.839
Fresh water							
Fresh water (P)	1.0000	100.0	10.726	2.062	2.050	1.406 (PS)	4.534
Fresh water (S)	1.0000	100.0	10.726	2.062	2.050	-1.406 (SB)	4.534
Fresh water tank top(P)	1.0000	100.0	0.500	5.585	8.500	1.500 (PS)	0.034
Totals for Fresh water			21.952	2.142	2.197	0.034 (PS)	9.101
Lightship			107.736	2.548	9.276	-0.018 (SB)	
Deadweight			68.935	1.812	8.250	0.024 (PS)	24.037
Displacement			176.671	2.261	8.876	-0.002 (SB)	24.037

Evaluation of criteria				
Minimum design criteria applicable to Tugs				
Description	Attained value	Criterion	Required value	Complies
Area 0° - 30°	0.1158 (mrad)	>=	0.0550 (mrad)	YES
Area 0° - 40°	0.1696 (mrad)	>=	0.0900 (mrad)	YES
Area 30° - 40°	0.0537 (mrad)	>=	0.0300 (mrad)	YES
Max. GZ at 30° or greater	0.337 (m)	>=	0.200 (m)	YES
Angle of max GZ	26.9 (Degr.)	>=	25.0 (Degr.)	YES
Initial metacentric height	0.819 (m)	>	0.150 (m)	YES
Severe wind and rolling criterion				YES
Wind pressure	51.4 (kg/m ²)			
Wind area	76.97 (m ²)			
Steady wind lever	0.066 (m)			
Deck immersion angle	12.99 (Degr.)			
Wind gust lever	0.100 (m)			
Ratio of areaA/areaB	0.462	<=	1.000	YES
Maximum allowed static heeling angle	4.3 (Degr.)	<=	16.0 (Degr.)	YES
Max allowed ratio static angle/deck immersion angle	0.332	<=	0.800	YES
The vessel complies with the stability criteria				

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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13.4.1 Evaluation of towing criteria:

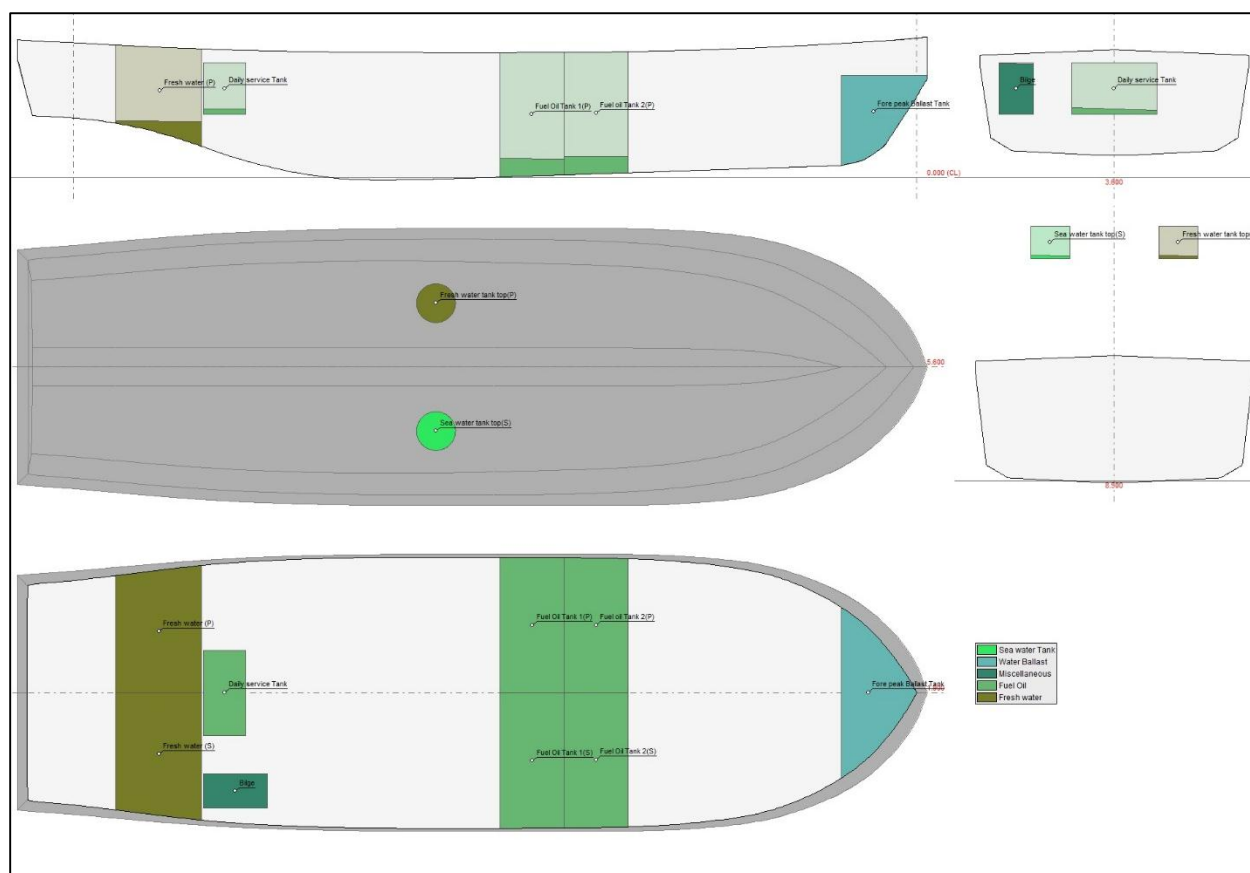
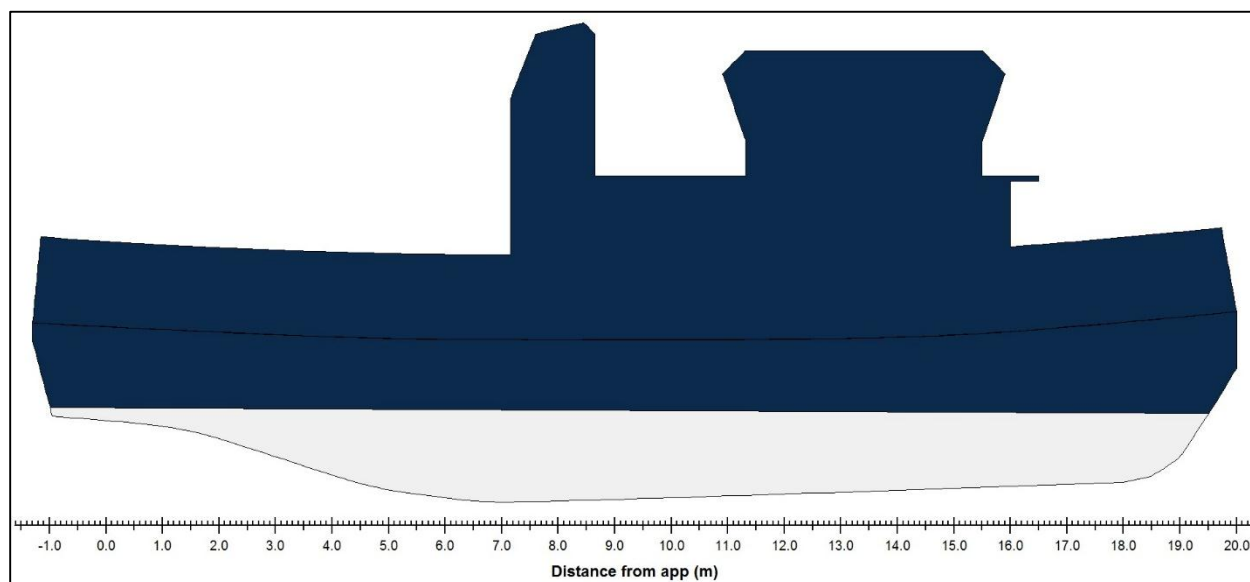
Maximum Bollard pull (T)	=	10.00	Ton
C.G. of Towline pull (C.G. _T)	=	04.40	m
V.C.B.	=	1.179	m
Lever (H = C.G. _T – V.C.B.)	=	3.221	m
Heeling Moment (0.7*T*H)	=	22.547	T-m
Displacement (Δ)	=	176.671	T
Heeling lever (0.7*T*H*cosθ/Δ)	=	0.128* Cosθ	m

Towing criteria				
Description	Attained value	Criterion	Required value	Complies
Area between heeling lever and righting lever calculated from the 1st intercept up to an angle 40° beyond the first intercept or up to the down flooding angle, whichever is less.	0.114	m-rad >=	0.09	m-rad YES
The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the down-flooding angle, whichever is less	2.067	- >=	1.40	YES
Note: The Towing Criteria for a Tug complies if one of the above two criteria is satisfied				
THE VESSEL COMPLIES WITH THE CRITERIA				

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
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13.5 Ballast Arrival condition

The below load case gives the Intact Stability Analysis of the vessel loaded with 10 %consumables and Ballast and other miscellaneous items.

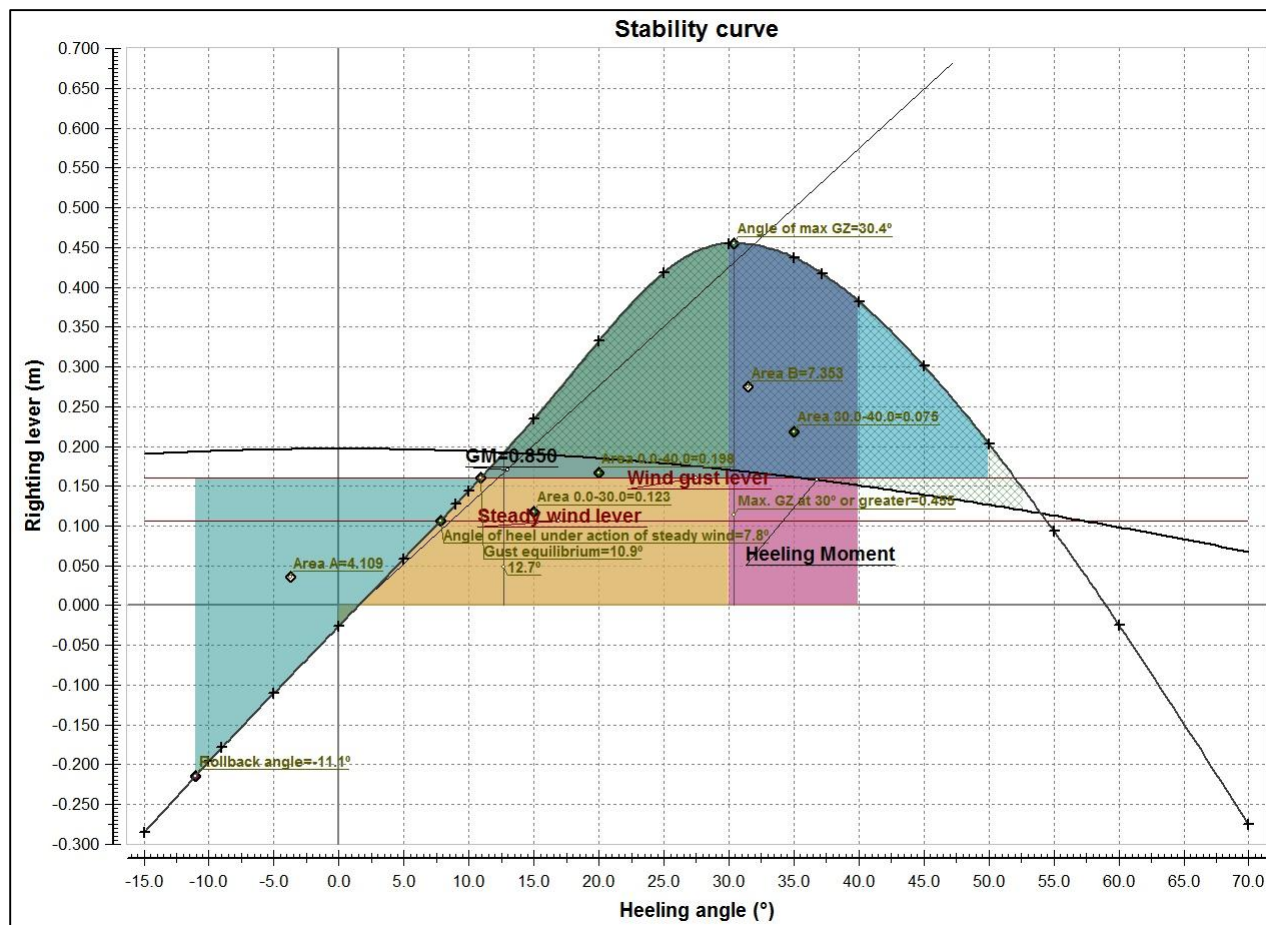


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Hydrostatic particulars			
List	-1.51 (SB) (Degr.)	KM	3.512 (m)
Draft aft pp	1.600 (m)	GM solid	1.097 (m)
Mean draft	1.553 (m)	GG'	0.247 (m)
Draft forward pp	1.505 (m)	GM liquid	0.850 (m)
Trim	-0.095 (m)	MCT	1.681 (t*m/cm)
Volume	121.10 (m ³)	Relative water density	1.0250
LCF	8.808 (m)	Immersion rate	1.150 (t/cm)

GZ values	
Heeling angle (Degr.)	GZ (m)
0.00 (CL)	-0.026
-5.00 (SB)	0.059
-9.00 (SB)	0.127
-10.00 (SB)	0.144
-15.00 (SB)	0.235
-20.00 (SB)	0.332
-25.00 (SB)	0.419
-30.00 (SB)	0.455
-35.00 (SB)	0.438
-37.20 (SB)	0.417
-40.00 (SB)	0.382
-45.00 (SB)	0.301
-50.00 (SB)	0.203
-55.00 (SB)	0.094
-60.00 (SB)	-0.024
-70.00 (SB)	-0.275

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Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Sea water Tank							
Sea water tank top(S)	1.0250	10.0	0.051	5.238	8.500	-1.500 (SB)	0.034
Totals for Sea water Tank			0.051	5.238	8.500	-1.500 (SB)	0.034
Water Ballast							
Fore peak Ballast Tank	1.0250	100.0	5.477	1.768	18.604	0.000 (CL)	6.636
Totals for Water Ballast			5.477	1.768	18.604	0.000 (CL)	6.636
Miscellaneous							
Bilge	1.0000	98.0	1.383	2.075	3.800	-2.300 (SB)	0.063
Crew and Misc.			2.000	4.000	12.300	0.000 (CL)	
Solid Ballast for list correction			0.845	0.326	8.250	2.313 (PS)	
Totals for Miscellaneous			4.228	2.636	8.710	-0.290 (SB)	0.063
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	10.0	1.072	0.245	10.726	1.255 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	10.0	1.072	0.245	10.726	-1.255 (SB)	3.573
Fuel oil Tank 2(P)	0.8500	10.0	1.047	0.298	12.222	1.221 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	10.0	1.047	0.298	12.222	-1.221 (SB)	3.569
Daily service Tank	0.8500	10.0	0.200	1.547	3.550	0.000 (CL)	0.555
Totals for Fuel Oil			4.438	0.329	11.109	0.000 (CL)	14.839

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Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Fresh water							
Fresh water (P)	1.0000	10.0	1.073	1.129	2.429	1.171 (PS)	4.534
Fresh water (S)	1.0000	10.0	1.073	1.129	2.429	-1.171 (SB)	4.534
Fresh water tank top(P)	1.0000	10.0	0.050	5.238	8.500	1.500 (PS)	0.034
Totals for Fresh water			2.195	1.222	2.567	0.034 (PS)	9.101
Lightship			107.736	2.548	9.276	-0.018 (SB)	
Deadweight			16.389	1.540	11.843	-0.075 (SB)	30.673
Displacement			124.125	2.415	9.615	-0.026 (SB)	30.673

Evaluation of criteria				
Minimum design criteria applicable to Tugs				
Description	Attained value	Criterion	Required value	Complies
Area 0° - 30°	0.1231 (mrad)	>=	0.0550 (mrad)	YES
Area 0° - 40°	0.1983 (mrad)	>=	0.0900 (mrad)	YES
Area 30° - 40°	0.0752 (mrad)	>=	0.0300 (mrad)	YES
Max. GZ at 30° or greater	0.455 (m)	>=	0.200 (m)	YES
Angle of max GZ	30.4 (Degr.)	>=	25.0 (Degr.)	YES
Initial metacentric height	0.850 (m)	>	0.150 (m)	YES
Severe wind and rolling criterion				YES
Wind pressure	51.4 (kg/m ²)			
Wind area	85.86 (m ²)			
Steady wind lever	0.107 (m)			
Deck immersion angle	22.19 (Degr.)			
Wind gust lever	0.160 (m)			
Ratio of areaA/areaB	0.559	<=	1.000	YES
Maximum allowed static heeling angle	7.8 (Degr.)	<=	16.0 (Degr.)	YES
Max allowed ratio static angle/deck immersion angle	0.352	<=	0.800	YES
The vessel complies with the stability criteria				

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13.5.1 Evaluation of towing criteria:

Maximum Bollard pull (T)	=	10.00	Ton
C.G. of Towline pull (C.G. _T)	=	04.40	m
V.C.B.	=	0.912	m
Lever (H = C.G. _T – V.C.B.)	=	3.488	m
Heeling Moment (0.7*T*H)	=	24.416	T-m
Displacement (Δ)	=	124.125	T
Heeling lever (0.7*T*H*cosθ/Δ)	=	0.197* Cosθ	m

Towing criteria				
Description	Attained value	Criterion	Required value	Complies
Area between heeling lever and righting lever calculated from the 1st intercept upto an angle 40° beyond the first intercept or upto the down flooding angle, whichever is less.	0.1271	m-rad >=	0.09	m-rad YES
The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the downflooding angle, whichever is less	1.564	- >=	1.4	YES
Note: The Towing Criteria for a Tug complies if one of the above two criteria is satisfied				
THE VESSEL COMPLIES WITH THE CRITERIA				

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14 Worked out Example

The trim and stability calculations have been carried out for different loading conditions. The calculation for any other loading condition can be carried out if we know the Wt, LCG, VCG and free surface moments of all the dead-weight items in that loading condition. As for example in loaded departure condition, all the items have been considered full. Therefore 98% of all the tank items, their LCG, VCG and free surface moments have been taken. Similarly 98% wt of stores, its LCG, VCG have been taken. The wt of crew & effect will be same for all the loading conditions.

The below example is given for the Load case "Ballast Departure condition (sec-13.4)".

Loading detail:

Description	Density (t/m ³)	Fill%	Weight (tonnes)	VCG (m)	LCG (m)	TCG (m)	FSM (Tonnes*m)
Sea water Tank							
Sea water tank top(S)	1.0250	98.0	0.503	5.577	8.500	-1.500 (SB)	0.034
Totals for Sea water Tank			0.503	5.577	8.500	-1.500 (SB)	0.034
Water Ballast							
Fore peak Ballast Tank	1.0250	0.0	0.000	0.000	0.000	0.000 (CL)	0.000
Totals for Water Ballast			0.000	0.000	0.000	0.000 (CL)	0.000
Miscellaneous							
Bilge	1.0000	10.0	0.141	1.547	3.800	-2.300 (SB)	0.063
Crew and Misc.			2.000	4.000	12.300	0.000 (CL)	
Solid Ballast for list correction			0.845	0.326	8.250	2.313 (PS)	
Totals for Miscellaneous			2.986	2.844	10.752	0.546 (PS)	0.063
Fuel Oil							
Fuel Oil Tank 1(P)	0.8500	98.0	10.504	1.488	10.747	1.536 (PS)	3.573
Fuel Oil Tank 1(S)	0.8500	98.0	10.504	1.488	10.747	-1.536 (SB)	3.573
Fuel oil Tank 2(P)	0.8500	98.0	10.264	1.525	12.246	1.525 (PS)	3.569
Fuel Oil Tank 2(S)	0.8500	98.0	10.264	1.525	12.246	-1.525 (SB)	3.569
Daily service Tank	0.8500	98.0	1.959	2.075	3.550	0.000 (CL)	0.555
Totals for Fuel Oil			43.494	1.532	11.130	0.000 (CL)	14.839
Fresh water							
Fresh water (P)	1.0000	100.0	10.726	2.062	2.050	1.406 (PS)	4.534
Fresh water (S)	1.0000	100.0	10.726	2.062	2.050	-1.406 (SB)	4.534
Fresh water tank top(P)	1.0000	100.0	0.500	5.585	8.500	1.500 (PS)	0.034
Totals for Fresh water			21.952	2.142	2.197	0.034 (PS)	9.101
Lightship			107.736	2.548	9.276	-0.018 (SB)	
Deadweight			68.935	1.812	8.250	0.024 (PS)	24.037
Displacement			176.671	2.261	8.876	-0.002 (SB)	24.037

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The above table is to be evaluated to obtain displacements and centre of gravities for the loading condition as shown below:

DESCRIPTION	WEIGHT	LCG	LMOM	TCG	TMOM	VCG	VMOM	FSM
Weight-1	W1	X1	W1X1	Y1	W1Y1	Z1	W1Z1	M1
Weight-2	W2	X2	W2X2	Y2	W2Y2	Z2	W2Z2	M2
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
Weight-n	Wn	Xn	WnXn	Yn	WnYn	Zn	WnZn	Mn
Total Weight	ΣWn	$\Sigma WnXn/\Sigma Wn$	$\Sigma WnXn$	$\Sigma WnYn/\Sigma Wn$	$\Sigma WnYn$	$\Sigma WnZn/\Sigma Wn$	$\Sigma WnZn$	ΣMn

$$\begin{aligned}\text{TOTAL DISPLACEMENT} &= \Sigma Wn \\ &= 176.671 \text{ T}\end{aligned}$$

$$\begin{aligned}\text{LCG} &= (\Sigma Wn \times Xn) / \Sigma Wn \\ &= 8.876 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{VCG} &= (\Sigma Wn \times Zn) / \Sigma Wn \\ &= 2.261 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{TCG} &= (\Sigma Wn \times Yn) / \Sigma Wn \\ &= -0.002 \text{ m}\end{aligned}$$

Once the displacement and Centroids are found, the draft values and other hydrostatic calculations are calculated from hydrostatic table for corresponding displacement at trim=0. The hydrostatics for the required displacement is calculated by interpolation.

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Trim: 0.000 (m)									
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	TpCm (t/cm)
1.400	103.907	106.504	9.852	0.817	9.337	130.64	3.611	25.95	1.068
1.450	109.192	111.922	9.822	0.846	9.110	135.48	3.569	27.19	1.102
1.500	114.636	117.501	9.784	0.876	8.956	139.62	3.537	27.71	1.128
1.550	120.184	123.189	9.743	0.906	8.868	143.07	3.505	27.63	1.146
1.600	125.800	128.945	9.704	0.936	8.843	145.85	3.462	27.03	1.156
1.650	131.457	134.743	9.667	0.966	8.828	148.54	3.422	26.40	1.166
1.700	137.171	140.600	9.631	0.995	8.816	151.23	3.385	25.80	1.175
1.750	142.923	146.496	9.598	1.025	8.808	153.90	3.352	25.24	1.183
1.800	148.718	152.436	9.567	1.054	8.805	156.55	3.322	24.69	1.192
1.850	154.545	158.408	9.539	1.083	8.810	159.14	3.291	24.14	1.199
1.900	160.407	164.417	9.512	1.112	8.822	161.70	3.261	23.58	1.205
1.950	166.303	170.461	9.488	1.141	8.835	164.27	3.235	23.07	1.212
2.000	172.232	176.538	9.466	1.169	8.847	166.84	3.213	22.59	1.218
2.050	178.188	182.642	9.445	1.198	8.860	169.43	3.194	22.14	1.225
2.100	184.178	188.782	9.426	1.227	8.874	172.02	3.178	21.73	1.232
2.150	190.202	194.957	9.409	1.255	8.888	174.63	3.165	21.35	1.239
2.200	196.265	201.171	9.393	1.283	8.902	177.24	3.154	20.99	1.245

Interpolated Hydrostatics									
Draft (m)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	MCT (t*m/cm)	TpCm (t/cm)
2.000	176.538	9.466	1.169	8.847	166.84	3.213	22.59	1.914	1.218
2.050	182.642	9.445	1.198	8.86	169.43	3.194	22.14	1.936	1.225
2.001	176.671	9.466	1.170	8.847	166.896	3.213	22.580	1.914	1.218

Trim calculation:

$$\begin{aligned}
 \text{Trim} &= \frac{(LCG - LCB) \times LBP}{(KMI - KG)} \\
 &= \frac{(8.876 - 9.466) \times 19.76}{(22.580 - 2.261)} \\
 &= 0.573 \text{ m by aft}
 \end{aligned}$$

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Hydrostatics for Trim=-0.50m									
Draft	Displ.	LCB	VCB	LCF	S	KMt	KMI	MCT	TpCm
(m)	(tonnes)	(m)	(m)	(m)	(m^2)	(m)	(m)	(t*m/cm)	(t/cm)
1.950	173.924	8.933	1.164	8.665	165.620	3.224	22.420	1.871	1.209
2.000	179.981	8.924	1.192	8.675	168.180	3.203	21.960	1.892	1.215
1.973	176.671	8.929	1.177	8.670	166.781	3.214	22.211	1.881	1.212

Hydrostatics for Trim=-0.75m									
Draft	Displ.	LCB	VCB	LCF	S	KMt	KMI	MCT	TpCm
(m)	(tonnes)	(m)	(m)	(m)	(m^2)	(m)	(m)	(t*m/cm)	(t/cm)
1.950	175.842	8.664	1.181	8.585	166.4	3.224	22.14	1.865	1.208
2.000	181.898	8.662	1.209	8.594	168.95	3.204	21.69	1.885	1.214
1.957	176.671	8.664	1.185	8.586	166.749	3.221	22.078	1.868	1.209

Hydrostatics for trim=-0.573 m										
Trim	Draft	Displ.	LCB	VCB	LCF	S	KMt	KMI	MCT	TpCm
(m)	(m)	(tonnes)	(m)	(m)	(m)	(m^2)	(m)	(m)	(t*m/cm)	(t/cm)
-0.750	1.957	176.671	8.664	1.185	8.586	166.749	3.221	22.078	1.868	1.209
-0.500	1.973	176.671	8.929	1.177	8.670	166.781	3.214	22.211	1.881	1.212
-0.573	1.968	176.671	8.851	1.179	8.645	166.772	3.216	22.173	1.877	1.211

So,

$$\text{Mean Draft } T_m = 1.968 \text{ m}$$

$$\text{Draft at LCF } T_{LCF} = T_m - \left(\frac{\frac{LBP}{2} - LCF}{LBP} \times TRIM \right)$$

$$= 2.00 \text{ m}$$

$$\text{Draft at Aft perpendicular } T_{AP} = T_{LCF} - \frac{LCF \times TRIM}{LBP}$$

$$= 2.255 \text{ m}$$

$$\text{Draft at Fwd perpendiculars } T_{FP} = T_{AP} + TRIM$$

$$= 1.682 \text{ m}$$

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Free Surface Correction: Provided a tank is completely filled with a liquid no movement of the liquid is possible and the effect on the ship's stability is precisely the same as if the tank contained solid material. When a quantity of liquid is withdrawn from the tank, the situation changes completely and stability of ship is adversely affected by what is known as "free surface effects". The adverse effect on the stability is referred to as „loss in GM" or as a virtual rise in VCG and is calculated as follows.

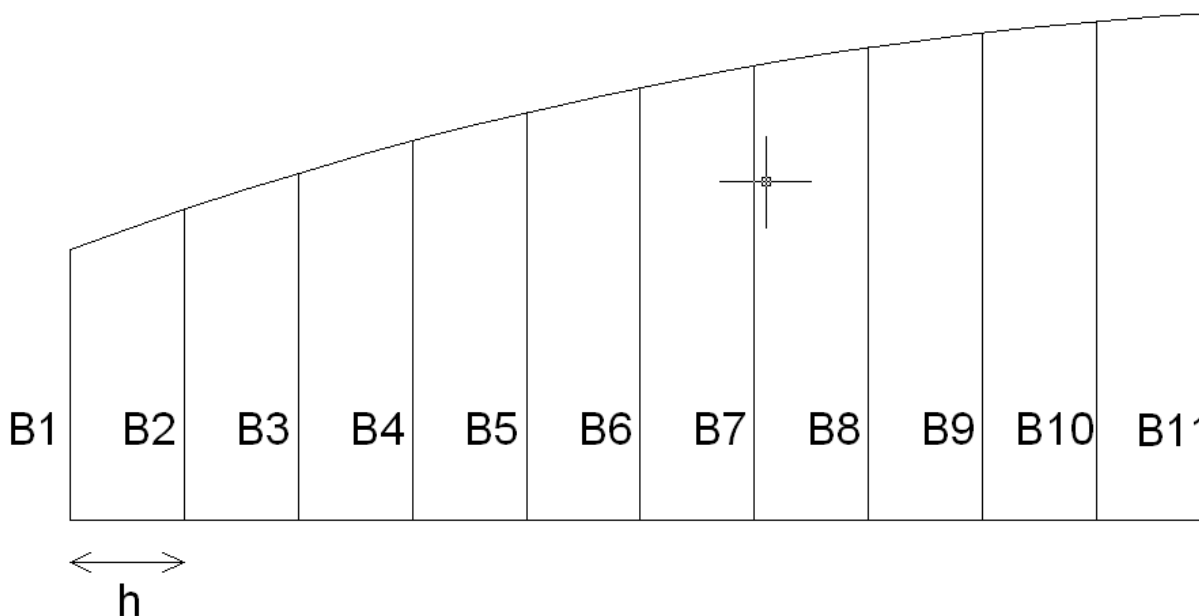
Loss in GM due to free surface effects (in meter) = Free surface moment (Tonnes Meters)/Displacement of Vessel (Tonnes)

Notes:

1. Slack must not be kept in all tanks. It must be collected in one tank.
2. The Master has to ensure that Actual KG < Maximum Permissible KG for corresponding displacement.

Example for computation of FSM

Tank FSM computation example for a random shape:



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Let the half breadths of a cross section be as shown above

The second moment of area is calculated as shown below using Simpsons rule:

S.no	B	B ³	SM	PRODUCT
01	B ₁	(B ₁) ³	1	1x(B ₁) ³
02	B ₂	(B ₂) ³	2	2x(B ₂) ³
03	B ₃	(B ₃) ³	4	4x(B ₃) ³
04	B ₄	(B ₄) ³	2	2x(B ₄) ³
05	B ₅	(B ₅) ³	4	4x(B ₅) ³
06	B ₆	(B ₆) ³	2	2x(B ₆) ³
07	B ₇	(B ₇) ³	4	4x(B ₇) ³
08	B ₈	(B ₈) ³	2	2x(B ₈) ³
09	B ₉	(B ₉) ³	4	4x(B ₉) ³
10	B ₁₀	(B ₁₀) ³	2	2x(B ₁₀) ³
11	B ₁₁	(B ₁₁) ³	1	1x(B ₁₁) ³

$$I = 2/9(h \times \Sigma \text{PRODUCT})$$

Free surface moment = $I \times \rho$ (density of the contents in the tank)

Here we have considered the maximum FSM for the tanks which are filled with their liquids for conservative approach.

Σ Free surface moment for all tanks = 24.037 t-m

$$\begin{aligned} \text{Free surface effect} &= \text{FSM} / \Delta \\ &= 24.037 / 176.671 \\ &= 0.136 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{KG corrected} &= 2.261 + 0.136 \\ &= 2.397 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{GM corrected} &= \text{KM} - \text{KG corrected} \\ &= 3.216 - 2.397 \\ &= 0.819 \text{ m} \end{aligned}$$

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Heel Calculation:

$$\begin{aligned}
 \text{HEEL} &= \text{TAN}^{-1}(\text{TCG} / \text{GM corrected}) \\
 &= \text{TAN}^{-1}(-0.002/0.819) \\
 &= 0.14 \text{ deg}
 \end{aligned}$$

Stability Criteria:

The purpose of cross curve is to enable stability curve to be drawn for the ship in any sailing condition.

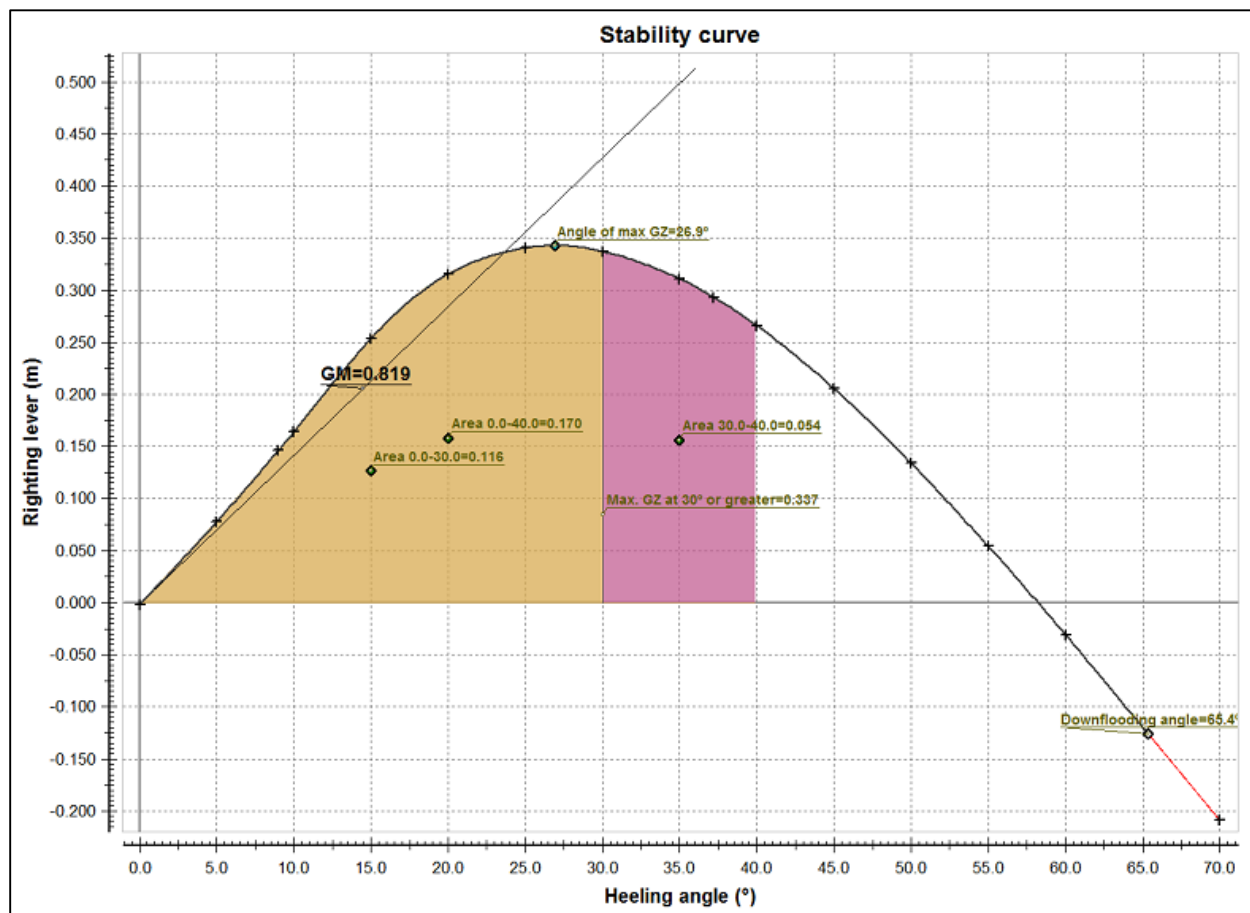
$$\text{GZ} = \text{KN} - \text{KG} \times \text{SIN} \theta - \text{GG}' \times \text{SIN} \theta - \text{TCG} \times \text{COS} \theta$$

The values for Trim = -0.562 m are evaluated as below.

Heeling angle (Degr.)	KN (m)	GZ values VCG sin(θ) (m)	GG' sin(θ) (m)	TCG cos(θ) (m)	GZ (m)
0.00 (CL)	0.000	0.000	0.000	0.002	-0.002
-5.00 (SB)	0.281	0.197	0.004	0.002	0.078
-9.00 (SB)	0.507	0.354	0.006	0.002	0.146
-10.00 (SB)	0.564	0.393	0.006	0.002	0.164
-15.00 (SB)	0.847	0.585	0.006	0.002	0.254
-20.00 (SB)	1.097	0.773	0.006	0.002	0.316
-25.00 (SB)	1.304	0.956	0.006	0.002	0.341
-30.00 (SB)	1.476	1.130	0.006	0.002	0.337
-35.00 (SB)	1.616	1.297	0.006	0.001	0.311
-37.20 (SB)	1.668	1.367	0.006	0.001	0.293
-40.00 (SB)	1.727	1.453	0.006	0.001	0.266
-45.00 (SB)	1.811	1.599	0.006	0.001	0.206
-50.00 (SB)	1.873	1.732	0.006	0.001	0.134
-55.00 (SB)	1.912	1.852	0.005	0.001	0.054
-60.00 (SB)	1.933	1.958	0.005	0.001	-0.031
-70.00 (SB)	1.920	2.125	0.004	0.001	-0.209

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A graph is plotted between angle of heel and GZ value:



The area under the curve is determined for the following regions in order to satisfy the stability criteria.

1. Area under the curve between 0 and 30 is 0.1158 m-rad (minimum required: 0.055m-rad)
2. Area under the curve between 0 and 40 is 0.1696 m-rad (minimum required: 0.09 m-rad)
3. Area under the curve between 30 and 40 is 0.0537 m-rad (minimum required: 0.03 m-rad)
4. Maximum GZ at an angle 30 or greater is 0.337 and the angle is 26.90 deg.
5. Initial metacentric height is 0.819 m

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Severe Wind and Rolling Criteria:

Wind Pressure = 51.4 kg/m²

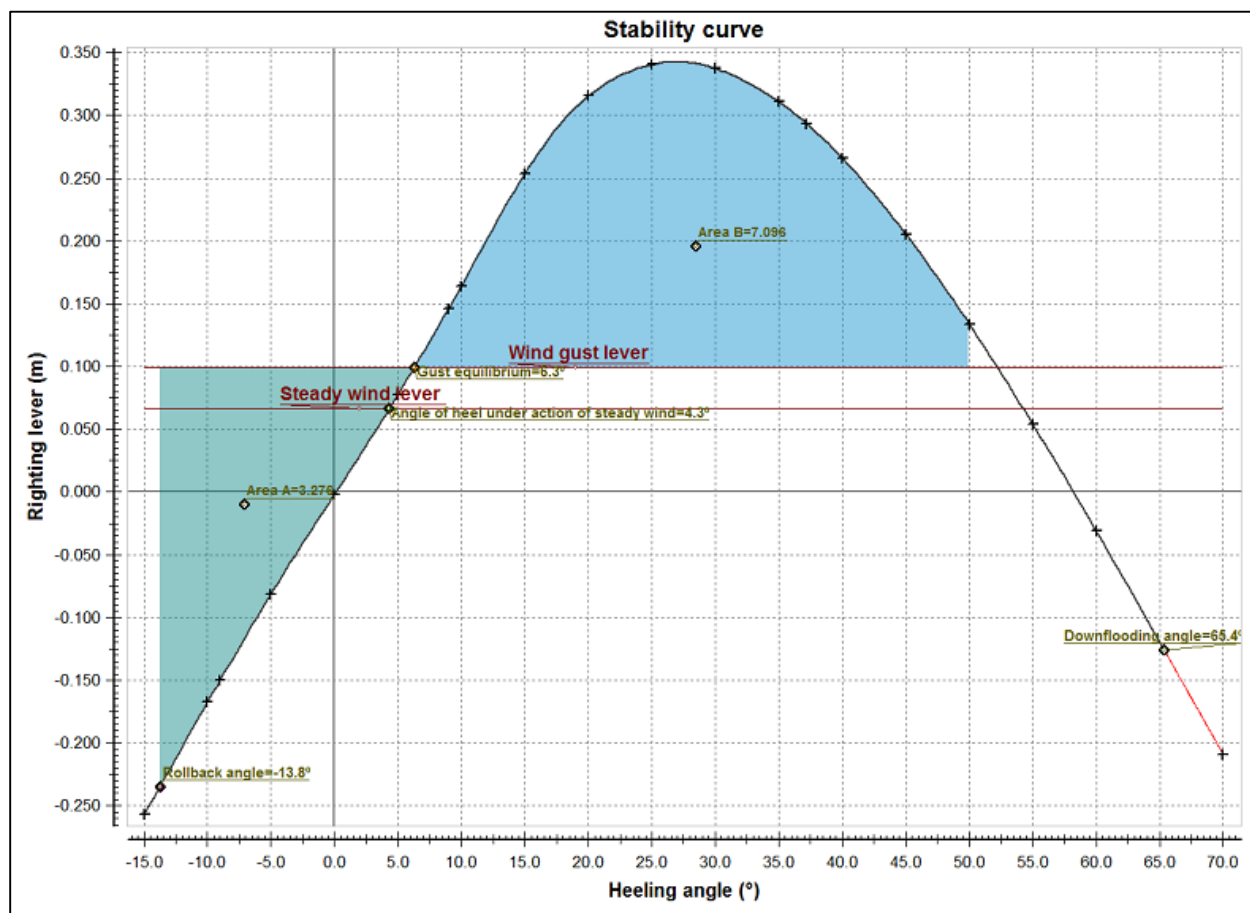
Wind Area = 76.97 m² (Interpolated Windage area table)

Wind Lever = 2.95 m (Interpolated Windage area table)

$$\begin{aligned}
 LW_1 &= (P \times A \times Z) / (1000 \times g \times \Delta) \text{ m} \\
 &= (504 \times 76.97 \times 2.95) / (1000 \times 9.81 \times 176.671) \\
 &= 0.066 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 LW_2 &= 1.5 \times LW_1 \\
 &= 1.5 \times 0.066 \\
 &= 0.100 \text{ m}
 \end{aligned}$$

Evaluation of Criteria:



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Deck immersion angle = 12.99 deg

Ratio of areaA/areaB = 0.462 (Maximum = 1)

Maximum allowed static heeling angle = 4.30 deg (Maximum 16.00 deg)

Max allowed ratio static angle/deck immersion angle = 0.332 (maximum 0.8)

Towing Criteria:

We calculate the heeling lever for every angle of heel as shown below

Maximum Bollard pull (T) = 10.00 Ton

C.G. of Towline pull (C.G._T) = 04.40 m

V.C.B. = 1.179 m

Lever (H = C.G._T – V.C.B.) = 3.221 m

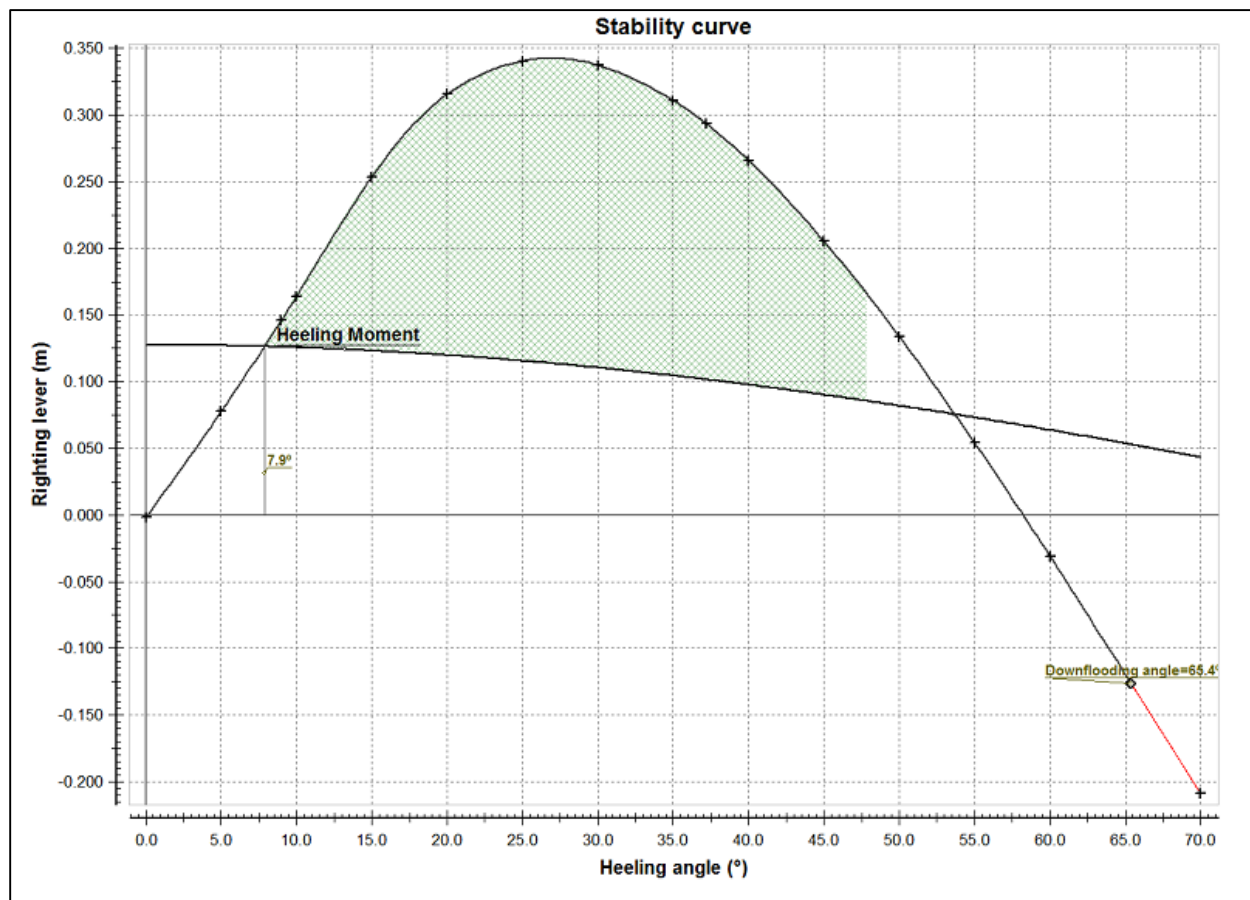
Heeling Moment (0.7*T*H) = 22.547 T-m

Displacement (Δ) = 176.671 T

Heeling lever (0.7*T*H*cosθ/Δ) = 0.128* Cosθ m

Calculation of Heeling Lever due to tow						
θ	cosθ	Bollard pull	H	disp	Heeling Lever	Area
Degr.	-	T	m	T	m	m2
0	1.000	10.000	3.221	176.671	0.128	0.000
5	0.996	10.000	3.221	176.671	0.127	0.011
10	0.985	10.000	3.221	176.671	0.126	0.022
15	0.966	10.000	3.221	176.671	0.123	0.033
20	0.940	10.000	3.221	176.671	0.120	0.044
25	0.906	10.000	3.221	176.671	0.116	0.054
30	0.866	10.000	3.221	176.671	0.111	0.064
35	0.819	10.000	3.221	176.671	0.105	0.073
40	0.766	10.000	3.221	176.671	0.098	0.082
45	0.707	10.000	3.221	176.671	0.090	0.090
50	0.643	10.000	3.221	176.671	0.082	0.098
55	0.574	10.000	3.221	176.671	0.073	0.105
60	0.500	10.000	3.221	176.671	0.064	0.111

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Area between heeling lever and righting lever calculated from the 1st intercept upto an angle 40° beyond the first intercept or up to the down flooding angle, whichever is less is 0.114 m-rad (minimum 0.09 m-rad)

or

The ratio of area under a righting lever curve to the area under a heeling lever curve upto an angle of 40° or the Down flooding angle, whichever is less is 2.067 (minimum 1.4)

THE VESSEL COMPLIES WITH THE CRITERIA

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15 Dead weight table

Displ. (SW) (tonnes)	Draft (m)	DWT(SW) (tonnes)	Draft (m)	Displ.(FW) (tonnes)	Draft (m)	DWT(FW) (tonnes)	Draft (m)
107.00	1.40	0.00	1.41	107.00	1.43	0	1.44
108.00	1.41	1.00	1.42	108.00	1.44	1.00	1.45
109.00	1.42	2.00	1.43	109.00	1.45	2.00	1.45
110.00	1.43	3.00	1.44	110.00	1.46	3.00	1.46
111.00	1.44	4.00	1.45	111.00	1.47	4.00	1.47
112.00	1.45	5.00	1.46	112.00	1.48	5.00	1.48
113.00	1.46	6.00	1.47	113.00	1.48	6.00	1.49
114.00	1.47	7.00	1.48	114.00	1.49	7.00	1.50
115.00	1.48	8.00	1.48	115.00	1.50	8.00	1.51
116.00	1.49	9.00	1.49	116.00	1.51	9.00	1.52
117.00	1.5	10.00	1.5	117.00	1.52	10.00	1.53
118.00	1.5	11.00	1.51	118.00	1.53	11.00	1.54
119.00	1.51	12.00	1.52	119.00	1.54	12.00	1.55
120.00	1.52	13.00	1.53	120.00	1.55	13.00	1.55
121.00	1.53	14.00	1.54	121.00	1.56	14.00	1.56
122.00	1.54	15.00	1.55	122.00	1.57	15.00	1.57
123.00	1.55	16.00	1.55	123.00	1.58	16.00	1.58
124.00	1.56	17.00	1.56	124.00	1.58	17.00	1.59
125.00	1.57	18.00	1.57	125.00	1.59	18.00	1.60
126.00	1.57	19.00	1.58	126.00	1.60	19.00	1.61
127.00	1.58	20.00	1.59	127.00	1.61	20.00	1.62
128.00	1.59	21.00	1.6	128.00	1.62	21.00	1.63
129.00	1.6	22.00	1.61	129.00	1.63	22.00	1.63
130.00	1.61	23.00	1.62	130.00	1.64	23.00	1.64
131.00	1.62	24.00	1.62	131.00	1.65	24.00	1.65
132.00	1.63	25.00	1.63	132.00	1.65	25.00	1.66
133.00	1.63	26.00	1.64	133.00	1.66	26.00	1.67
134.00	1.64	27.00	1.65	134.00	1.67	27.00	1.68
135.00	1.65	28.00	1.66	135.00	1.68	28.00	1.69
136.00	1.66	29.00	1.67	136.00	1.69	29.00	1.70
137.00	1.67	30.00	1.68	137.00	1.70	30.00	1.70
138.00	1.68	31.00	1.68	138.00	1.71	31.00	1.71
139.00	1.69	32.00	1.69	139.00	1.72	32.00	1.72
140.00	1.69	33.00	1.7	140.00	1.72	33.00	1.73
141.00	1.7	34.00	1.71	141.00	1.73	34.00	1.74
142.00	1.71	35.00	1.72	142.00	1.74	35.00	1.75
143.00	1.72	36.00	1.73	143.00	1.75	36.00	1.76

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

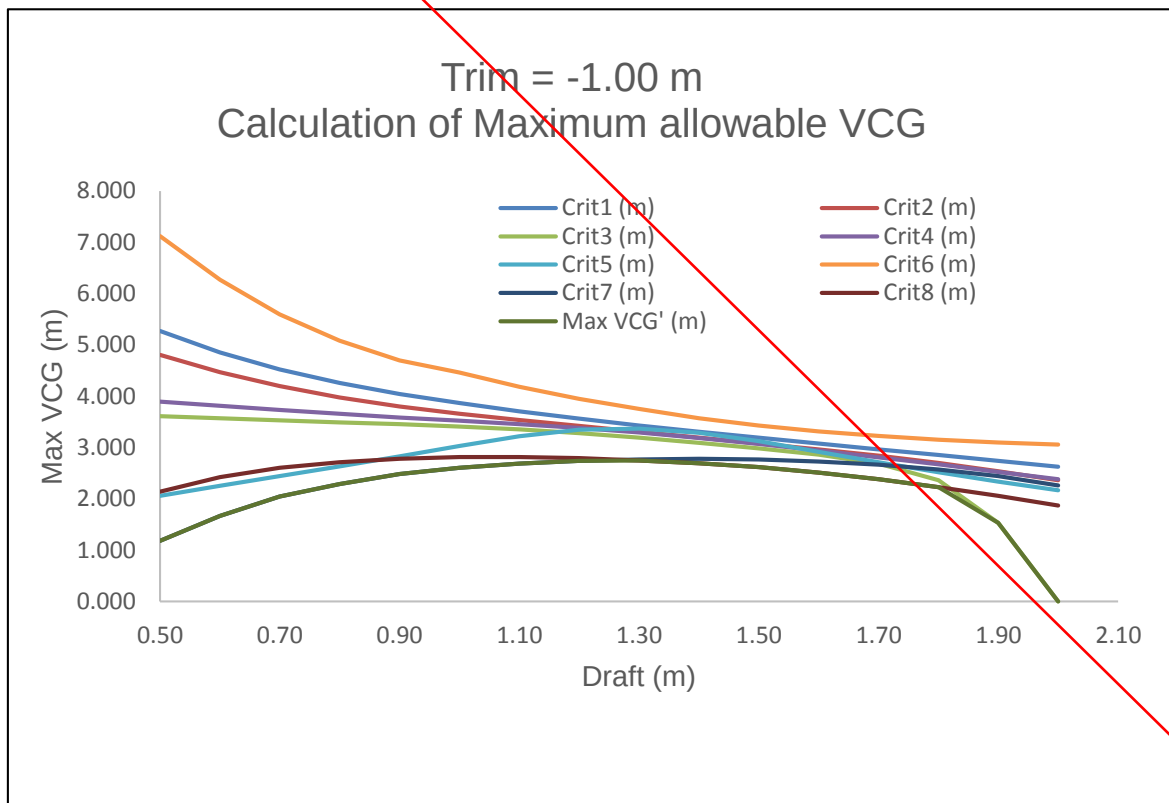
Displ. (SW) (tonnes)	Draft (m)	DWT(SW) (tonnes)	Draft (m)	Displ.(FW) (tonnes)	Draft (m)	DWT(FW) (tonnes)	Draft (m)
144.00	1.73	37.00	1.74	144.00	1.76	37.00	1.77
145.00	1.74	38.00	1.74	145.00	1.77	38.00	1.77
146.00	1.75	39.00	1.75	146.00	1.78	39.00	1.78
147.00	1.75	40.00	1.76	147.00	1.79	40.00	1.79
148.00	1.76			148.00	1.79	41.00	1.80
149.00	1.77			149.00	1.80	42.00	1.81
150.00	1.78			150.00	1.81	43.00	1.82
151.00	1.79			151.00	1.82	44.00	1.83
152.00	1.8			152.00	1.83	45.00	1.83
153.00	1.8			153.00	1.84	46.00	1.84
154.00	1.81			154.00	1.85	47.00	1.85
155.00	1.82			155.00	1.85	48.00	1.86
156.00	1.83			156.00	1.86	49.00	1.87
157.00	1.84			157.00	1.87	50.00	1.88
158.00	1.85			158.00	1.88	51.00	1.89
159.00	1.85			159.00	1.89	52.00	1.89
160.00	1.86			160.00	1.90	53.00	1.90
161.00	1.87			161.00	1.91	54.00	1.91
162.00	1.88			162.00	1.91	55.00	1.92
163.00	1.89			163.00	1.92	56.00	1.93
164.00	1.9			164.00	1.93	57.00	1.94
165.00	1.9			165.00	1.94	58.00	1.95
166.00	1.91			166.00	1.95	59.00	1.95
167.00	1.92			167.00	1.96	60.00	1.96
168.00	1.93			168.00	1.96	61.00	1.97
169.00	1.94			169.00	1.97	62.00	1.98
170.00	1.95			170.00	1.98	63.00	1.99
171.00	1.95			171.00	1.99	64.00	2.00
172.00	1.96			172.00	2.00	65.00	2.00
173.00	1.97			173.00	2.01	66.00	2.01
174.00	1.98			174.00	2.01	67.00	2.02
175.00	1.99			175.00	2.02	68.00	2.03
176.00	2.00			176.00	2.03	69.00	2.04
177.00	2.00			177.00	2.04		

K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

16 Max VCG

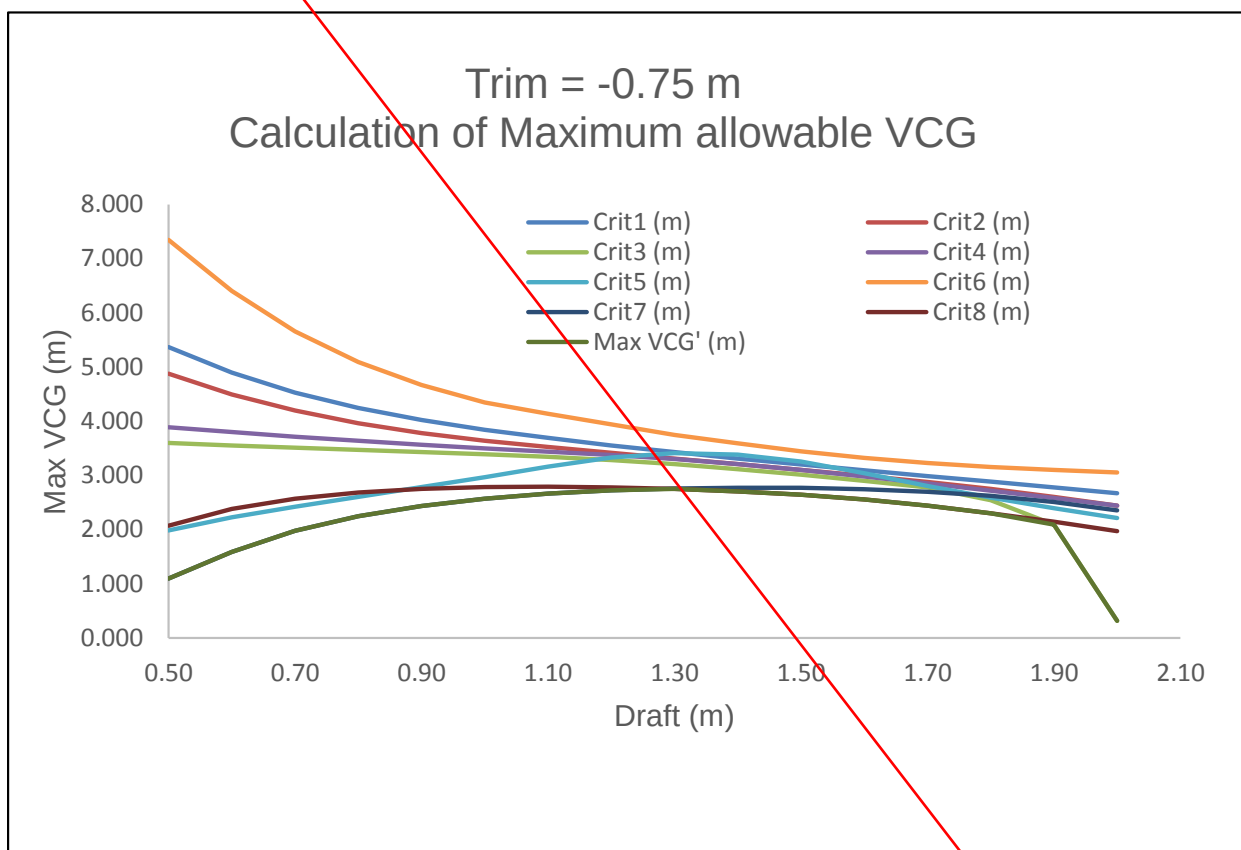
Minimum design criteria applicable to Tugs	
Criterion	description
Crit1	Area 0° - 30°
Crit2	Area 0° - 40°
Crit3	Area 30° - 40°
Crit4	Max. GZ at 30° or greater
Crit5	Angle of max GZ
Crit6	Initial metacentric height
Crit7	Severe wind and rolling criterion
Crit9	Area Between GZ curve and heeling arm curve(Tug criteria)

Initial trim: -1.000 (by stern)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
1.40	3.309	3.195	3.093	3.190	3.290	3.573	2.779	2.694	2.694	1.030
1.50	3.191	3.081	2.983	3.071	3.126	3.430	2.767	2.619	2.619	0.962
1.60	3.078	2.965	2.864	2.944	2.917	3.316	2.731	2.514	2.514	0.953
1.70	2.965	2.840	2.688	2.809	2.712	3.226	2.665	2.383	2.383	0.994
1.80	2.855	2.698	2.362	2.671	2.521	3.156	2.573	2.231	2.231	1.075
1.90	2.742	2.539	1.531	2.529	2.337	3.100	2.443	2.060	1.531	1.720
2.00	2.625	2.362	0.000	2.386	2.169	3.059	2.265	1.868	0.000	3.210



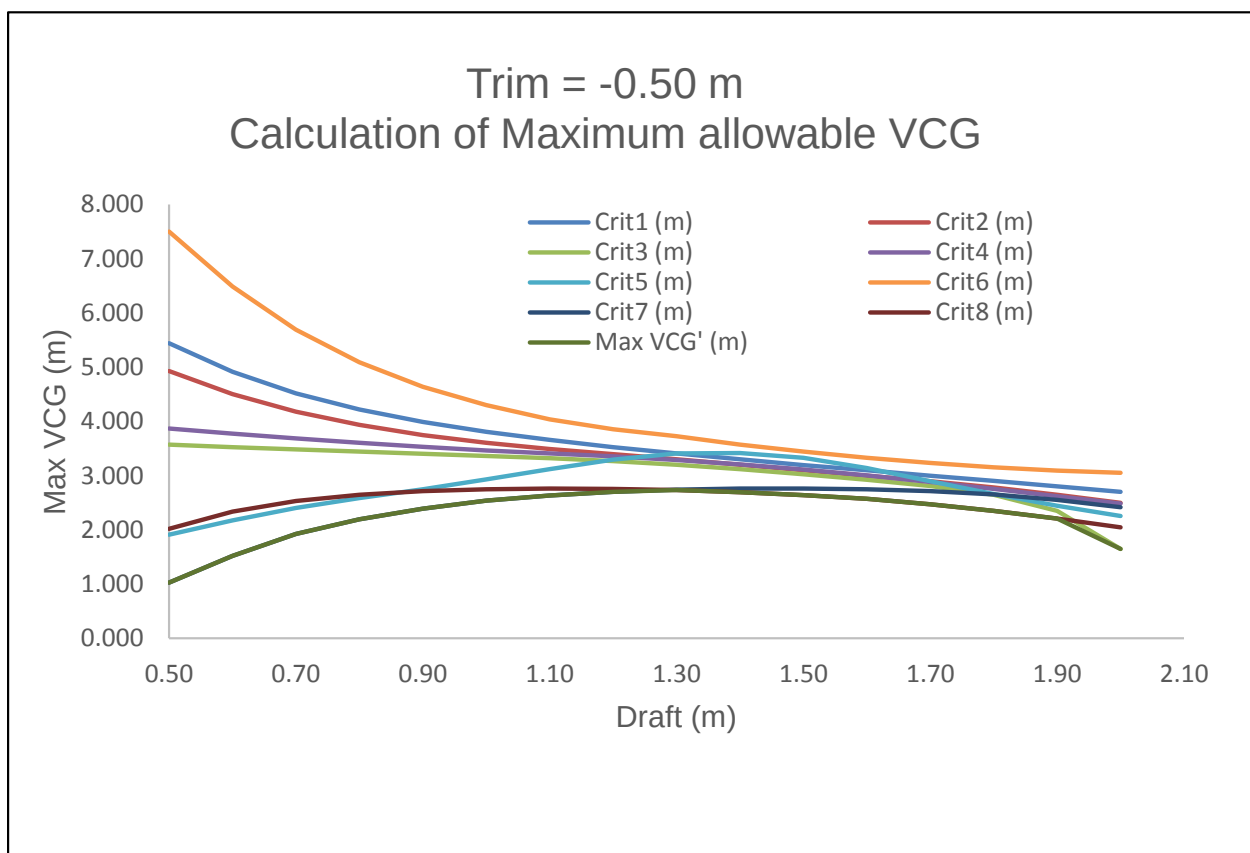
K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Initial trim: -0.750 (by stern)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
1.40	3.311	3.206	3.115	3.209	3.382	3.589	2.772	2.702	2.702	1.038
1.50	3.200	3.099	3.013	3.102	3.254	3.442	2.771	2.640	2.640	0.953
1.60	3.094	2.990	2.901	2.981	3.040	3.323	2.746	2.553	2.553	0.920
1.70	2.989	2.879	2.776	2.853	2.805	3.228	2.699	2.438	2.438	0.940
1.80	2.884	2.749	2.542	2.719	2.586	3.154	2.623	2.300	2.300	1.004
1.90	2.779	2.604	2.093	2.581	2.391	3.097	2.510	2.144	2.093	1.154
2.00	2.669	2.442	0.314	2.440	2.215	3.054	2.351	1.969	0.314	2.890



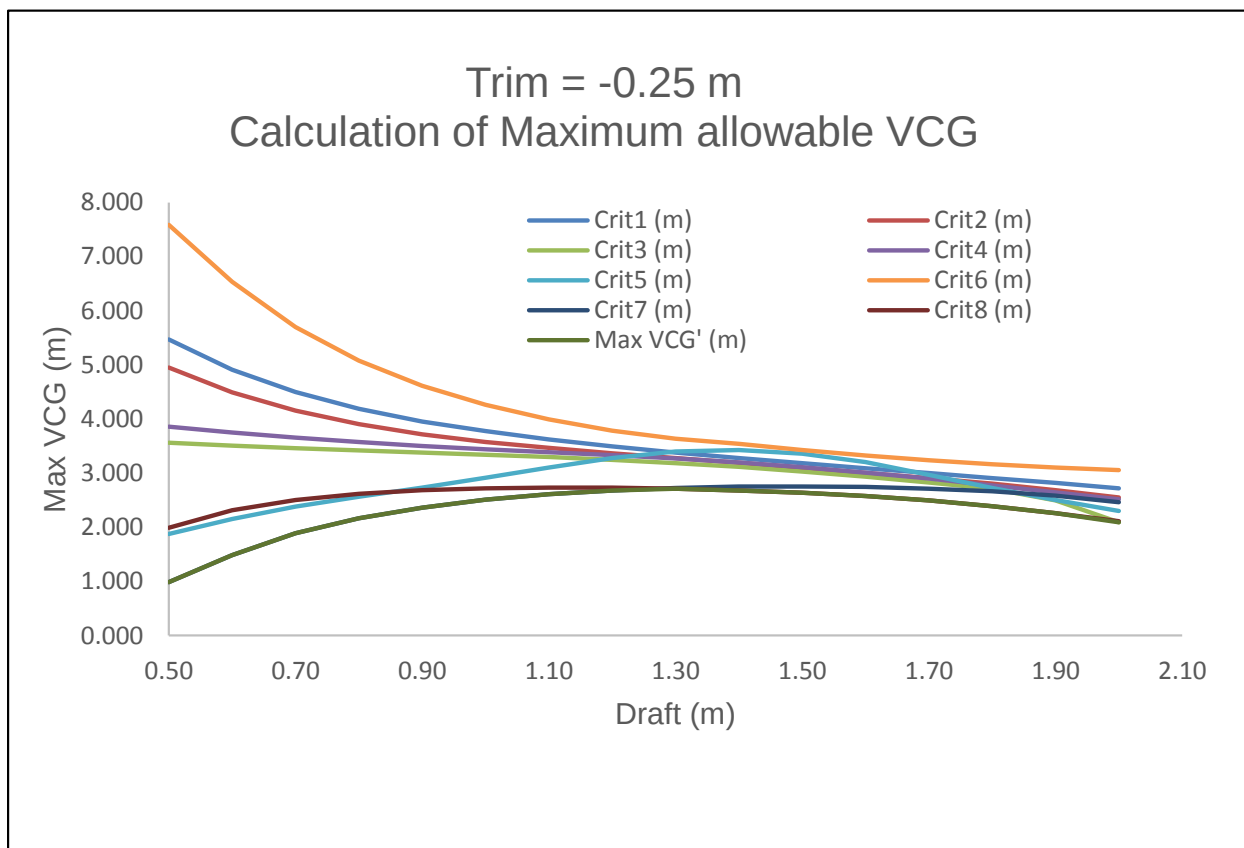
K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Initial trim: -0.500 (by stern)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
1.40	3.302	3.204	3.123	3.212	3.420	3.573	2.763	2.697	2.697	1.027
1.50	3.199	3.105	3.029	3.117	3.334	3.447	2.766	2.644	2.644	0.953
1.60	3.100	3.003	2.924	3.005	3.145	3.332	2.749	2.575	2.575	0.908
1.70	3.001	2.899	2.811	2.883	2.899	3.234	2.714	2.477	2.477	0.907
1.80	2.903	2.785	2.651	2.754	2.660	3.157	2.655	2.354	2.354	0.953
1.90	2.805	2.652	2.354	2.621	2.444	3.097	2.558	2.212	2.212	1.036
2.00	2.702	2.504	1.653	2.484	2.260	3.053	2.419	2.050	1.653	1.550



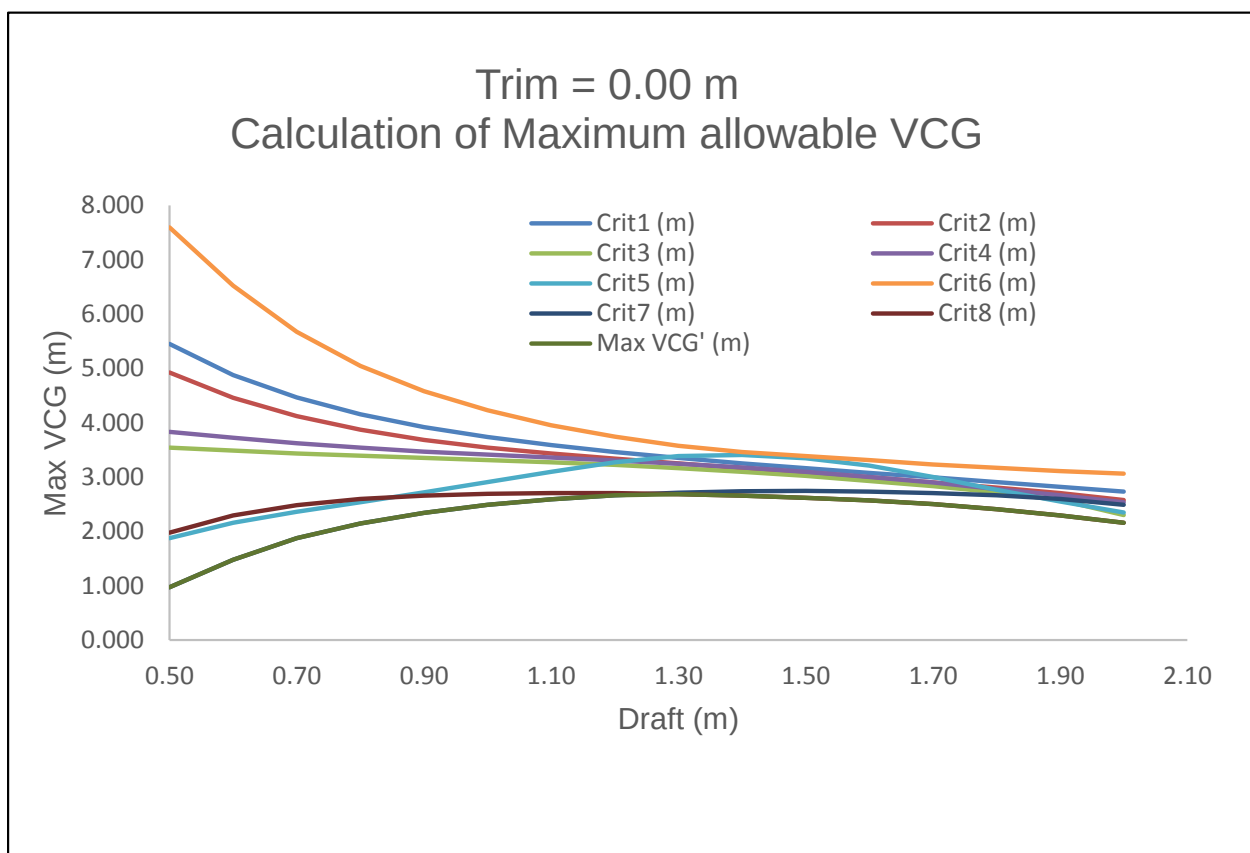
K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

Initial trim: -0.250 (by stern)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
1.40	3.282	3.190	3.117	3.201	3.427	3.544	2.751	2.682	2.682	1.013
1.50	3.187	3.099	3.031	3.114	3.358	3.429	2.755	2.637	2.637	0.942
1.60	3.094	3.005	2.934	3.013	3.204	3.328	2.745	2.580	2.580	0.899
1.70	3.004	2.907	2.828	2.899	2.971	3.242	2.716	2.499	2.499	0.894
1.80	2.912	2.806	2.713	2.776	2.724	3.163	2.668	2.391	2.391	0.922
1.90	2.820	2.686	2.496	2.649	2.502	3.102	2.586	2.262	2.262	0.990
2.00	2.723	2.551	2.091	2.516	2.304	3.056	2.465	2.114	2.091	1.115



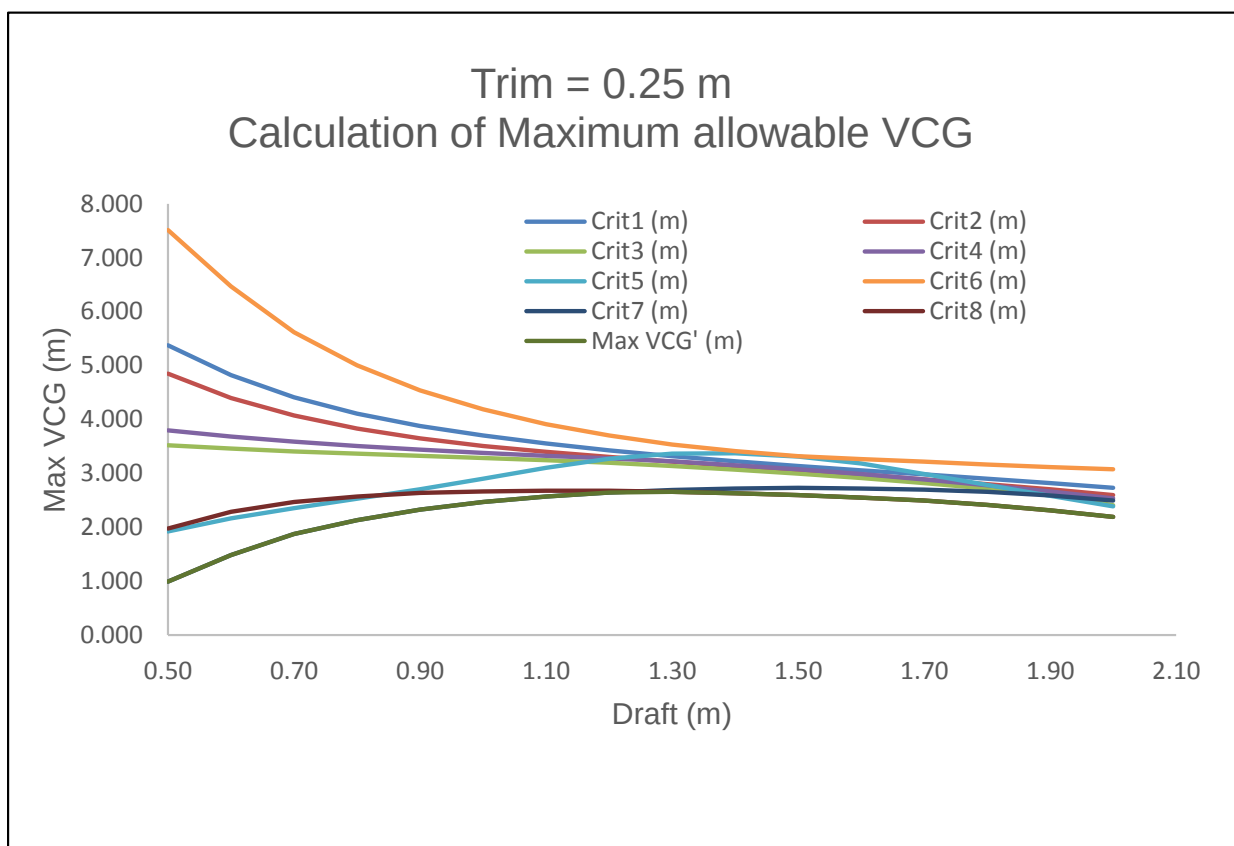
K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

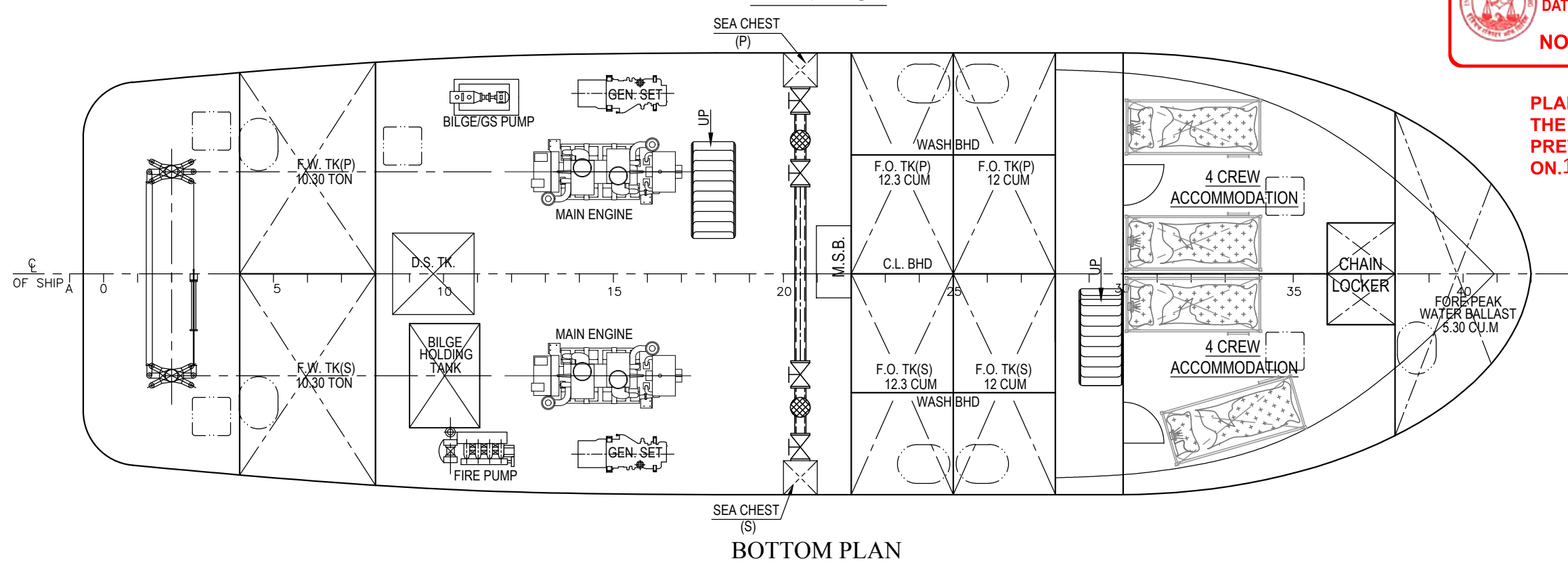
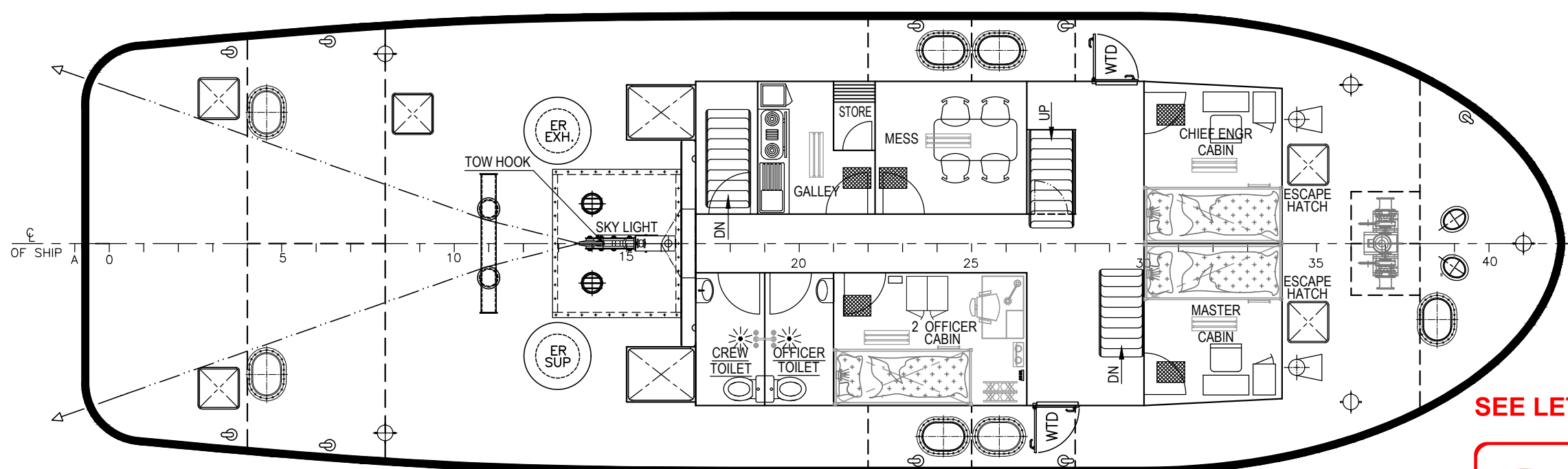
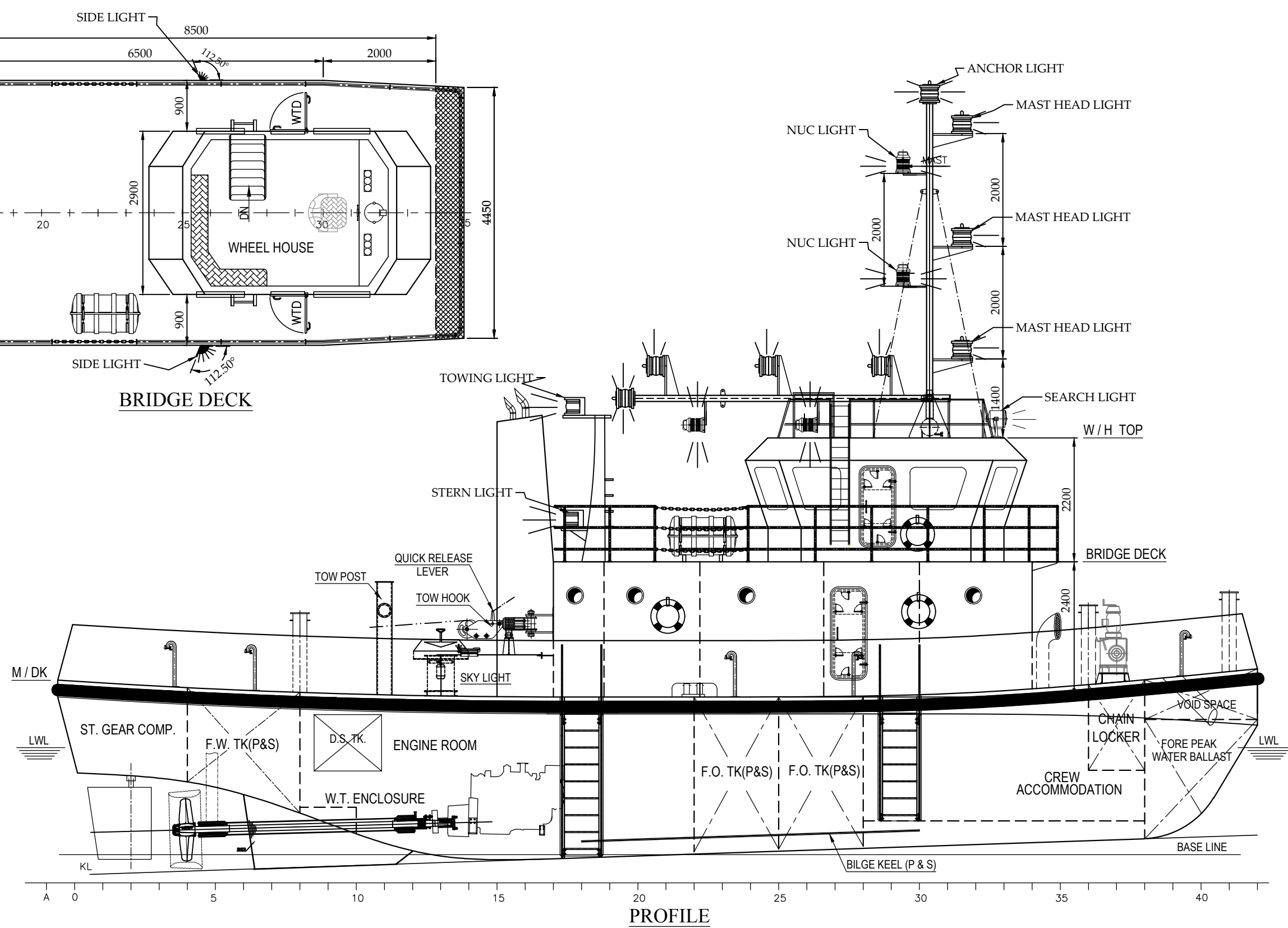
Initial trim: 0.00 m										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
1.40	3.253	3.166	3.099	3.181	3.407	3.460	2.739	2.659	2.659	0.952
1.50	3.165	3.082	3.021	3.100	3.347	3.387	2.744	2.620	2.620	0.917
1.60	3.080	2.995	2.932	3.006	3.214	3.312	2.735	2.571	2.571	0.891
1.70	2.996	2.904	2.832	2.899	3.003	3.235	2.710	2.505	2.505	0.880
1.80	2.911	2.810	2.723	2.784	2.769	3.171	2.667	2.412	2.412	0.910
1.90	2.824	2.705	2.576	2.663	2.553	3.111	2.597	2.296	2.296	0.965
2.00	2.733	2.581	2.300	2.535	2.351	3.063	2.492	2.162	2.162	1.051



K.B. SHIPPING & CO.	21.30M TUG BOAT	SBM-16-1011-708
5/23/2017	FINAL INTACT STABILITY BOOKLET	Rev No: 0

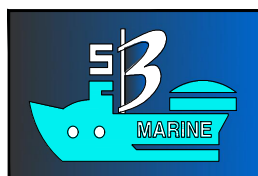
Initial trim: 0.250 (by stem)										
Draft (m)	Crit1 (m)	Crit2 (m)	Crit3 (m)	Crit4 (m)	Crit5 (m)	Crit6 (m)	Crit7 (m)	Crit8 (m)	Max VCG' (m)	Min GM' (m)
1.40	3.223	3.138	3.074	3.155	3.377	3.413	2.723	2.634	2.634	0.930
1.50	3.137	3.057	2.999	3.076	3.314	3.321	2.731	2.598	2.598	0.874
1.60	3.058	2.975	2.915	2.987	3.186	3.265	2.722	2.552	2.552	0.863
1.70	2.980	2.891	2.821	2.887	2.993	3.221	2.697	2.496	2.496	0.876
1.80	2.901	2.802	2.720	2.778	2.786	3.164	2.657	2.417	2.417	0.898
1.90	2.820	2.709	2.610	2.663	2.583	3.117	2.593	2.315	2.315	0.953
2.00	2.733	2.597	2.409	2.541	2.387	3.075	2.500	2.191	2.191	1.034

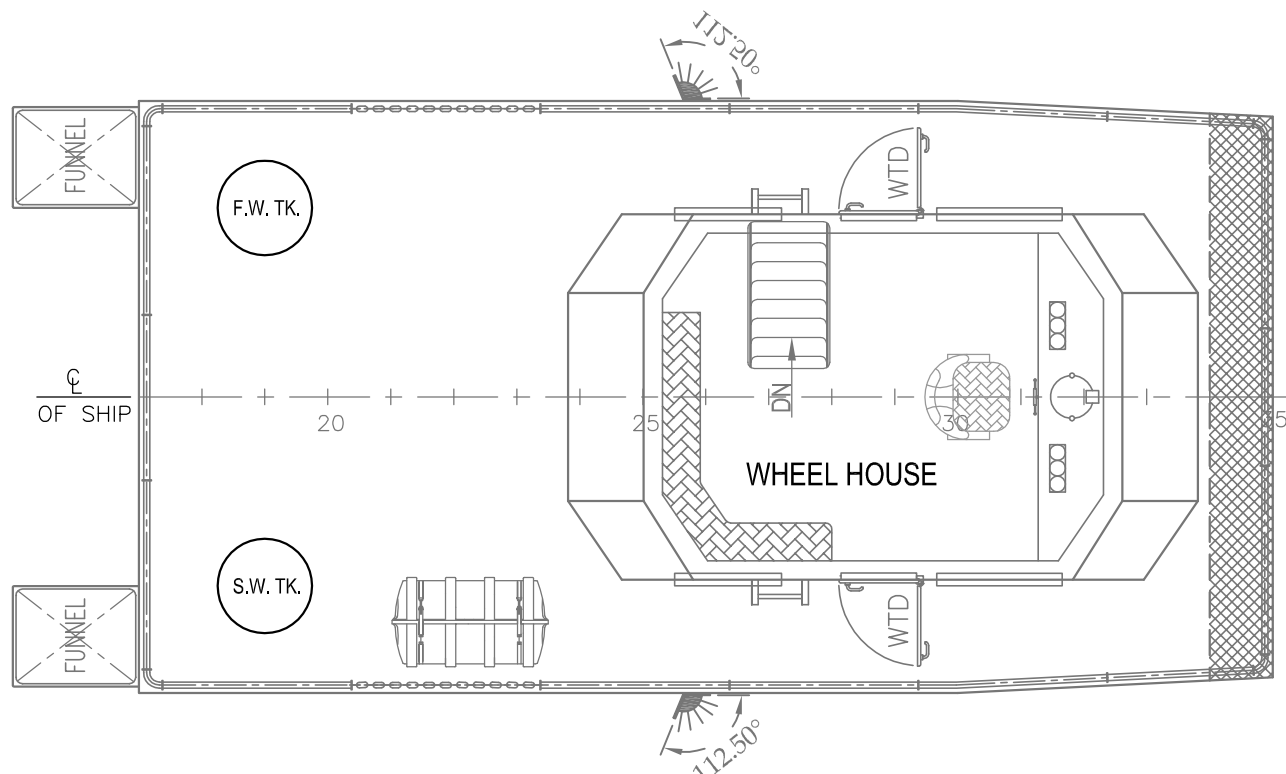
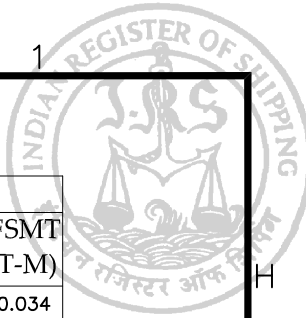




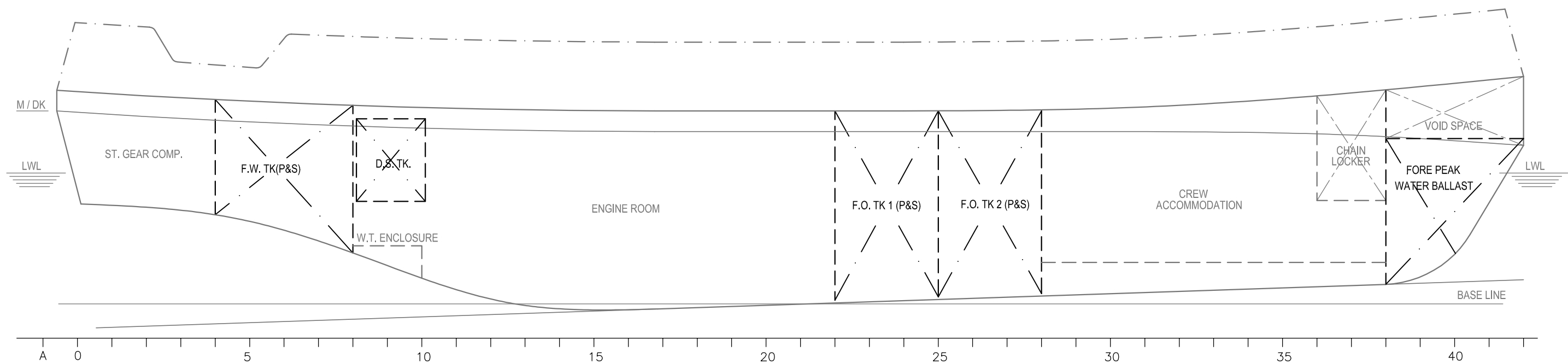
PRINCIPAL PARTICULARS

LENGTH OVERALL	:-	21.30M
LENGTH B.P.	:-	19.76 M
BREADTH	:-	06.50 M
DEPTH	:-	02.80 M
DRAFT(DESIGN)	:-	01.90 M
DRAFT(SCANTLING)	:-	02.00M
FRAME SPG.	:-	500 MM
BOLLARD PULL	:-	10 T
ENGINE POWER	:-	2 x 400 BHP
DESIGN SPEED	:-	10 KNOTS
COMPLEMENT	:-	12 PERSONS
CLASS NOTATION	:-	[ㄣ] SUL TUG [ㄣ] I Y INDIAN RIVER-SEA VESSEL TYPE - IV
CLASS	:-	INDIAN REGISTER OF SHIPPING
FUEL OIL	:-	48.60 CU. MTR
FRESH WATER	:-	20.60 T

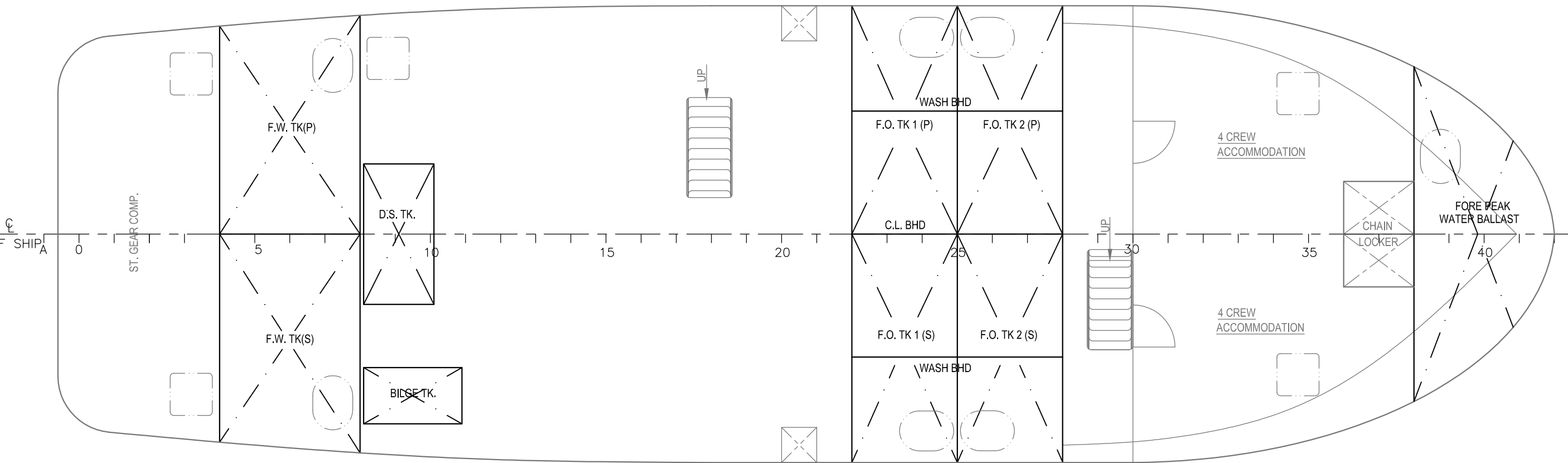
01	01.12.16	IRS COMMENT INCORPORATED										SAGAR	MS										
Rev.	Date	Description										Drawn	Review										
<u>Issued for</u> :														<u>Client Approval</u>									
Client		<u>KB SHIPPING & CO.</u>																					
Project		<u>21.30 M TUG BOAT</u>																					
Drawing Title		<u>GENERAL ARRANGEMENT PLAN</u>																					
Drawn	Checked	Review	Date	Scale	Paper	Rev #	Drawing No :-										Sheet						
PSS	VG	SC	20.07.16	1:75	A2	01	S	B	M	-	1	6	-	1	0	1	1	-	0	0	1	1 Of 1	
 S.B. MARINE CONSULTANT Tel :- 91 22 40244296 , Fax :- 25862318 E-mail : info@sb-marine.com Web : www.sbmarineconsultant.com							SHIPYARD : CHISTI MARINE ENGINEERING WORKS, SIKKA																
							YARD NO : 002																
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BRIDGE DECK



PROFILE



BOTTOM PLAN

SEA WATER TANK								
TANK NO.	ITEMS	SP. GR.	VOLUME (M) ³	WEIGHT (T)	VCG (M)	LCG (M)	TCG (M)	FSMT (T-M)
1	Sea Water Tank Top(S)	1.025	0.500	0.513	5.585	8.500	-1.500	0.034

WATER BALLAST TANK								
TANK NO.	ITEMS	SP. GR.	VOLUME (M) ³	WEIGHT (T)	VCG (M)	LCG (M)	TCG (M)	FSMT (T-M)
1	Fore Peak Ballast Tank	1.025	5.343	5.477	1.768	18.604	0.000	6.636

MISCELLANEOUS								
TANK NO.	ITEMS	SP. GR.	VOLUME (M) ³	WEIGHT (T)	VCG (M)	LCG (M)	TCG (M)	FSMT (T-M)
1	Bilge Tank	1.000	1.411	1.411	2.087	3.800	-2.300(SB)	0.063

FUEL OIL								
TANK NO.	ITEMS	SP. GR.	VOLUME (M) ³	WEIGHT (T)	VCG (M)	LCG (M)	TCG (M)	FSMT (T-M)
1	Fuel Oil Tank 1(P)	0.850	12.610	10.718	1.515	10.747	1.525 (PS)	3.573
2	Fuel Oil Tank 1(S)	0.850	12.610	10.718	1.515	10.747	-1.525 (SB)	3.573
3	Fuel oil Tank 2(P)	0.850	12.321	10.473	1.552	12.247	1.514 (PS)	3.569
4	Fuel Oil Tank 2(S)	0.850	12.321	10.473	1.552	12.247	-1.514 (SB)	3.569
5	Daily service Tank	0.850	2.352	1.999	2.087	3.550	0.000 (CL)	0.555

FRESH WATER								
TANK NO.	ITEMS	SP. GR.	VOLUME (M) ³	WEIGHT (T)	VCG (M)	LCG (M)	TCG (M)	FSMT (T-M)
1	FRESH WATER (P)	1.000	10.726	10.726	2.062	2.050	1.406 (PS)	4.534
2	FRESH WATER (S)	1.000	10.726	10.726	2.062	2.050	-1.406 (SB)	4.534
3	FRESH WATER TANK TOP (P)	1.000	0.500	0.500	5.585	8.500	1.500 (PS)	0.034

PRINCIPAL PARTICULARS

LENGTH OVERALL	:-	21.30M
LENGTH B.P.	:-	19.76 M
BREADTH	:-	06.50 M
DEPTH	:-	02.80 M
DRAFT(SCANTLING)	:-	02.00M
FRAME SPG.	:-	500 MM
BOLLARD PULL	:-	10 T
ENGINE POWER	:-	2 x 400 BHP
DESIGN SPEED	:-	10 KNOTS
COMPLEMENT	:-	12 PERSONS
CLASS NOTATION	:-	[5] SUL TUG [5] I Y INDIAN RIVER-SEA VESSEL TYPE - IV
CLASS	:-	INDIAN REGISTER OF SHIPPING

Note:
LCG is measured from Fr-2.
TCG is measured from CL.
VCG is measured from BL.
Transverse measurements towards port is +ve.
Oil Tanks used are to be filled 98 %.
Maximum FSM observed for a tank is depicted in capacity table.

1	23.05.17	AS FITTED	AB	SIVA
Rev.	Date	Description	Drawn	Review

Issued for	:	Approval
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Client	KB SHIPPING & CO.
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Project	21.30 M TUG BOAT
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Drawing Title	TANK CAPACITY PLAN
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Drawn	Checked	Review	Date	Scale	Paper	Drawing No :-														Sheet	
PSS	VG	SC	20.07.16	1:60	A2	S	B	M	-	1	6	-	1	0	1	1	-	0	0	2	1 Of 1



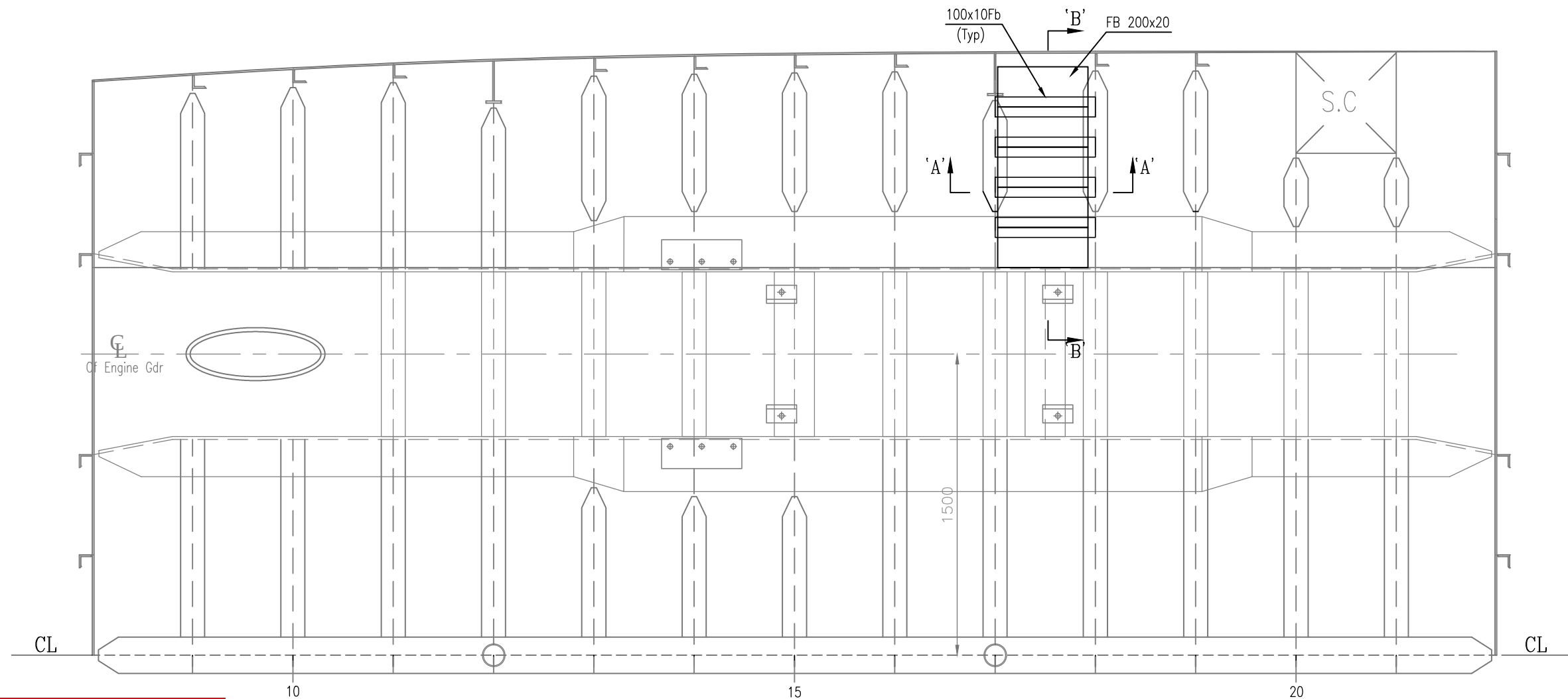
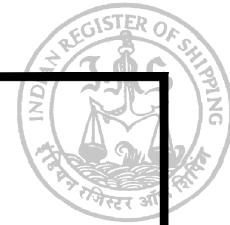
S.B. MARINE CONSULTANT

Tel :- 91 22 40244296 , Fax :- 25862318
E-mail : info@sb-marine.com
Web : www.sbmarsineconsultant.com

SHIPYARD :
CHISTI MARINE ENGINEERING WORKS, SIKKA

YARD NO :
002

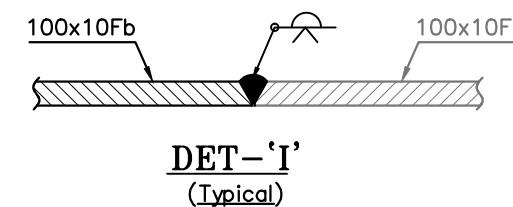
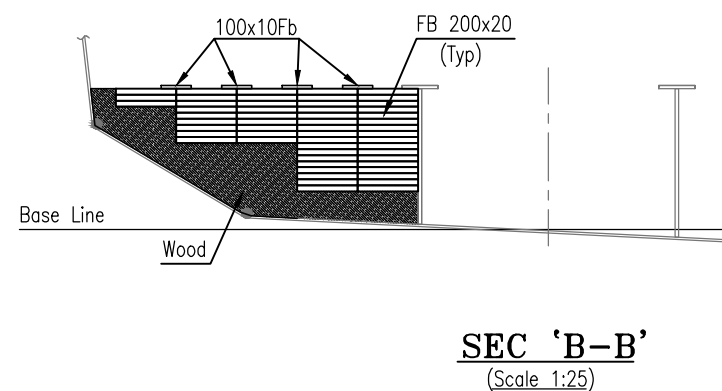
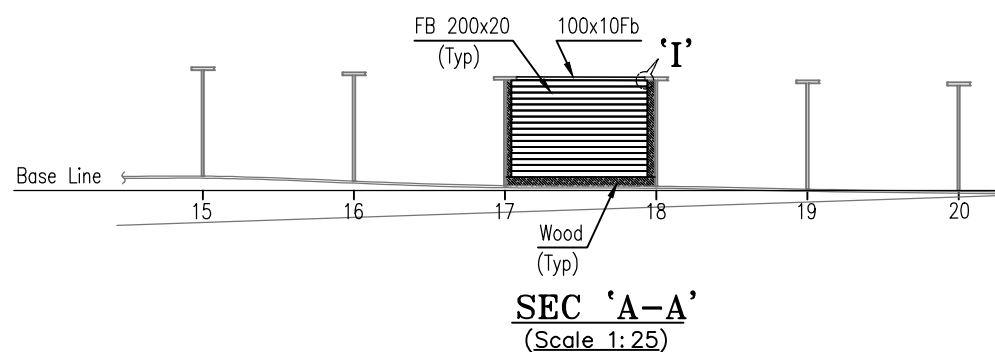
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BOTTOM PLAN
(Scale 1:25)



SEE LETTER E-49927-62739



FLAT BARS DIMENSION:

EACH BAR DIM=450 X 200 X 20 MM
NUMBER OF BARS = 55 NO'S
WEIGHT OF EACH BAR = 14.15 KG
TOTAL WEIGHT = 14.15 X 55 = 778 KG

Description	Weight(T)	LCG (m)	TCG (m)	VCG (m)
Extra Weight added for List Correction				
Flat bars(450x 200 x 20)	0.778	8.250	2.300	0.330
Wood at bottom	0.045	8.250	2.505	0.197
Wood at sides	0.006	8.250	2.300	0.330
Securing bars(100 x 10)	0.016	8.250	2.431	0.476
Total Extra weight added	0.845	8.250	2.313	0.326
Approved Lightship	107.740	9.276	-0.018	2.548



S.B. MARINE CONSULTANT

Tel :- 91 22 40244296 , Fax :- 25862318
E-mail :- info@sb-marine.com
Web :- www.sbmarineconsultant.com

Client :- **KB SHIPPING & CO.**

Project :- **21.30 M TUG BOAT**

Drawing Title :- **SOLID BALLAST ARRANGEMENT**

Issued for :- **Approval**

Drawn	Checked	Review	Date	Scale	Paper	Drawing No :-	Sheet
HARSHAD	SURESH	SIVA	10.04.17	1:25	A3	S B M - 1 6 - 1 0 1 1 - 1 3 2	1 Of 1

Rev.	Date	Description	Drawn	Review

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21.30 M TUG BOAT

INCLINING EXPERIMENT REPORT



SEE LETTER E-49483-61968

AS AMENDED ON PAGE 6, 7, 12 & 13.

THE FOLLOWING LIGHTSHIP PARTICULARS ARE APPROVED:
 LIGHTSHIP WEIGHT: 107.736 T
 LIGHTSHIP VCG: 2.548 M ABOVE BASELINE
 LIGHTSHIP LCG: 9.276 M FORWARD OF AP (FR. 2)
 LIGHTSHIP TCG: 0.018 M OFF CENTERLINE TOWARDS STARBOARD

0	30-Mar-17		SIVA	MNSS	SC
Rev	Date	Description	Prepared	Checked	Reviewed
Client:		Title:			
K.B. SHIPPING & CO.		INCLINING EXPERIMENT REPORT			
		21.30 M TUG BOAT			
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		30-Mar-17	Page 1	16-1011	
		Document:SBM-16-1011-706-0			Rev:0
		SB Marine, Marine and Offshore Structure Consultant Tel: +91 22 40244296 Web: www.sbmarineconsultant.com			

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	

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K.B. SHIPPING & CO.	21.30 M TUG BOAT	
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1 General Particulars

Name of the Vessel : 21.30m Tug
 Owner : KB SHIPPING
 Type : Tug Boat
 Flag : Indian
 IMO Number : 9823376
 Class : Indian Register of Shipping
 Yard Number : Chisti/002
 Engine : 2 x 400 BHP
 Class Notation : [SUL TUG, IY,
INDIAN RIVER SEA VESSEL TYPE 4

2 Principle Particulars

Length O.A. : 21.30 M
 Length B.P : 19.76 m
 Breadth (MLD) : 06.50 M.

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	

3 References

	Frame Number	Longitudinal Position (m)
Aft Perpendicular	2	0.000
Forward Perpendicular	#41 + 260mm	19.760
Mid Ship	#21 + 380 mm	9.880
Aft Draft Mark	8	3.000
Forward Draft Mark	38	18.000
Mid Draft Mark	22	10.000

- LCG, LCB and LCF are measured from '2' Frame, forward of '2' frame is positive and aft of '2' frame is negative.
- VCB and VCG are measured from Baseline, with above base line being positive.
- TCG is measured from centerline, with port being positive and starboard being negative.

4 Frame Spacing

Frame Spacing : 500 mm

5 Inclining Condition

Place of Experiment	:	Sikka
Date of Experiment	:	29-03-2017
Weather	:	Slight breeze
Wind	:	2-4 Knots
Water Condition	:	Calm
Specific Gravity of Water	:	1.023 T/m ³
Temperature of water	:	27.000 °C
Depth of the Water	:	4.00m
Condition of the vessel	:	completed

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
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6 Persons Witnessed

1. Owner rep	:	Mr. Amit garg	KB SHIPPING
		Mr. K.B.shailesh	KB SHIPPING
2. Class/Statutory	:	Mr. Vikas khare	IRS
3. Designer/Consultant	:	P. Siva	SB Marine

7 Details of Inclining Weights

Number of weights	:	120 No's each of 20 kg
Type of Weights	:	Iron Blocks
Mode of Shifting Operation	:	Manual Handling

Description	Weight	LCG	TCG	VCG
	(T)	(m)	(m)	(m)
W1	0.600	1.500	-2.000	3.360
W2	0.600	-0.500	2.000	3.460
W3	0.600	4.500	-2.000	3.250
W4	0.600	2.600	2.000	3.310

8 Details of Pendulum

Number of Pendulums : 2 No's

	Location	Length
	L. Pos. (m)	(m)
Pendulum (Aft)	3.500	4.050
Pendulum (Forward)	16.500	4.050

9 Density of water

Density of water = 1.023 T/m³

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	

10 Draft Mark Readings

	Port (m)	Starboard (m)	Mean (m)
Aft Draft Mark	1.620	1.620	1.620
Forward Draft Mark	1.220	1.220	1.220
Mid Draft Mark	1.440	1.440	1.440

11 Draft calculations

Location of Aft Draft Mark	3.000	m
Location of Forward Draft Mark	18.000	m
Location of Mid Draft Mark	10.000	m
Length Between Draft Marks	15.000	m
Location of Aft Perpendicular	0.000	m
Location of Forward Perpendicular	19.760	m
Location of Midship	9.880	m
Length between Perpendiculars	19.760	m
Trim between perpendiculars	-0.527	m
Heel	0.000	m

Note: Draft mark locations are measured from AP

	Port (m)	Starboard (m)	Mean (m)
A.P	1.700	1.700	1.700
F.P	1.173	1.173	1.173
Midship(From Fwd and aft Draft marks)	1.437	1.437	1.437

Moulded draft at midship	1.427 1.437	m	(AFTER KEEL THK. CORRECTION)
Trim @ LBP	-0.527	m	
Heel	0.000	m	

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	

12 Hydrostatics Properties for the Vessel in Inclined Condition

Volume of Displacement	112.29	m ³
Displacement	114.87	T
KMt	3.645	m
KMI	29.170	m
VCB	0.860	m
LCB	9.020	m

13 Sequence of Shifting

Shift No.	Position of Weight	
	Port	Stbd
0	W2 W4	W1 W3
1	W4	W1 W2 W3
2		W1 W2 W3 W4
3	W1	W2 W3 W4
4	W1 W3	W2 W4
5	W1 W2 W3	W4
6	W1 W2 W3 W4	
7	W2 W3 W4	W1
8	W2 W4	W1 W3

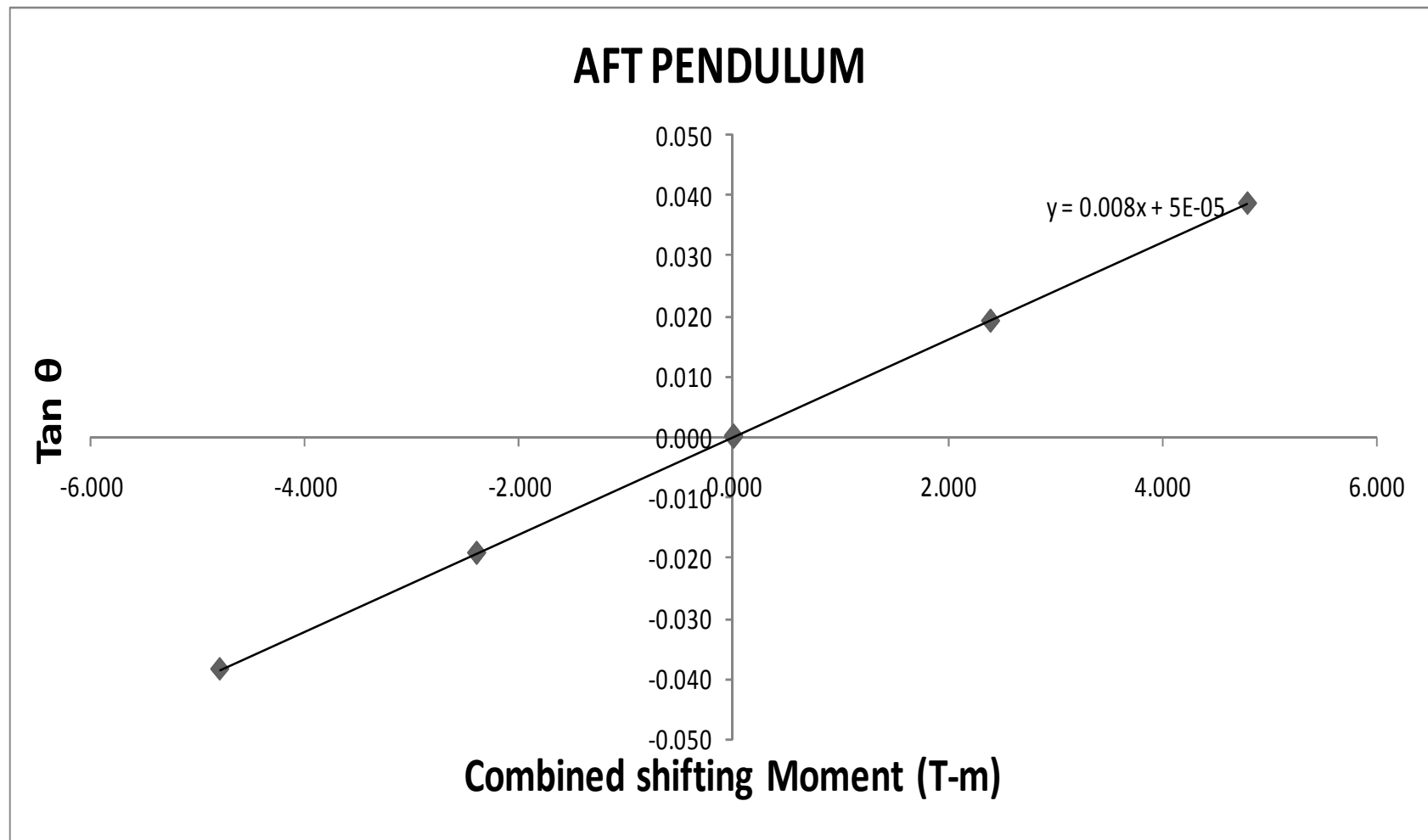
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	INCLINING EXPERIMENT REPORT	

AFT PENDULUM

Length of the Pendulum 4.050 m
Location of the Pendulum 3.500 m

Shift No.	Weight Shift	Net Weight moved (T)	Distance moved (m)	Deflection per Shift	Total Deflection (mm)	Tan θ -	Shifting Moment Ton-m	Combined Moment (Ton-m)	GM ₇ (m)
0	-	0.000						0.000	
1	W2 to STRB	0.600	4.000	78	78	0.019	2.400	2.400	1.085
2	W4 to STRB	0.600	4.000	79	157	0.039	2.400	4.800	1.078
3	W1 to PORT	0.600	-4.000	-79	78	0.019	-2.400	2.400	1.085
4	W3 to PORT	0.600	-4.000	-78	0	0.000	-2.400	0.000	-
5	W2 to PORT	0.600	-4.000	-78	-78	-0.019	-2.400	-2.400	1.085
6	W4 to PORT	0.600	-4.000	-78	-156	-0.039	-2.400	-4.800	1.085
7	W1 to STRB	0.600	4.000	78	-78	-0.019	2.400	-2.400	1.085
8	W3 to STRB	0.600	4.000	79	1	0.000	2.400	0.000	-

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	



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	INCLINING EXPERIMENT REPORT	

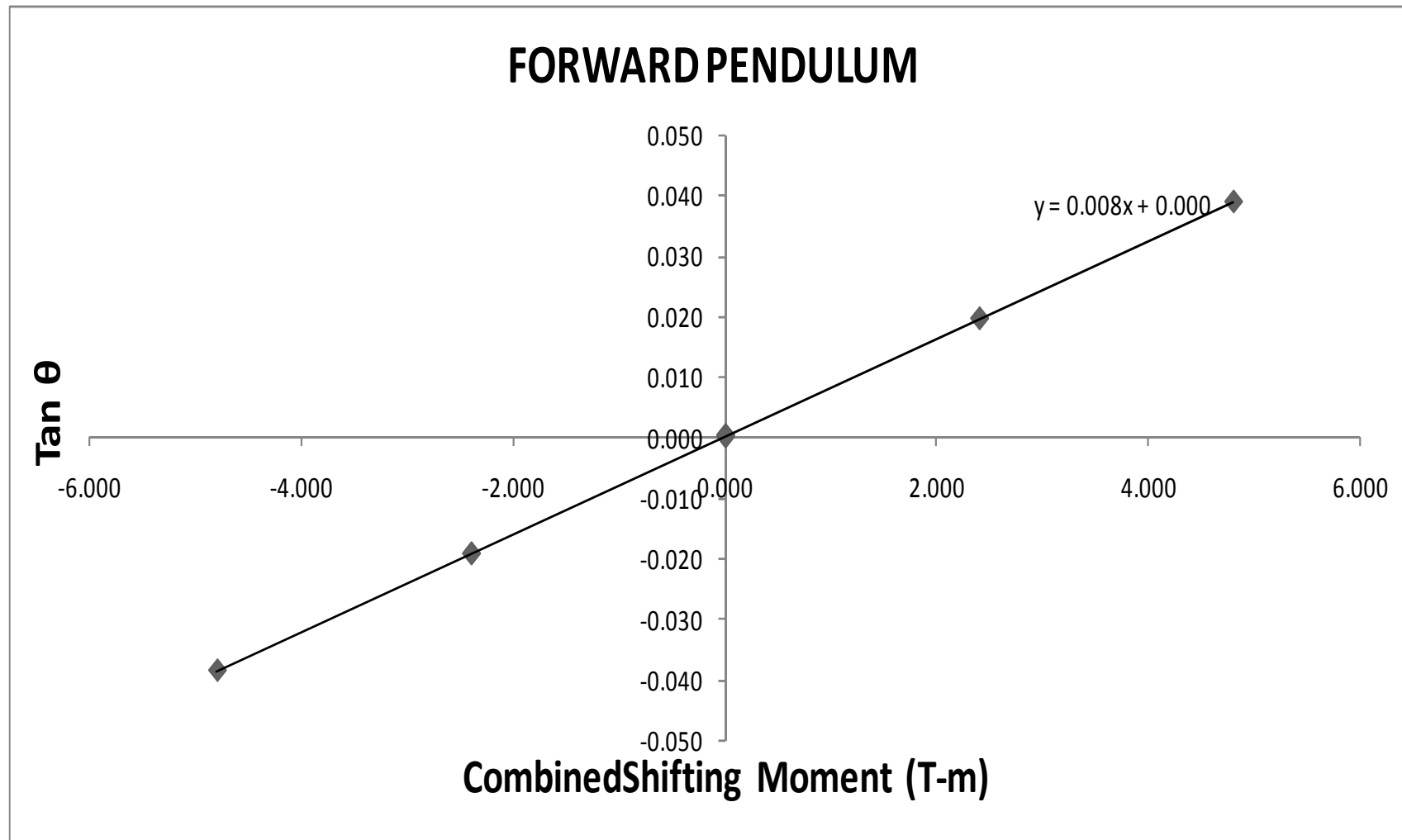
FORWARD PENDULUM

Length of the Pendulum 4.050 m

Location of the Pendulum 16.500 m

Shift No.	Weight Shift	Net Weight moved	Distance moved	Deflection per Shift	Total Deflection	Tan θ	Shifting Moment	Combined Moment	GM _T
		(T)	(m)		(mm)	-	Ton-m	(Ton-m)	(m)
0	-	0.000				0.000		0.000	
1	W2 to STRB	0.600	4.000	79	79	0.020	2.400	2.400	1.071
2	W4 to STRB	0.600	4.000	78	157	0.039	2.400	4.800	1.078
3	W1 to PORT	0.600	-4.000	-78	79	0.020	-2.400	2.400	1.071
4	W3 to PORT	0.600	-4.000	-78	1	0.000	-2.400	0.000	-
5	W2 to PORT	0.600	-4.000	-79	-78	-0.019	-2.400	-2.400	1.085
6	W4 to PORT	0.600	-4.000	-78	-156	-0.039	-2.400	-4.800	1.085
7	W1 to STRB	0.600	4.000	78	-78	-0.019	2.400	-2.400	1.085
8	W3 to STRB	0.600	4.000	79	1	0.000	2.400	0.000	-

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	



K.B. SHIPPING & CO.	21.30 M TUG BOAT	
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14 Calculation VCG of the vessel in Inclined Condition

GMt measured from Aft Pendulum Readings	1.084	m
GMt calculated from Forward Pendulum Readings	1.079	m
Mean GM of the vessel in Inclined Condition	1.081	m 1.103
KMt	3.645	m 3.69
VCG (KMt- GMt) of the vessel in inclined condition	2.564	m 2.586

15 Solid Weights to Go On board

Sl. No.	Description	Weight	LCG	TCG	VCG
		(T)	(m)	(m)	(m)
1		0.000	0.000	0.000	0.000
		0.000	0.000	0.000	0.000

16 Solid Weights to Go Off board

Sl. No.	Description	Weight	LCG	TCG	VCG
		(T)	(m)	(m)	(m)
1	Inclining Weight 1 (W1)	0.600	1.500	-2.000	3.360
2	Inclining Weight 2 (W2)	0.600	-0.500	2.000	3.460
3	Inclining Weight 3 (W3)	0.600	4.500	-2.000	3.250
4	Inclining Weight 4 (W4)	0.600	2.600	2.000	3.310
5	Pendulum + Oil Trough (Aft)	0.150	3.500	0.000	3.800
6	Pendulum + Oil Trough (Fwd)	0.150	16.500	0.000	3.981
7	Persons Onboard (10 No's)	0.825	4.000	0.000	3.950
8	counter weight for list correction	0.600	9.000	2.800	3.600
		4.125	4.015	0.407	3.543

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	

17 Liquid Weights to Go Off board

Sl. No.	Description	Sounding	Volume	Weight	LCG	TCG	VCG
		(mm)	(m3)	(T)	(m)	(m)	(m)
1	Fresh water Tank (P)	0	0	0.000	0.000	0.000	0.000
2	Fresh water Tank (S)	0	0	0.000	0.000	0.000	0.000
3	Daily service Tank	200	0.392	0.333	3.550	0.000	1.587
4	Fuel oil Tank 1(P)	0	0	0.000	0.000	0.000	0.000
5	Fuel oil Tank 1(S)	0	0	0.000	0.000	0.000	0.000
6	Fuel oil Tank 2(P)	0	0	0.000	0.000	0.000	0.000
7	Fuel oil Tank 2(S)	0	0	0.000	0.000	0.000	0.000
8	Fresh water Tank Top (P)	-	0.200	0.200	8.500	1.500	5.360
9	Sea water Tank Top (S)	0	0	0.000	0.000	0.000	0.000
10	Fore peak tank	0	0	0.000	0.000	0.000	0.000
				0.533	5.407	0.563	3.003

18 Weights to Relocate

There are no weights to Re-Locate.

19 Light Ship Calculation:

Sl. No.	Description	Weight	LCG	TCG	VCG
		(T)	(m)	(m)	(m)
1	Vessel in Inclined Condition	112.4 114.869	0.065 8.975	2.586 0.000	2.564
2	Solid Weights to Go On board	0.000	0.000	0.000	0.000
3	Solid Weights to Go Off board	-4.125	4.015	0.407	3.543
4	Liquid Weights to Go Off board	-0.533	5.407	0.563	3.003
5	Weights to Relocate	0.000	0.000	0.000	0.000
	Total	110.211 107.736	9.177 9.276	-0.018 -0.018	2.525 2.548

Lightship Weight **107.736** ~~110.211~~ T
 LCG **9.276** ~~9.177~~ m
 TCG **-0.018** m
 VCG **2.548** ~~2.525~~ m

K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	

20 Appendix

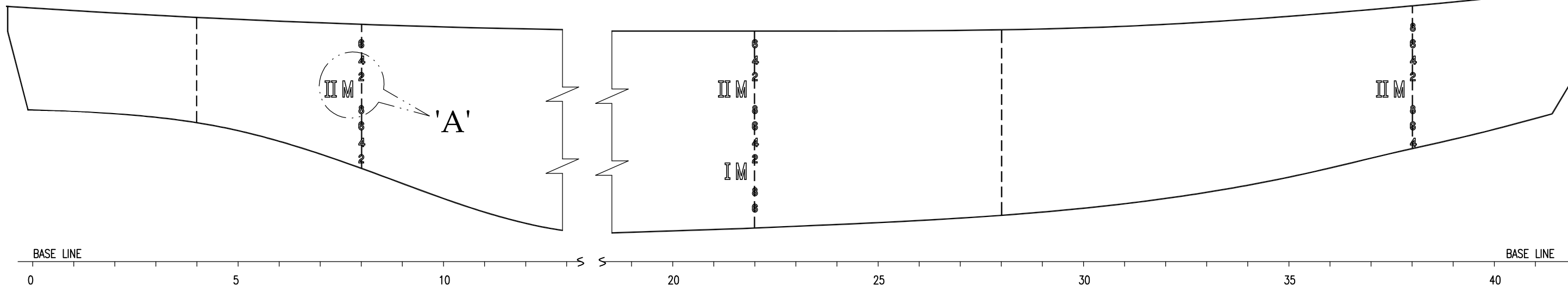
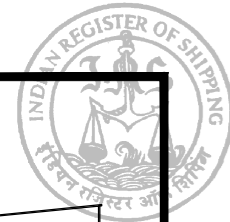
20.1 Trimmed Hydrostatics:

Water Density = 1.023 T/m³

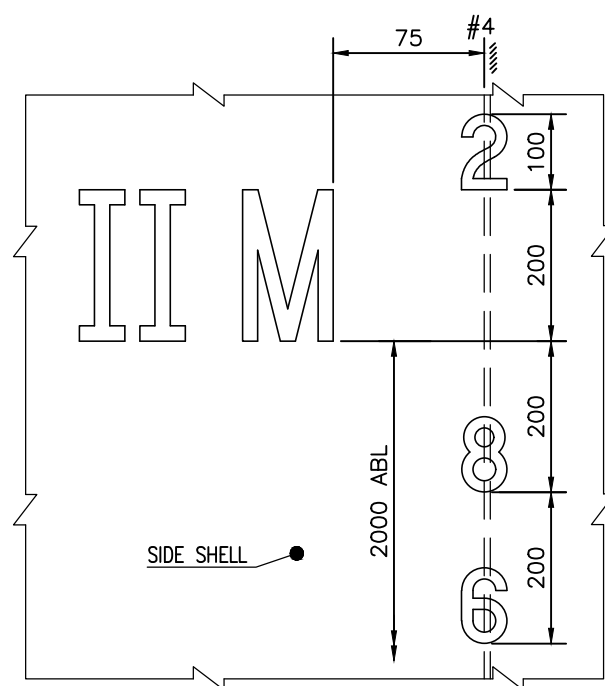
Trim is according to LBP = 19.76 m

Hydrostatics at Trim = -0.527 m

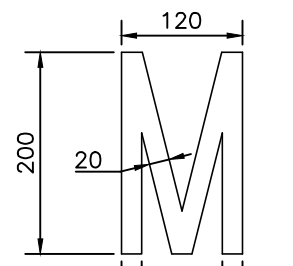
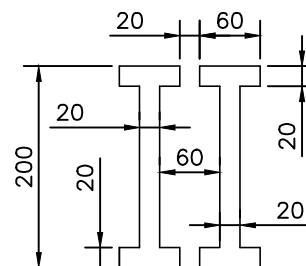
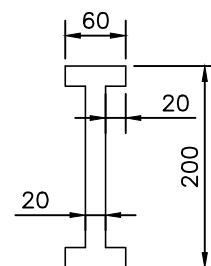
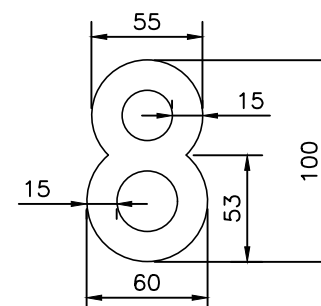
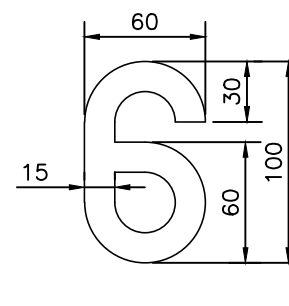
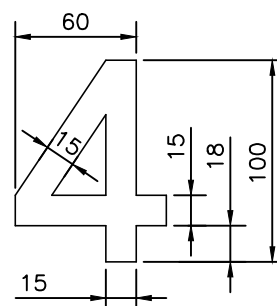
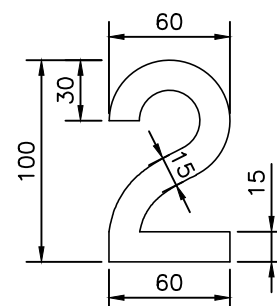
Trim: -0.527 (m)									
Draft (m)	Vol. mould (m ³)	Displ. (tonnes)	LCB (m)	VCB (m)	LCF (m)	S (m ²)	KMt (m)	KMI (m)	TpCm (t/cm)
1.407	108.956	111.462	9.033	0.842	8.599	149.61	3.683	29.66	1.133
1.417	110.070	112.601	9.028	0.848	8.597	150.15	3.670	29.49	1.135
1.427	111.173	113.730	9.024	0.854	8.595	150.69	3.657	29.33	1.137
1.437	112.286	114.868	9.020	0.860	8.594	151.23	3.645	29.17	1.139
1.447	113.411	116.019	9.014	0.866	8.592	151.77	3.632	29.00	1.141
1.457	114.526	117.160	9.012	0.872	8.591	152.31	3.620	28.84	1.143
1.467	115.635	118.294	9.009	0.878	8.589	152.85	3.608	28.69	1.145
1.477	116.751	119.437	9.005	0.884	8.588	153.38	3.596	28.53	1.146
1.487	117.879	120.590	9.001	0.890	8.587	153.91	3.585	28.37	1.148
1.497	118.997	121.734	8.997	0.896	8.586	154.45	3.573	28.22	1.150
1.507	120.120	122.883	8.993	0.902	8.585	154.98	3.562	28.06	1.152



PROFILE



DETAIL - 'A'
(SCALE 1:10)



LETTERING DETAIL
(SCALE : NTS)

NOTE:

- 1) ALL DRAFT MARKS ARE TO BE MEASURED FROM UNDERSIDE OF KEEL.
- 2) ALL MARKS TO BE CUT FROM 6 MM THK. PLATE & WELDED CONTINUOUSLY ON THE SHELL.
- 3) ALL DIMENSIONS ARE IN MM.
- 4) ALL MARKS TO BE AFFIXED ON BOTH PORT AND STARBOARD SIDES.
- 5) MARKS ARE TO BE PAINTED WITH TWO COATS OF PRIMER. TWO COATS OF ANTI CORROSIVE PAINTS AND TWO COATS OF WHITE FINISH.

MAIN PARTICULARS :

TYPE OF VESSEL	= TUG BOAT
LENGTH O.A.	= 21.30 M
LENGTH B.P.	= 19.76 M
BREADTH (MLD.)	= 6.50 M
DEPTH (MLD.)	= 2.80 M



S.B. MARINE CONSULTANT

Tel :- 91 22 40244296 , Fax :- 25862318
E-mail :- info@sb-marine.com
Web :- www.sbmarineconsultant.com

Client :- **KB SHIPPING & CO.**

Project :- **21.30 M TUG BOAT**

Drawing Title :- **DISPOSITION OF DRAFT MARK**

SHIPYARD :-

CHISTI MARINE ENGINEERING WORKS, SIKKA

Yard No :-

002

Issued for :- **Approval**

Drawn	Checked	Review	Date	Scale	Paper	Rev #	Drawing No :-	Sheet
Kunal	VG	SC	19.08.16	1:50	A3	0	S/BM-116-110111-11161	1 Of 1

Rev.	Date	Description	Drawn	Review

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K.B. SHIPPING & CO.	21.30 M TUG BOAT	
	INCLINING EXPERIMENT REPORT	

Inclining Experiment Prerequisites for Part-I and Part-II

Inclining Experiment Pre-requisites and Reporting Format
Part I

Pre requisites for Conducting Inclining Experiment
(To be provided /confirmed by Builder prior commencement of inclining experiment)

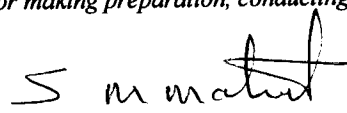
Name Of Builder/owner rep conducting trial	KIB SHIPPING
Hull Number/Vessel name	KB-4
Type of Vessel (Cargo/passenger)	Tug
Location of Vessel	Sikka
Date of Proposed inclining experiment /draft survey	29 th March 2017
Draft of Vessel and trim	1.44 m 5 0.4m trim by aft about draft mark
Depth of Water	4.00 m
Anticipated tide on date of inclining test	4.50 m
Weather prediction for proposed date	Calm
Availability of Approved inclining experiment procedure and resolving issues/comments mentioned in appraisal letter.	Yes/No
Availability of following reports from builder/Owner/Owners rep	Yes/No
Declaration of weights to go on from builder (not to exceed 2% of estimated light ship weight)	Yes
Declaration of weights to go off from builder - tank contents - Loose items (weights to go off not to exceed 4% of light ship weight excluding ballast used for trimming vessel)	Yes
Stability verification: (a) List not to exceed 0.5 deg (b) Min GM 0.20m (c) It will be assured that abrupt changes in the water-plane will be avoided as the ship heels in the intended draught and trim	Yes
Facilities/Arrangement for conduct of experiment	Yes/No
Calibration of weights used and its details	Yes
Availability of communication equipment	Yes



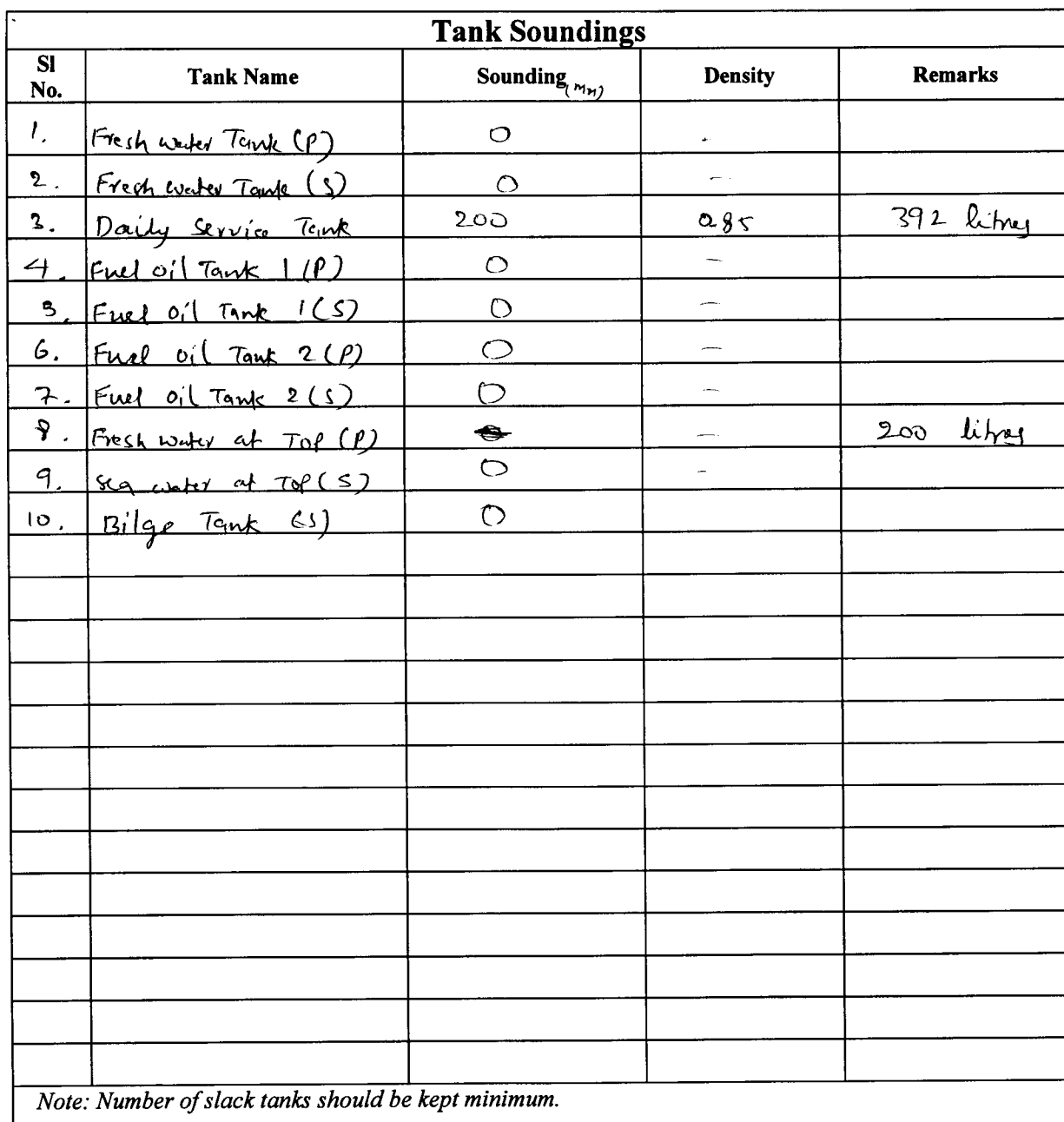
Availability of boat to take draft reading	Yes
Availability of draft mark plan/approved document like Prelim TSB/Final TSB for confirming location of draft marks	Yes
Confirmation of strength of deck for placing inclining weights	Yes
Whether bilges are pumped out	Yes
Safety Requirements- lighting, gas freeing, accessibility of areas requiring inspection as part of inclining experiment	Yes
Availability of sounding tapes, hydrometer etc	Yes
Securing of weights/objects which may move during inclining experiment	Yes
Closing/isolating cross connected tanks	Yes - No cross connected tanks
Availability of at least two pendulum, with sufficient length to give 75 mm deflection in single shift and 150mm in combined shift	Yes
Pendulum to located in protected location	Yes
Removal of gangway during draft reading/experiment	Yes
Permanent ballast used ? ; if any and its details	No
Arrangement for checking slackness of mooring ropes during exp(detailing personnel for the purpose)	Yes
Number of slack tanks kept to a minimum and draining of tanks that can have large free surface effect	Yes
Number of people not participating in the Test will be reduced to minimum	Yes
A trial is conducted to see whether the proposed weight give an inclination of the ship between 1 & 4 degree and and pendulum lengths give atleast 150 mm pendulum deflection for combined shift.	Yes/No

NOTE : The responsibility of the IRS Surveyor is to verify that the test is conducted according to accepted procedures and that all basic measurements and data are correctly taken and recorded. The responsibility for making preparation, conducting the inclining test and calculating the results rest with Builder / consultant

Confirm Readiness for test

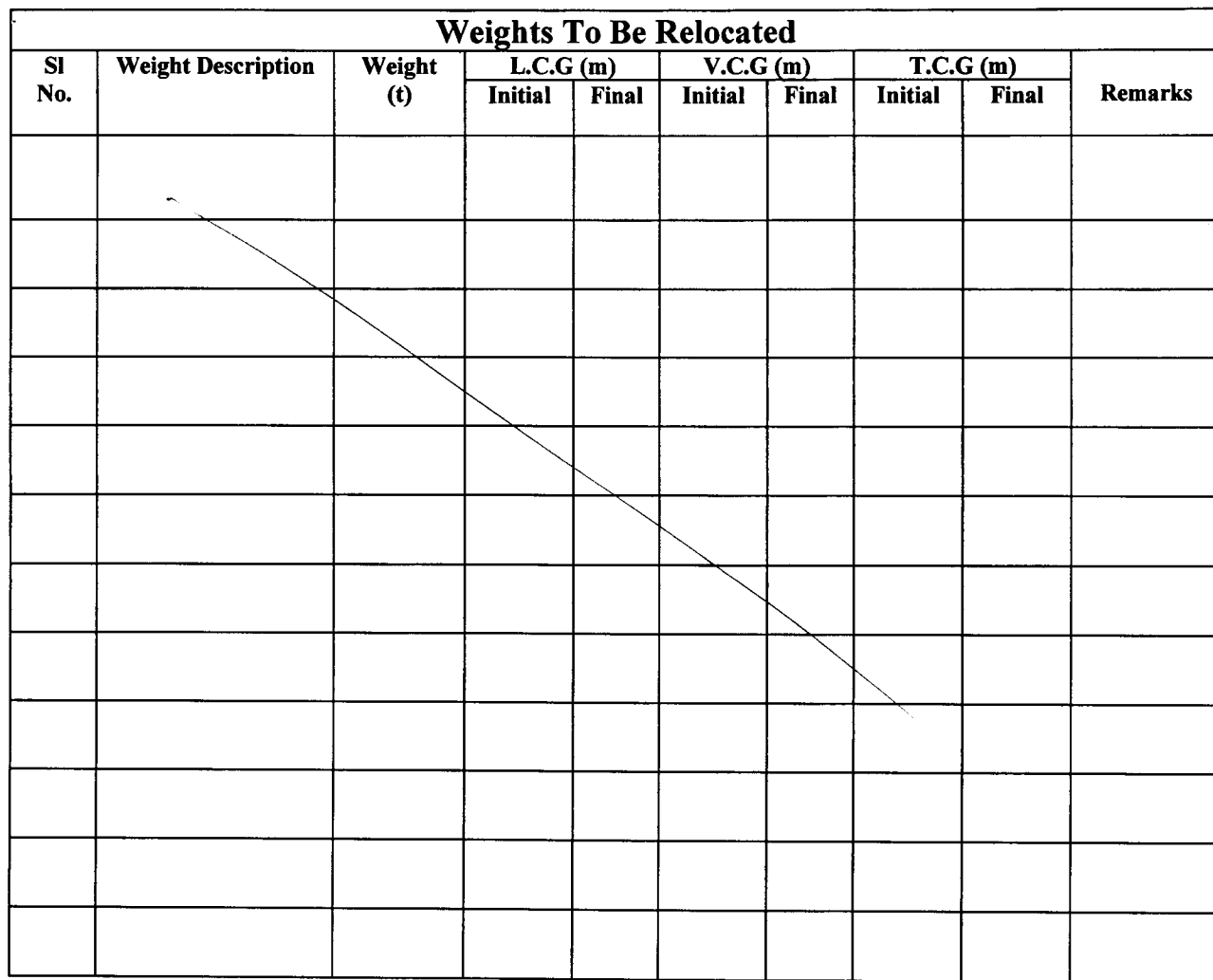

Name & Signature of Shipyard /Owners rep

[illegible]



Notes:

- *Preferably, all tanks should be either full or empty. The number of tanks containing liquids should be kept to a minimum.*
- *When a large number of slack tanks are present, the allowance of free surface correction in determining the lightship KG will be at the discretion of the head office.*
- *Ballast water tanks shall be either pressed full (100%) or completely empty (0%). Partial fillings in ballast water tanks are not acceptable. However in order to adjust initial heel/trim partial filling of any one ballast water tank is permitted however while calculating lightship KG, free surface correction of that ballast water tank will not be accepted.*
- *Soundings and density of liquids in tanks should be taken. Shapes of tanks, which are partly filled, are to be known in order to determine the free liquid surface effect.*
- *Adequate measures are to be taken to preclude air pockets in completely full tanks. All connections between tanks are to be closed and all empty tanks are to be adequately dried.*



TCG: Transverse center of Gravity from centerline (port is $^{+ve}$ & starboard is $^{-ve}$)

Inclining Experiment Pre-requisites and Reporting Format

Part II



REPORTING FORM FOR INCLINING EXPERIMENT

Attending Surveyor should witness inclining experiment only if inclining procedure is approved by H.O and Part-I submitted by builder is found satisfactory

General Particulars

Name of Vessel	KB-4
Builder/ Owner	KB SHIPPING & CO
Yard Number/ IR Number	CHIST/002 / 49848
Type of vessel Cargo/Passenger	TUG
Date(dd/mm/yyyy);Local Time of inspection/test	29 th march 2017; 3:00 PM
Wind / Weather condition	calm
Location of Test	Ch SIKKA
Depth of water where vessel is moored, m	4.00 m
Method of reading Draft(By Boat-Y/N)	Yes

Length Between the perpendiculars (m)	19.76
Breadth (m)	6.50
Depth (m)	2.80

Location of Draft Marks	Frame Number	Draft Reading (Port side),m	Draft Reading (STBD Side),m
Aft	8	1.62	1.62
Midship	22	1.44	1.44
Forward	38	1.22	1.22

Note : Recommended to attach Photographs of draft.

Inclining Weights

Type of weights used	Iron blocks			
Number of weights Used	120 No's each of 20 kg			
	W1	W2	W3	W4
Inc. weight (Tons)	0.60	0.60	0.60	0.60
Position(w.r.t frame) ^{to Fr. 2}	1.50	-0.50	4.50	2.60
Distance of shift (d_i)	4.00	4.00	4.00	4.00
Dimensions (l x b x d)	0.3 x 0.3 x 0.80	0.3 x 0.3 x 0.8	0.3 x 0.3 x 0.8	0.3 x 0.3 x 0.80

Note: l, b, d represents the dimensions of the inclining weights along longitudinal, transverse and vertical directions. d_i represents the distance of shift i.e., the distance between the centers of respective inclining weights before and after shifting (See Fig. 3).

combined

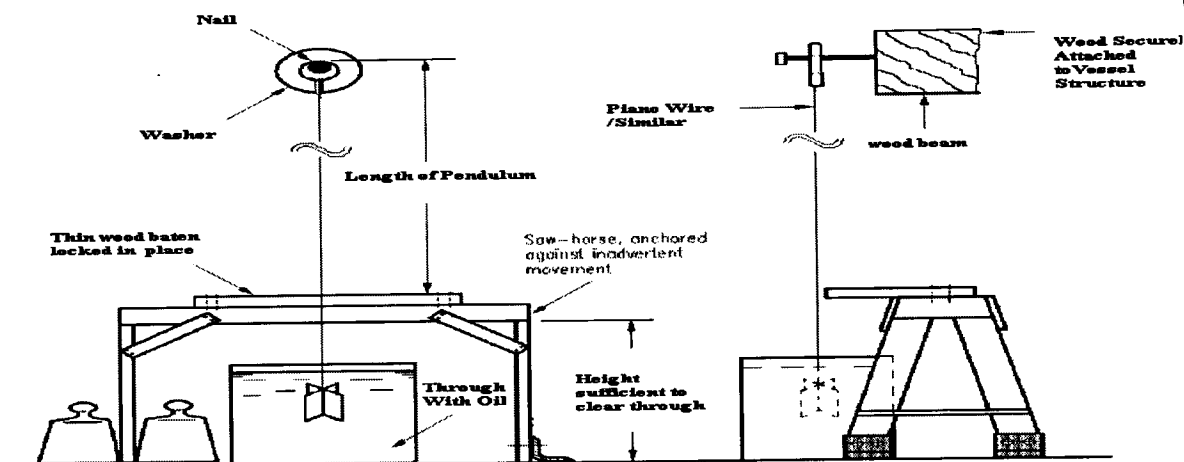


Fig 1: Recommended set-up of Pendulum

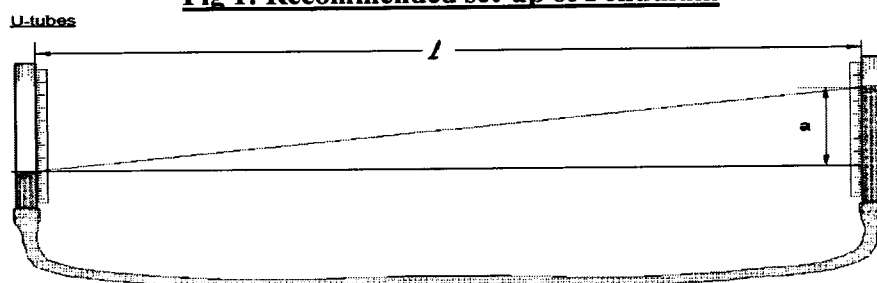


Fig 2: Use of U-Tube

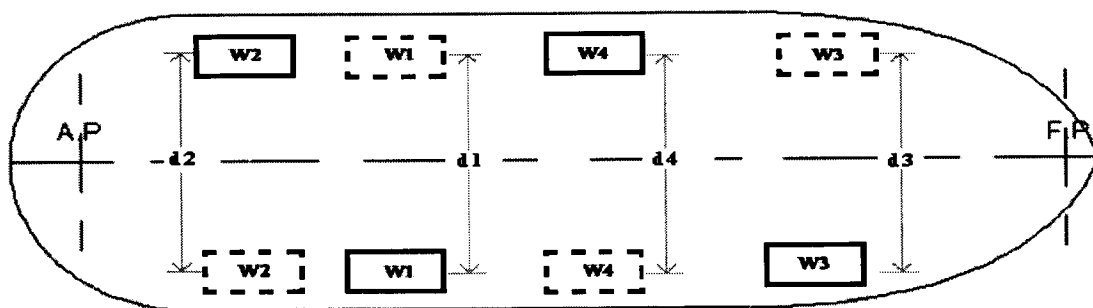


Fig.3 – Initial Arrangement of Inclining Weights

Shift No:	Port Side	Starboard Side
0	W2,W4	W1,W3
1	W4	W1,W2,W3
2		W1,W2,W3,W4
3	W1	W2,W3,W4
4	W1,W3	W2,W4
5	W1,W2,W3	W4
6	W1,W2,W3,W4	
7	W2,W3,W4	W1
8	W2,W4	W1,W3

Table.1 – Sequence of Inclining Weight Shift

Note : All the four Inclining weights should be kept at different longitudinal positions along the length of the ship. They should not come on top of each other while shifting.

Details of Pendulum

No. of Pendulums used		
	Location (w.r.t frame)	Length (m)
Aft	3.50 m from Fr. 2	4.05
Forward	16.50 m from Fr. 2	4.05
Midship		
Note: Minimum 2 number of pendulums should be used, wherever impracticable one pendulum can be replaced by U-tube. (See Fig 1&2)		

Initial Arrangement and Sequence of Weight Shift

Verification of Initial Arrangement and Sequence followed (Refer Fig.3 & Table.1)	Yes/No
--	--------

Pendulum deflection readings

Shift No:	Forward (mm)	Aft (mm)	Mid
0	0	0	
1	79	78	
2	78	79	
3	-78	-79	
4	-78	-78	
5	-79	-78	
6	-78	-78	
7	78	78	
8	79	79	

Density of Water

Location	Forward (t/m3)	Aft(t/m3)	Mid(t/m3)
Port	1.023	1.023	1.023
Starboard	1.023	1.023	1.023

Verification of weights to go on / go off	SEE PART 1
Note: See Part 1 (the sheet for recording the above data) The Surveyor should examine the condition of the ship to confirm that the reported data on weights to go on/go off are generally in order. However, it is not necessary to verify the accuracy of the weights to go on/go off in detail.	

Verification of the consistency of the results by plotting heeling moments Vs. Tangent of heel angle	SEE ATTACHMENT 2
Note: During the test recording of each deflection corresponding to steps 1-8, the Surveyor shall monitor the plot of heeling moment Vs Tangent of heel angle to verify the consistency of the result. (See attachment 2 for sample calculation) The consistency check has to be monitored at least at one location of pendulum and record attached. The preparation of the graph is the responsibility of the builder /consultant, however monitoring authority is the attending surveyor. Conclusion of Inclining Experiment is subject to satisfactory verification of the plot by the attending surveyor. If any readings are found not following the trendline, those steps may be repeated until satisfactory plot is obtained.	

Comments on Inclining Test

See Attachment

- Attachment 1: Calculation sheet
- Attachment 2: Graph to Plot the heeling moment Vs Tangent of heel angle (Separate plot by shipbuilder/consultant containing similar data would also be acceptable)

Attending surveyor to forward these forms Part 1 and Part 2 together with completed attachments 1 and 2 to HO.




Name & designation of Attending Surveyor: VIRAS ICHARE (SURVEYOR-1)

29/03/2017

Attachment 1

Calculation Sheet

Pendulum <u>APL</u>										
Shift No.	Shift Sequence		Weight	Distance Shifted	Single Moment	Combined Moment	Pendulum Deflection Per Shift	Total Deflection	Pendulum Length	Total Tan @
	Port	Starboard	(T) w	(m) d	(T-m) w*d	(T-m)	(mm) 	De (mm)	Le (mm)	De/Le
0	w2,w4	w1,w3	0							
1	w4	w1,w2,w4	0.6	4.00	2.40	2.40	78	78	4050	0.019
2	0	w1,w2,w3,w4	0.60	4.00	2.40	4.80	79	157	4050	0.039
3	w1	w2,w3,w4	0.60	4.00	-2.40	2.40	-79	78	4050	0.019
4	w1,w3	w2,w4	0.60	4.00	-2.40	0.00	-78	0	4050	0.00
5	w1,w2,w3	w4	0.60	4.00	-2.40	-2.40	-78	-78	4050	-0.019
6	w1,w2,w3,w4	0	0.60	4.00	-2.40	-4.80	-78	-156	4050	-0.039
7	w2,w3,w4	w1	0.60	4.00	2.40	-2.40	78	-78	4050	-0.019
8	w2,w4	w1,w3	0.60	4.00	2.40	0.00	79	1	4050	0.00

\$ - Minimum combined deflection should be at least 150 mm & Combine angle of heel due to inclining should be between 1degree to 4 degree.

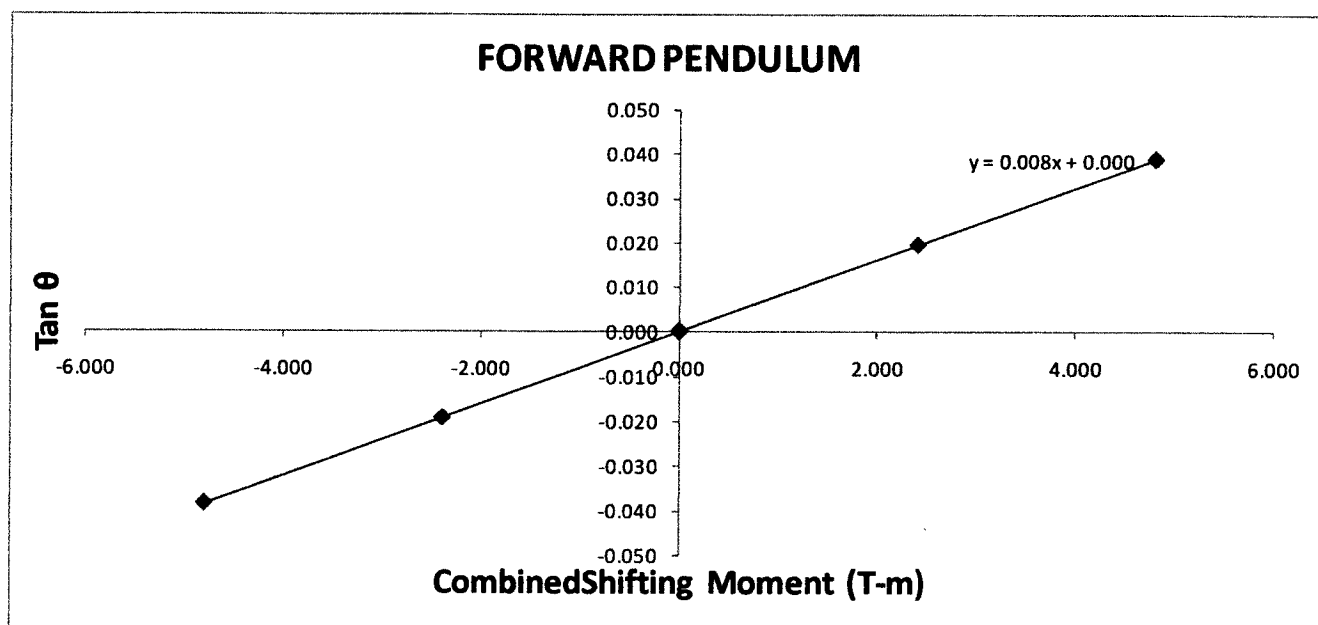
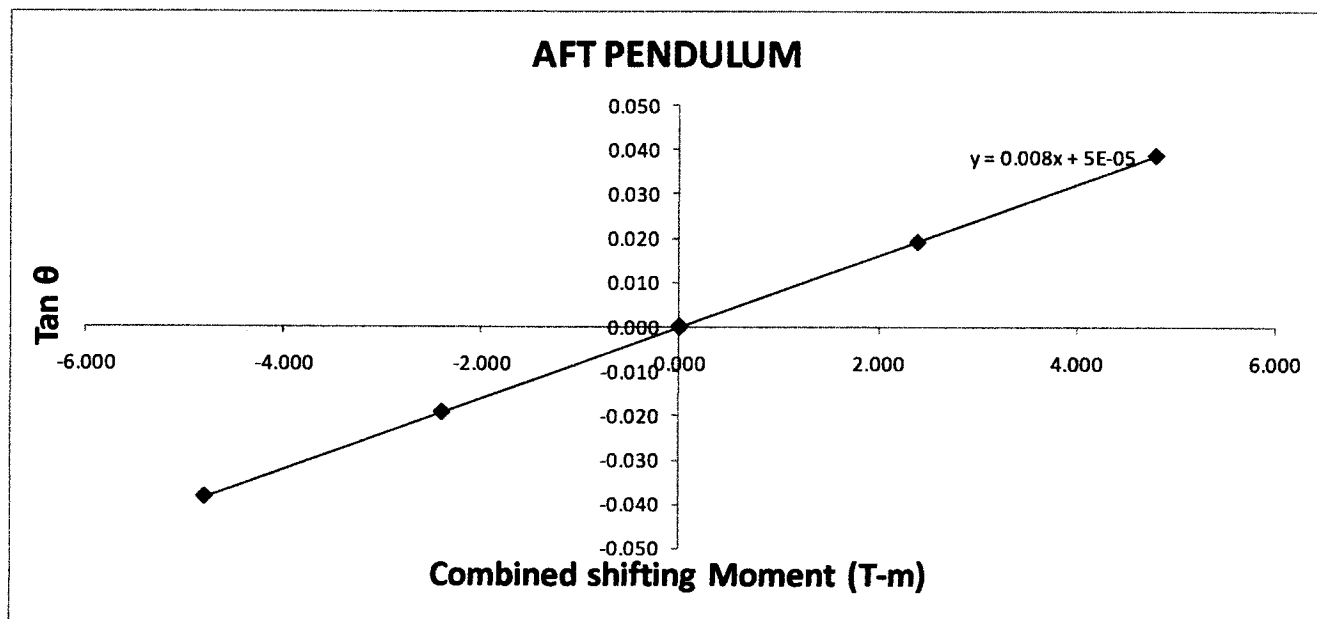
Attachment 1

Calculation Sheet

Pendulum Forward

Shift No.	Shift Sequence		Weight	Distance Shifted	Single Moment	Combined Moment	Pendulum Deflection Per Shift	Total Deflection	Pendulum Length	Total Tan @
			(T)	(m)	(T-m)	(T-m)	(mm)	(mm)	(mm)	
	Port	Starboard	w	d	w*d			De	Le	De/Le
0	w2,w4	w1,w3								
1	w4	w1,w2,w4	0.60	4.00	2.40	2.40	79	79	4050	0.02
2	0	w1,w2,w3,w4	0.60	4.00	2.40	4.80	78	157	4050	0.039
3	w1	w2,w3,w4	0.60	-4.00	-2.40	2.40	-79	79	4050	0.02
4	w1,w3	w2,w4	0.60	-4.00	-2.40	0.00	-78	1	4050	0.00
5	w1,w2,w3	w4	0.60	-4.00	-2.40	-2.40	-79	-78	4050	-0.019
6	w1,w2,w3,w4	0	0.60	-4.00	-2.40	-4.80	-78	-156	4050	-0.039
7	w2,w3,w4	w1	0.60	4.00	2.40	-2.40	78	-78	4050	-0.019
8	w2,w4	w1,w3	0.60	4.00	2.40	0.00	79	1	4050	0.00

\$ - Minimum combined deflection should be at least 150 mm & Combine angle of heel due to inclining should be between 1degree to 4 degree.



Sample calculation & plot of Heeling moment Vs Tangent of heel angle.

Pendulum Forward										
Shift No.	Shift Sequence		Weight	Distance Shifted	Single Moment	Combined Moment	Pendulum Deflection Per Shift	Total Deflection	Pendulum Length	Total Tan @
	Port	Starboard	w	d	w*d			De	Le	De/Le
0	w2,w4	w1,w3	0	0	0	0	0	0	3750	0
1	w4	w1,w2,w4	1.5	5.3	7.95	7.95	77	77	3750	0.021
2	0	w1,w2,w3,w4	1.5	5.3	7.95	15.9	78	155 ^s	3750	0.041
3	w1	w2,w3,w4	1.5	5.3	7.95	7.95	77	78	3750	0.021
4	w1,w3	w2,w4	1.5	5.3	7.95	0	79	-1	3750	0
5	w1,w2,w3	w4	1.5	5.3	7.95	-7.95	80	-81	3750	-0.022
6	w1,w2,w3,w4	0	1.5	5.3	7.95	-15.9	78	-159 ^s	3750	-0.042
7	w2,w3,w4	w1	1.5	5.3	7.95	-7.95	79	-80	3750	-0.021
8	w2,w4	w1,w3	1.5	5.3	7.95	0	79	-1	3750	0

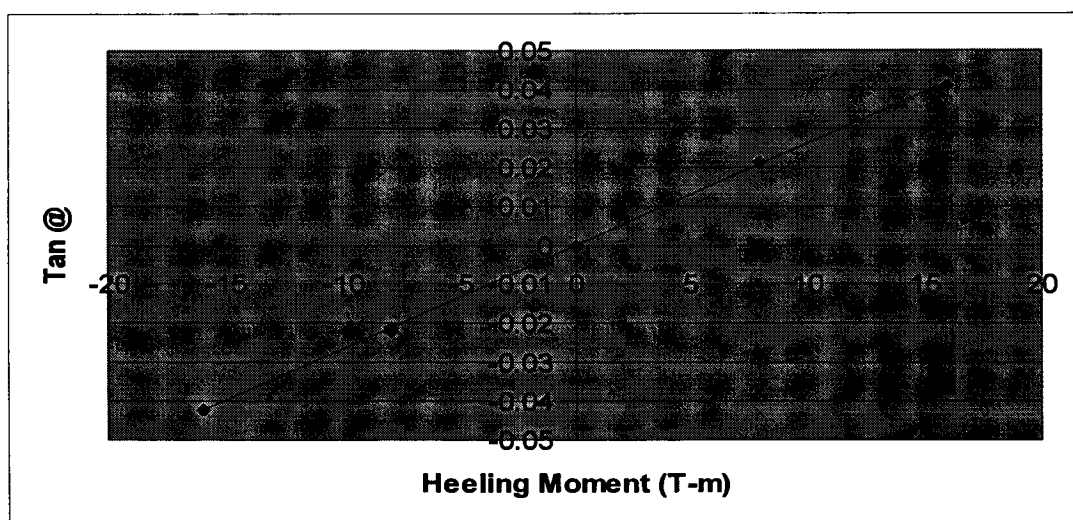


Fig .4. Plot of Heeling moment Vs Tangent of heel angle

Check:

- Whether a reasonable straight line plot is obtained / not (See Fig.4). If there is any discrepancies in the plot (See Fig 5.1-5.4) .
- \$ - Minimum combined deflection should be at least 150 mm & Combine angle of heel due to inclining should be between 1degree to 4 degree.

Note:

- Single Moment is the product of Inclining weight and the distance moved for the individual shifts
- Combined moment is the total residual moment from " 0 " Position to Port or Starboard Side for the shift
- Pendulum Deflection is the deflection of the pendulum for a single shift from the equilibrium position of previous shift.
- Total Deflection is the total residual deflection from "0" position for the shift.
- **Sign Convention :** (+) Starboard and (-) Port

Corrective measures for inclining experiment

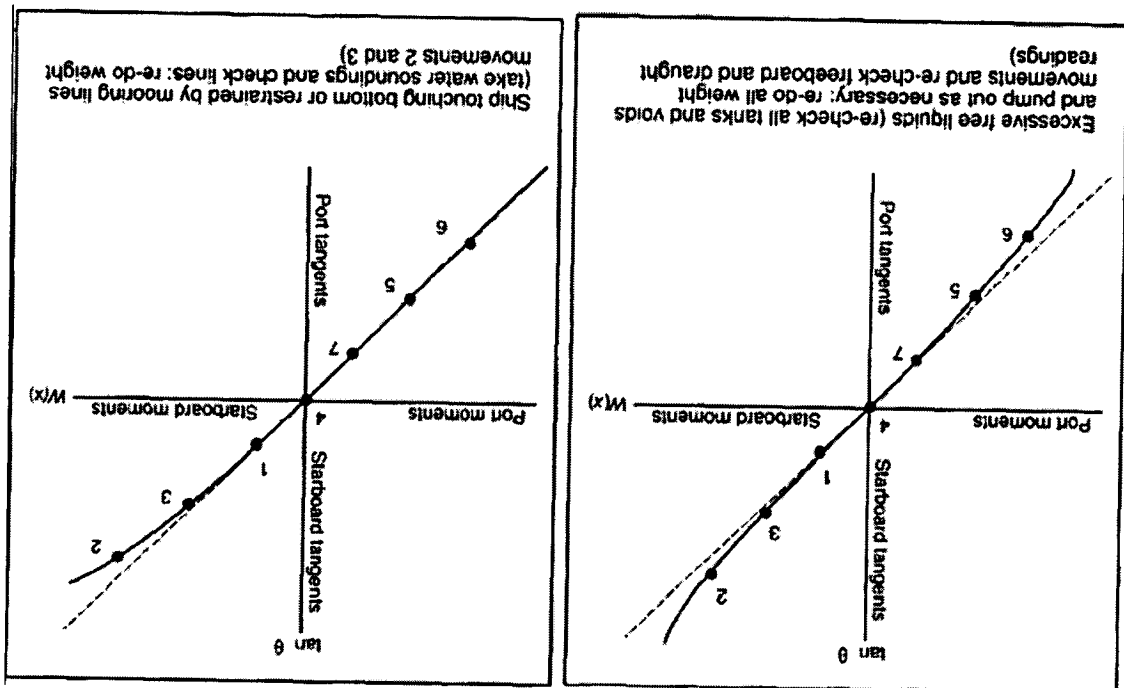


Fig. 5.1

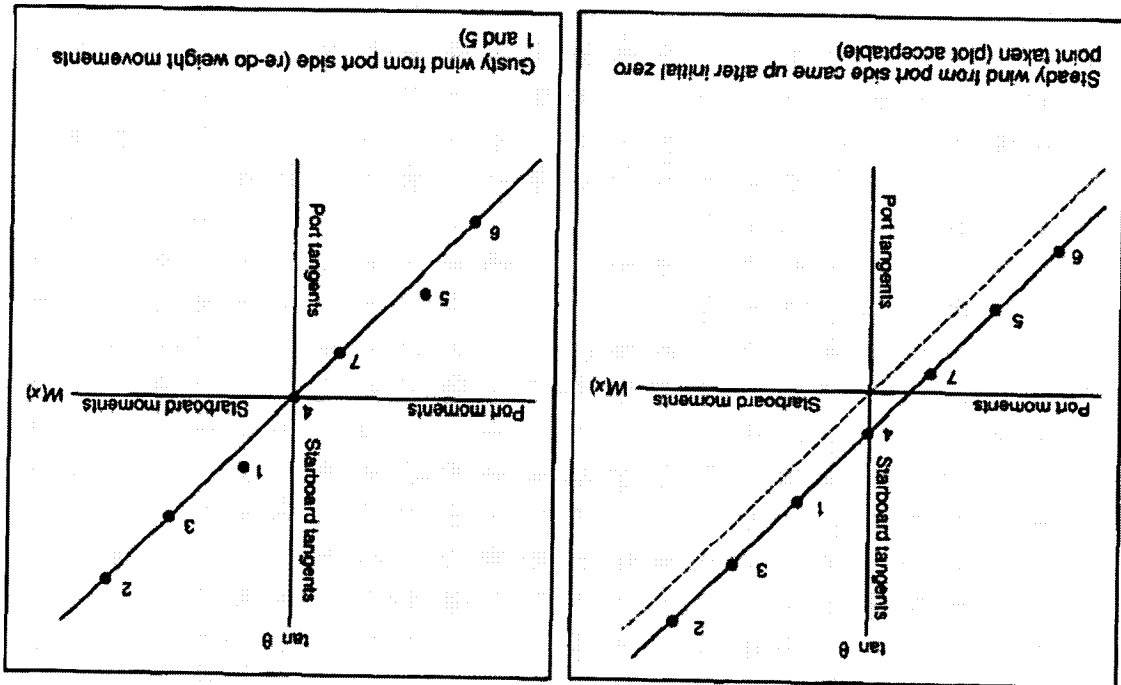


Fig. 5.2

Fig. 5.3

Fig. 5.4

- Note : The surveyor should follow the remedial measures as shown in Figures 5.1 – 5.4, if a linear plot is not obtained.